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Technical Assistance to the National Planning Authority, including a Feasibility Study for Establishing a Comprehensive Iron and Steel Industry in Uganda

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Abbreviations

Abbreviation	Meaning
%Fe	Percent Iron
AfCFTA	African Continental Free Trade Area
ASM	Artisanal and Small-Scale Miners
BF	Blast Furnace
BIF	Banded Iron Formation
BOF	Basic Oxygen Furnace
CA	Contracting Authority
Capex	Capital expenditures
CCM	Continuous Casting Machine
CDP	Continuous Development Program
COG	Coke Oven Gas
DGSM	Directorate of the Geological Survey and Mines
DRI	Direct Reduced Iron
DRIOC	Direct Reduced Iron Ore Company
DRC	Democratic Republic of the Congo
DRP	Direct Reducing Plant
DSGE	Dynamic General Equilibrium Model
EAC	East African Community
EAf	Electric Arc Furnace
EPC	Engineering, Procurement & Construction
ESP	Electrostatic Precipitator
EU	European Union
Fe met	Metallized Iron
Fe Tot	Total Iron
GDP	Gross Domestic Product
GIS	Geographical/Geospatial Information System
HBI	Hot Briquetted Iron
HDRI	Hot Direct Reduced Iron
HRC	Hot Rolled Coil
I&S	Iron and Steel
IMCU	International Mining Company of Uganda
IOT	Internet of Things
LF	Ladle Furnace
MGR	Meter Gauge Rail
MIC	Made in China
MoSTI	Ministry of Science, Technology and Innovation
MOU	Memorandum of Understanding
NDP	National Development Plan
NG	Natural Gas
NPA	National Planning Authority
Opex	Operating expenditures

RHF	Rotary Hearth Furnace
RKF	Rotary Kiln Furnace
ROI	Return on Investment
SADC	Southern African Development Community
SGR	Standard Gauge Rail
SME	Small and Medium Enterprise
SMI	Sino Minerals International
TPA	Tonnes Per Annum
SMIC	Sino Minerals Investment Company
SMS	Steel Melting Shop
TIEG	Transport Infrastructure Expertise Group
ToR	Terms of Reference
UBOS	Uganda Bureau of Statistic
UGSM	Uganda Geological Survey and Mining
UNBS	Uganda National Bureau of Standards
VRF	Vertical Reactor Furnace

EXECUTIVE SUMMARY

I. Introduction

The European Union funded a project on behalf of the Ugandan Government with an overall objective of supporting the sustainable development of the iron and steel value chain in Uganda; providing specific recommendations to ensure success in the development of the value chain. The project also included assessing the country's total demand for natural gas with an aim of providing information for the natural gas pipeline from Tanzania to Uganda, which gas would serve as a reductant for iron ore. The specific objectives were:

1. to help bridge the gaps along Uganda's iron and steel value chain – from iron ore mining ("Phase 1") to iron ore smelting ("Phase 2"), to steelmaking and refining ("Phases 3 and 4"), to steel rolling and marketing ("Phase 5 and 6") and,
2. to assess the country's total demand for natural gas, both for industrial – including, but not restricted to, iron ore-smelting – and domestic use.

This report examines the viability of Uganda's commitment to build/develop an iron and steel (integrated) industry with capacity to meet the domestic and part of the regional and global market.

Underpinning this research are the following analytical outcomes, recommendations and conclusions.

II. Scope

Uganda wishes to develop a full iron and steel value chain using its iron ore deposits rather than importing raw material inputs for steelmaking, hot rolling and cold drawing. The iron and steel industry is widely known as a backbone of industrialisation in any economy. It is therefore prudent for Uganda to pursue this agenda in order to support the process(es) of industrialisation and economic development. Uganda's iron ore deposits in the south western region are potentially among the highest quality iron ore deposits in the world. Given this abundant natural resource, Uganda can therefore advance its plans to tap into this resource to feed into the iron and steel value chain, thereby halting the threat facing Uganda's steel manufacturing industry – i.e. scarcity of scrap metal currently used as raw material for liquid steel production and import of expensive semi-finished steel for steel products' manufacturing, which makes Uganda's steel products expensive and the industry uncompetitive when compared to other players in the region. This research is therefore designed to determine the options available for the development of Uganda's iron and steel industry and markets, and the extent to which it can go to utilise its natural iron resource for liquid steel production in the short and long term.

III. Market Evolution Analysis

Analysis of the upstream and downstream supply and demand scenarios indicated potential for the establishment of an iron and steel industry value chain that can sustain the steel requirements of Uganda, with a potential for exporting some of the steel products in the region, particularly the land locked countries.

Supply Side Analysis:

Iron ore is known to exist in Uganda, with the south eastern region hosting hematite deposits and the eastern region hosting magnetite deposits. The hematite deposits are known to be of larger quantities and superior quality as compared to the magnetite deposits and therefore the focus of this research was on the hematite deposits as the main source of raw material for the value chain. Field and literature research established the following characteristics and recommendations for the hematite deposits:

- A reconnaissance survey of South Western Uganda confirmed the existence of hematite deposits that visually show characteristics of high-quality iron ore (55-68% Fe) suitable for the proposed DRI method of iron oxide reduction.
- From surficial investigations, the deposits are mainly comprised of thin beds of mineralization ranging from 5 to +/-100m. The full geometry of the deposits could not be established due to limited exploration data for the region.
- The deposits occur in the form of massive hematite, which is amenable to mining and direct feed to the ore processing plants, and specularite hematite (a metamorphic platy variety), which normally requires mineral processing for amenability to reduction processes. Both ore types have potential for high fines content on mining such that sintering and/or pelletization may be required prior to feeding into iron making processes.
- Most of the visited areas have been parcelled out to explorers and miners, most of which are SME category. Iron industry being a bulk commodity industry requires proper regulation to ensure that licensees are credible, with the financial muscle to fully optimize the resource. A review of the current policies as to who should own a license is necessary if the resource is to be exploited in an efficient manner that benefits the communities and the country.
- It could be vital for the government to consider the iron deposit(s) a national asset and therefore invest in value addition by taking the responsibility of implementing modern exploration to quantify the resource. By so doing the government can then put measures as to who can be issued with a license to economically develop the deposits and maximize benefit to the country. With known quantities and quality of the deposits, including preliminary metallurgical tests, the government can lock in some value by putting potential investors on tender using a benchmark price that takes into account the amount of investment committed by the government at mineral resource development stage, or alternatively the government can use that value to negotiate equity in prospective Public Private Partnership deals.
- Government has to start planning for local content skills requirements and development to ensure maximum benefit to the country. At present there is limited scope on the availability of credible geologists, mining and process engineers, and other support skilled personnel that allows the establishment of a successful mining operation that can efficiently supply the iron making processes with quality raw material feed.
- The fact that the ore is outcropping points to the possibility of open pit mining method as opposed to the more costly underground mining. However, the variations in deposit thicknesses may have a bearing on the amount of overburden to be stripped when mining, a scenario which further emphasises the need for mapping to ascertain lateral and width extent of the deposit lenses to be in a position to select the more prospective areas of the deposits that can be developed at lower costs.

- Coal, which is readily available in Tanzania can be an important reductant in the short term with players using coal based DRPs to produce sponge or iron briquettes for steel production. The main constrain is transport logistics from Tanzania to the iron reduction plants, which should be addressed at national level.
- In the event of having increased demand for steel products, gas from Tanzania is the most reliable source of reductant should the industry ramp up production to allow gas based DRI processes.
- Limestone and Dolomite, which form part of the flux feed in the furnaces are also available in Uganda.

Demand Side Analysis:

Uganda has several steel players of varying production capacities and operating at different stages of the downstream Iron and Steel (I&S) value chain. There is a huge variety of different sizes of plants among the steel producers and producing different mix of products. The following constitute key outcomes and recommendations of the research:

- Of the key players identified, most of them are steel processors using semi-finished steel sourced from abroad. A few have moved into liquid steel production by induction furnaces using scrap metal which is sourced either locally or from regional countries in Africa. Two steel companies, Dongsong and Tembo Steel, are at development stage of DRPs to produce locally made DRI for the steel industry.
- Presently in Uganda there is an over installed capacity in the steel players plants of final products production compared to the actual country needs. The total amount of final products produced in 2018 was about one third (half a million versus 1.5 million tonnes) of their installed plants capacity. This difference between produced quantities and installed capacity is important and its value really impacting the country's iron and steel industry development in the coming years. In order to reach the stage at which it will be necessary to install new steel plants, it is mandatory to reach a steel demand at least at the level of the installed plants capacity.
- By inference, the 0.5Mt/year of steel products equate to the current market demand for steel in Uganda. As there is a critical shortage of scrap metal in the country to continue sustaining the liquid steel processes, Uganda has an opportunity to produce 0.5Mt/y of iron ingots/briquettes/sponge from its iron ore resource to feed into the existing steel industry. This opportunity would equate to at least 800,000t/y of iron ore mining for raw material supply to the iron making steel plants.
- Actual steel processing is limited to long products with low carbon content (barbed wire, nails, meshed wire, hollow sections, etc.) and some cold rolled coils for roofing material or galvanized sheets, all considered not high-level steel products. The low quality of final products is not properly supporting the development of industrialization.
- Only eight plants have capability of producing finished products by liquid steel and only seven are actually using it (namely Roofing Steel, Steel & Tube, Tembo Steel in both Lugazi and Iganga plants, Tian Tang, Pramukh Steel and Yogi Steel for about 45% of finished products). The rest of the plants are importing semi-finished products (Plate Hot and Cold Rolled coils, Billets and Wire rod coils), which consequently makes the final products rather expensive and not competitive on the international market due to high transport costs. This import level confirms the immediate need to increase internal liquid steel production, an opportunity area for Uganda's steel industry to close the gap.

- Products for construction works (heavy profiles and structural steel) are almost absent in Uganda due to unavailability of special steel products (High Alloy) that require dedicated production plants. Whilst developing this steel sector is unforeseen in the near future due to high tech requirements, its existence can cut down on imports and significantly promote heavy infrastructure industry, and therefore economic growth, like the planned new dams, upgrade of Entebbe airport and other new airports and aerodromes, new SGR (railway) projects etc. Only re-bars or small profiles can be utilized for these projects with minor direct impact on the local industry.
- Uganda's steel demand is expected to gradually grow in line with the country's projected demographics (population growth and urban migration) and economic growth, which, together with the planned infrastructures - expected to lower transport costs - and a natural gas pipeline in place, can the country consider developing new more efficient steel producing methods. For example, gas-based DRI with vertical reactors, can be envisioned to match-up with the increased steel demand by the year 2028. In the meantime, the current scenario of growth through private investment should continue to feed the market until such a time the environment is considered economic to construct a large process plant. Within this period, the government can actively support the current steel industry by providing the necessary infrastructure, policies and regulations that are favourable for the local development of the industry. By developing such measures today, the government would have created a competitive environment for the industry to thrive until such a time the steel demand warrants bigger investments in the industry. The Government should therefore play an important role in the short term by putting in place measures like introduction of taxation systems of imported semi-products for those products which can be produced locally from liquid steel, and at the same time facilitating investment on DRI facilities with RKF's in key players' plants by implementing tax concession systems, loan instrument facilities at attractive rates and efficient utility systems at competitive rates..
- Given that other EAC countries have similar scenarios to Uganda's steel industry i.e. importing semi-finished steel products as raw feed for their steel plants, there is opportunity for Uganda to ramp-up capacity for export to the region only if quality of products is improved and costs of production significantly reduced by improved infrastructure development. The current higher cost of importing raw steel as compared to other EAC countries with port facilities can be curtailed by local production, provided all the other drivers in the value chain (infrastructure, gas pipeline, competitive utility etc.) are addressed by the government, thereby making Uganda's steel competitive on the regional market.

IV. Iron Ore Reduction Options

Available iron making technologies were analysed taking into cognisance the existing market of the Uganda steel industry.

1. Currently the production of steel in the world can be divided in two process routes namely reduction processes, utilising iron ore as the main raw material feed (BFs, DRPs); and melting furnaces utilising scrap, cast iron and DRI products (induction furnaces, EAFs). Electrical energy and a stable grid are basic requirements for these processes.
2. Blast furnace is not a viable option even in the lower production plants (mini-blast furnaces) due to high capex on establishing the integrated plant and high operating costs.
3. DRI can be viable now only with carbon-based plants and in future, when natural gas is made available, with gas-based reactors.

Direct Reduced Iron Making Routes:

DRI processes can either be coal based or gas based and are generally put in place at lower capex than the BF process making them a more viable option than the BF process. Technology continues to advance in this process route providing options for the Uganda steel industry development.

1. Rotary Kiln Furnace

RKF are limited in capacity from 50,000 to 200,000 tonnes per annum. Production can however be doubled by installing two units on one common charging system, thus having the yearly production capacity reaching up to 400,000 t/y, which makes this process suitable for the Ugandan environment considering that the steel industry is young and growing requiring limited amount of steel products for its market.

Lump hematite ore (4-18 mm) or hematite pellets (9-30 mm) and a relatively coarse fraction of non-coking coal (6–20 mm) fluxed with fine limestone or dolomite, are required as raw material feed. Given the ore characteristics of hematite in South West Uganda, investment in pelletisation plant may be required to adopt this reduction process.

Considering the current scenario in Uganda steel industry and that the proposed natural gas from Tanzania is still in the pipeline, the industry is encouraged to adopt this route. Suitable coal is readily available in Tanzania and limestone/dolomite is available in Uganda.

2. Rotary hearth furnace

The main benefit in RHF is the wide capacity range from 20,000t/y to 700,000t/y (optimum 300,000 to 500,000t/y plants). The RHF can be charged with any type of hematite or magnetite without particular quality restriction, thus widening the scope of iron ore source to the magnetite deposits of Eastern Uganda. In addition, any coal, even with high volatiles, is suitable for this process, making the low-grade coal deposits of Tanzania potential raw material source.

The main economic potential advantages of RHF plants are the ability to use a wide variety of iron oxides and high volatile coal - low-cost iron ore fines and inexpensive steam coal. Moreover, steel plant wastes and coke breeze and/or carbon bearing wastes can also be used aiming toward the ultimate goal of becoming a zero-waste production facility.

The main downside is maintenance, which is difficult, and is associated with high general expenses. As a result, Return on Investment (ROI) can take longer to be realised.

3. Vertical Reactor Furnace

VRF are gas based DRI processes normally requiring large quantities of gas in the order of 300–350 Nm³ per tonne of DRI. Typical production capability is in the range of 300,000 t/year (mini module) up to 1.000.000 t/year (maxi module); but a more optimized plant in terms of cost of installation and production should be in the order of 500.000 t/year.

These processes require higher quality hematite than RKF or RHF. Laboratory tests are necessary to characterize the hematite in terms of friability, impurity etc. to assure their possible use in the reactor. Pellets and lump iron ore can both be used, but pelletized ore is the only option if the iron ore is friable. If iron ore quality is not adequate, then a beneficiation plant is required.

VRF plants are very high and must be supported by steel structures. The erection is complex due to the need for massive site welding of components of the vertical reactor, and due to a very high level of erection. Crane sizes of 700-800 tonnes is required to be available on site for erection of over 120 meters of heavy weights, a logistical constraint for many countries requiring this technology. Capex and Opex are generally on the higher side for these processes.

Whilst it is a challenge to erect these plants, they are generally more efficiency and environmentally friendly, and the iron produced is of higher quality than that produced by other DRI methods.

The requirement for natural gas for the VRF plants implies it cannot be utilized in the Uganda steel industry in the short to medium term but can become the best option should the natural gas from Tanzania becomes available for Uganda's consumption.

Besides NG requirement, stable electric energy (mainly from the grid) and increased market for steel products are necessary for establishment of VRF.

V. Iron and Steel Value Chain

1. Infrastructure and Logistics

Uganda has over the years established a steel manufacturing industry and installed capacity to date is now around 1.5 million tonnes. At one time Uganda's private players were into small scale mining of iron ore for export. For the steel manufacturing industries, over 60% of inputs for steel manufacturing were and are still being imported. These scenarios point to a paradox or irony in that the expected scenario was to mine the ore, produce steel semis via liquid steel and supply the rolling mills. The reasons given for the non-commercial mining of the high-grade iron ore for direct export of the commodity before the ban was lack of infrastructure among others. There is no commercial mining of the iron ore for local liquid steel production from iron ore raw material because there is no established iron ore reduction and smelting industry in the country because of lack of infrastructure for road, rail, ferries and barges transport systems, high voltage power transmission lines and no infrastructure to bring in natural gas, the preferred reductant that would make DRI based iron and steelmaking a competitive venture. Chapter 7 was dedicated to infrastructure and logistical issues and it examines these issues and provides some recommendations. Some of the issues that need to be considered include inadequate transportation means for a medium steel industry, which result in transport costs that are too high.

The other constraint is that there is not enough power for the furnaces. At present the energy required to produce liquid steel from scrap is about 630-640 kWh/t, which is normally 75% electrical and 25% chemical for the modern EAF. As a reference the power of a medium size EAF (100 ton/day) is 120-160MVA. Also, stability of electric energy is a major issue due to the actual status of the grid distribution. Stable energy is required to the steel industry that needs to operate on continuous production basis. Technologies that allow refining of liquid steel and thus production of alloyed qualities require processes like EAFs, which require high peaks on the electricity grid due to the presence of arc instability mainly in the first phases of the scrap melting process. The induction furnace is the alternative because the electricity requirement from the grid is constant but in this process is not possible to refine the steel, which therefore calls for high scrap quality charge.

It is therefore necessary that the state should put in place all the requisite infrastructure singularly or through PPP. This includes better roads for mining and haulage of the ores to relevant sites. The relevant transportation infrastructure projects that are presented in the document "6.4-Corridor-Diagnostic Study-NCC-EA-v1" need to be actualized **in phases**, as described in chapter 7, starting with those that would enable mining of iron ore, manufacturing of iron and steel, and trade of iron ore (if ban on export of iron ore is lifted), DRI, Steel and other by-products.

The maps in Figures 22, 23 and 24 show that infrastructure and logistics have an influence on the selection of the suggested site of the proposed iron and steel complex in Kabale. Some of the factors considered include:

- Distance from Iron Ore Deposits: Rugando, Muko, and Buhara

- Proximity to limestone deposit
- Distance from proposed Gas Line
- Proximity to Power Lines
- Proximity to Road and Rail Networks
- Proximity to Local Markets
- Proximity to export markets

2. Value Chain Analysis

Value chain analysis points to competitiveness, sustainability and profitability of the iron and steel industry once the manufacturing model shifts towards value addition of local ores. Uganda should therefore consider taking the necessary steps to establish iron ore mining, iron ore reduction and steel production using the recommended DRI process.

Establishing an I & S value chain from iron ore mining to the markets constitutes a strategic position in the steel sector and the wider industrialization for a country, which conforms to Uganda Vision 2040. A compilation of Uganda's current I & S value chain, as tabulated in Table 4, shows the current situation for steel companies in Uganda and their positions on the value chain. From the table, it is evident that exploration activity is minimal, with only Sino Minerals Investment Company Ltd. (SMIC) having carried out credible exploration activity among the 16 licensed mining companies operating in the region. In the mining phase, activity was observed from SMIC and Direct Reduced Iron Ore Company (DRIOC), whilst International Mining Company of Uganda (IMCU) is guarding its site for possible resumption of mining in the event of the government lifting the ban on iron ore export. **In summary, iron ore mining in Uganda is next to being inactive, with only artisanal miners and small-scale processors like DRIOC being active in the region.** SMIC is the only company expected to establish a more professional mining operation soon. **In the iron making phase (DRI production) only Steel Rolling are operating at Jinja using hematite ore supplied by DRIOC.** In this analysis, it can be summarized that the lack of iron reduction and its subsequent smelting in steel melting shop (The "Gap") is the major reason why Uganda's steel manufacturing industries continue to face the existential threat from scrap scarcity and imported semi steel products at high cost. It is therefore prudent to conclude that this scenario should not happen for a country which is endowed with premium grade iron ores and high potential for electricity generation, coupled with availability of reductants in the region like the coal and natural gas available in neighbouring Tanzania.

As a response to the said threat the analysis considers actions mainly by the existing private sector and national plans being considered for the establishment of a viable and sustainable iron and steel industry in Uganda. In summary, the following two phased approach should be considered for the development of the value chain:

• **Private Sector Backward Integration and New Investments (Phase 1)**

In order to quickly arrest further imports of billets and coil rods which can be made in Uganda, the three existing companies (Tembo Lugazi, Steel Rolling and Pramukh) and the two upcoming companies (Kigezi and Dong Song), already licensed for DRI production, should be encouraged and supported to meet implementation targets. The companies also need support by creating an enabling business environment including infrastructure like rail, rail spurs, wide roads and adequate electrical energy supply.

• **Establishment of Steelworks in Kabale**

With increased demand, Uganda can then consider the mover to establish a steelworks complex, initially to meet the 1.5 Million tonnes per annum production as estimated by the economic models. This goes together with, among other things, having the suggested infrastructure in place.

The analysis goes further to examine the competitiveness of these two phases and it has shown that manufacturing iron and steel using the iron ore in the Southwest is competitive, assuming detailed geological exploration to assess the ore bodies will point towards low priced iron ore as is being achieved currently. The geology will confirm whether mining can be done sustainably over the life cycle of the assets at the metallurgical complex or even over human and plant generations to come. The value chain analysis further touches on the markets which are indeed the driving engine for steel manufacturing or any business. Further details from market study will confirm viability or lack of it just as geology will confirm sustainability/viability. The analyses made were all on the assumption of the products finding the suggested profitable homes.

Based on the assumptions of holding market conditions constant, section 8.4 looks at Local, Regional and International markets, which generally shows the potential for Uganda steel to be competitive in the domestic and regional markets. The product mix is one critical area requiring careful planning as some of the steel imports are currently not being produced by any manufacturer in Uganda. There must be efforts to bring the required technology, develop skills and produce imported steels products equivalence from within the country. Areas of competitive advantage that can propel the Uganda steel industry should be identified for implementation based on cost leadership and differentiation strategies.

Finally, there is the acknowledgement of the technical and political will witnessed during the various visits to the stakeholders. It is to be acknowledged that over and above the hard issues of getting the purely technical issues going, it is necessary to start skills and supplier development programs so as to enhance the policy of Local Content as spelt out by the Government of Uganda.

VI. Conclusions

Whilst there is potential for huge iron ore deposits in South West Uganda, consumption of steel products within the country is generally low to warrant the development of an integrated iron and steel industry in the current situation. The private players active in the industry should continue to develop the industry as determined by the market forces until such a time when the demand for steel products increases to levels that warrant the development of gas based DRPs.

The economic model predicts a levelling effect of consumption primarily due to population growth surpassing economic growth as detailed in section 5.3.2. This stagnation of consumption can be overcome by government intervention putting up measures to grow the industry like infrastructure development, road rail network and water transportation system development. Tariffs on electricity and policies that protect the industry must be reviewed and aligned with those of other countries to facilitate competitiveness. The gas pipeline development should be considered a priority to ensure availability of iron oxide reductant at such a time the market warrants the installation of a DRI plant, which could be in five to eight years as predicted by the economic model.

With the right measures put in place to promote the industry, steel can be produced at competitive cost thereby making it marketable to at least the regional countries in EAC but also capturing a potential export market in inland confined regions with similar transportation impact of cost as Uganda, like DRC, Sudan and some of the SADC countries.

1. INTRODUCTION

Uganda Vision 2040, which was officially launched in 2013, is a 30-year vision being implemented by the National Planning Authority (NPA) and other Government Institutions through three 10-year plans and six 5-year National Development Plans (NDPs). It is aimed at transforming the country from a predominantly peasant and low-income country to a competitive middle-income country by the 2040s through increased sustainable production, productivity and value addition in key growth opportunity areas of which the Iron and Steel (I&S) Value Chain has been identified as one.

The first and second National Development Plans (NDPI & II) emphasize the importance of developing iron and steel industry in Uganda as one of the key growth opportunity areas, and to support this initiative, the Government of Uganda, with the financial assistance of the European Union (EU), has moved to conduct a comprehensive analysis in this key growth opportunity area. This project is therefore implemented to explore the viability of the Iron and Steel value chain in Uganda for both the domestic and the global market. It involves assessing the current status of the market, including existing steel making infrastructure, and recommending proposals for the sustainable development of the sector.

At present, Uganda has about fifteen main iron and steel companies producing downstream steel products for the Ugandan market, and a few (about 100,000 t/year) for export to East African Community (EAC) countries. Most of the raw material, in the form of semi-processed steel, for these steel industries is imported from abroad, mainly Japan and South Africa. In addition to imported raw material, few companies produce intermediate steel from scrap metal by induction furnaces for use as raw material for final steel products manufacture.

Whereas Uganda has iron ore deposits mainly in the south western and eastern parts of the country, which remains hugely untapped, the existing industry has not taken full advantage of this natural resource as a source of raw material for its downstream steel industry. Some companies, like Guangzhou Dongsong Energy Uganda Limited and Tembo Steel Ltd, have projects in the pipeline for iron ore reduction to produce intermediate steel products as raw material for the existing industry and/or to produce final steel products. Several companies are into iron ore exploration and mining, with the aim of selling the ore to upcoming iron ore processing companies or to develop captive mines for their own steel processing facilities. Research shows that most of the players in the exploration and mining of iron ores range from artisanal to SME to intermediate in size but are nowhere near any significant economic utilisation of the “huge iron ore deposits” as predicted in previous studies.

It is this gap the government of Uganda would like to address in order to develop a complete iron and steel value chain, which is expected to partly fulfil the objectives derived from Vision 2040 – with some of the envisioned benefits being creation of support and service industries; growth in job creation, GDP, and foreign currency earnings from exports of steel products; and savings in foreign currency when raw steel imports are curtailed by the development of the upstream industry. In any case, the current laws of Uganda prohibits the export of any unprocessed metal ores/minerals prior to beneficiation implying that in the current scenario, the economic value of iron ore deposits of Uganda can only be realised with the development of a full value chain from upstream mining to downstream production of steel products. The feasibility study was therefore commissioned to determine the viability of mining and processing iron ore deposits of Uganda, the viability of developing steelworks plants and steel products, and the impact of that development on the existing steel industry and the economy of Uganda.

Most of the focus of this study has been concentrated on the hematite deposits of the south west for reasons that the deposits are more abundant and are of higher quality than the magnetite deposits of eastern Uganda. The hematite deposits are expected to be more economic to mine and process and

more amenable to iron oxide reduction processes. To sustain the proposed industry, the study also looks into policy issues with the aim of developing industry policies that supports economic growth of the country for the benefit of its people, including but not limited to sustainability considerations in areas of environment, social and communities, gender, and local content development in key skills shortage-areas required for the industry.

Whilst the study explores options available for the reduction of hematite ore, the Ugandan government's proposed method of reduction is gas based mainly due to the availability of natural gas in the region for use as a reductant in iron making. Given that the hematite deposits are of high-grade quality (55-68 %Fe), the DRI method stands out as an efficient reduction method capable of achieving the required result at lower costs.

However it is necessary to highlight that the gas-based DRP (Midrex, HYL) have a production capability in the range of 300,000 t/year (mini module) up to 1,000,000 t/year (maxi module); but a more optimized plant in terms of cost of installation and production should be in the order of 500,000 t/year. In accordance with Uganda steel market, this solution seems not suitable for the present needs. However, if the country preconditions are fulfilled (adequate electricity and transport network) by the time in which gas will be available, then the scenario of exporting steel to neighbouring countries can materialize and bigger centralized plants can be considered for the steel industry in Uganda.

In the immediate future, the technology should be RKF coal fed process to produce liquid steel. This is due to the small market size and absence of natural gas, which is expected to be the reductant raw material for the DRI processes should the market increase to levels that are economic to implement this technology. Natural gas is abundant in Tanzania (onshore and offshore), Lake Kivu in Rwanda, and the recently discovered gas in the Albertine region of North West Uganda. The massive gas reservoirs in Tanzania of about 57.3 trillion ft³, with projected lifespan under current and planned use of 80 years, are already under production. It comes as no surprise that the Ugandan government has leaned towards the Tanzanian source given that it is readily available should the development of the iron and steel value chain start soon. A Memorandum of Understanding (MOU) is already in place between the two governments (Uganda and Tanzania) for a joint program to pipe the gas from Tanzania to Uganda for the benefit of both countries. In addition to the supply of gas for the iron and steel furnaces, Uganda would like to take that advantage to extend the use of gas to other industries and the domestic market. It is in light of this that a parallel study to this research was commissioned; to determine the holistic gas demand in Uganda, including the required infrastructure and logistics to harness it to the end user.

2. OBJECTIVE AND METHODOLOGY

2.1. Objective

The objective of this report is fourfold:

- To provide analysis and overview of the geology, and characterisation of the iron ore deposits, of South West Uganda as a basis for the subsequent analysis of the units of the iron and steel value chain, including proposed mining method, processing requirements, metallurgical infrastructure type and location, and transport logistics.
- To provide quantitative market evolution analysis in the geographical areas of interest as defined in the Terms of Reference (ToR) of this project. This should enable the NPA to understand the general steel market dynamics in perspective of the strategy to develop a complete sustainable iron and steel industry in Uganda.
- To provide the rationale for the choice of infrastructures and their locations including transport logistics.
- To provide a work plan to the NPA for the delivery of the units of the iron and steel value chain, including sequencing, scheduling and expected costs for implementation in its NDP III programme.

2.2. Methodology

The report has been prepared based on both desk and field research, with several procedures being implemented in order to meet the objectives.

2.2.1. Geology and Mining

Available geological literature was collected from various sources, mainly from the Directorate of the Geological Survey and Mines (DGSM) and analysed in order to understand the mode of iron ore mineralization in Uganda - in particular, iron ore deposits of South West Uganda.

A field excursion to the iron ore deposits of South West Uganda was implemented in order to: ground truth the information obtained from literature research; carryout a detailed reconnaissance study so as to benchmark the likely scenarios required in the development of the value chain; and to establish the likely mining and processing requirements of the deposits. As part of the excursion, the land surrounding the ore deposits was assessed for suitability for infrastructure development and logistics taking into consideration the environment and social issues.

2.2.2. Market Evolution Analysis

Literature collected from the NPA, industry stakeholders and internet-based platforms was used to draw baseline market economics of the steel industry, with a focus on Uganda as the primary market of the steel products but not negating the EAC countries and the rest of the world.

Analysis of production and trade statistics (World Steel Statistics, UN trade statistics, specific import/export data from Uganda Bureau of Statistics (UBOS)) provided benchmarks as to how much steel Uganda consumes per capita and as a country. Market assessment was made in terms of “finished steel”, mainly classified into Hot rolled long steel and Cold rolled/coated flat steel.

Tours and visits to existing steel industries enabled capture of real quantitative data on raw material input to the industry, product types, capacities and efficiencies, with a component on business improvement and training. Steel companies visited are mainly domiciled in Kampala, Jinja and Tororo.

Definition of products: For the current state of development, the steel market is segmented into “flat product” and “long product” market for carbon steel qualities. Raw material requirements for those products are therefore determined by backward integration, with the result being used in a top down approach to determine possible economic parameters of establishing the upstream and intermediate industry units of the value chain. Whilst the evaluation of Uganda steel market use both real industry data and literature research, the market component for export is predominantly based on literature research.

In the event of Uganda developing the iron and steel value chain, its immediate target global market is the EAC countries, which, like Uganda, imports raw steel for its existing steel industries, an opportunity for Uganda to have a share of the regional raw steel cake by exporting part of its semi processed steel products to EAC. In long term, Uganda foresees itself meeting a percent of the regional market demand, which, again by backward integration, is factored in and added to the domestic raw steel demand to arrive at the total required quantities of iron ore production from the upstream side. The figure is then used to advise the NPA and Government on the quantitative figures on iron ore mine production required to feed the steel making plants, the size of the steel plants, the quantities of reductants and other steel making inputs required, the type of the intermediate steel products to be produced; and the exploration procedures necessary to ascertain whether the iron ore deposits of South West Uganda can be economically mined to meet this raw steel market.

Sustainability of the industry forms a key component of the research, which necessitated the need to review the existing draft steel policy and provide advice to the NPA and Government on best practice that promote investment in the steel industry. Incorporated in this component are visits to iron and steel industry stakeholders in South Africa to draw analogy lines between the proposed Uganda policies and those of South Africa (SA), which enabled its (SA) iron and steel industry to thrive in this competitive market. Similarly, for the Uganda steel policy, a review will inform the research whether Ugandan policies are designed to promote investment and growth in the steel industry.

Consumption forecasts enables the determination of the likely steel demand in future, which then advises the government on the required industry growth rate if the steel industry is to stay at par with demand. Analyses of trends of relevant economic indicators such as population demographics, employment, literacy, GDP, energy consumption etc. as provided by the UBOS, enabled consumption forecasting and therefore steel demand in future. The research will therefore advise on the likely trends in the growth of the industry based on these economic indicators and provide economic and capability models that can be used for such predictions. This will advise the government on the size of steel products expected in a certain time in future, which again by backward integration, will inform the industry on the quantities of raw steel required, and the size of the steel industry and mining operation to meet that demand.

2.2.3. Infrastructure and Logistics

The field trip to South West Uganda included a component on area assessment for the purposes of establishing suitability for developing an integrated iron and steel infrastructure so as to draw lines of comparison with other potential locations nearer to the existing downstream steel industries, which are likely to be the main consumers of the raw steel products should the proposed integrated steel industry materialize.

The research is expected to fully evaluate other considerations when deciding on the most optimal location for the steel infrastructure development including social and environmental considerations when factoring in transport logistics for: iron ore transportation from source to the steelworks; and also transport logistics from the steelworks to the market.

A road map on the proposed infrastructural development will advise the government of Uganda on the required infrastructure and sustainable transport systems for delivering the commodity at each stage of the value chain from one business unit to the next. Other key consideration is for Uganda to invest in infrastructure that promote investment in all sectors of the economy, which will, directly or indirectly, impact on the ramp-up of the iron and steel value chain.

A similar analysis for the gas demand is expected as it forms an integral part of the iron and steel value chain, as further explained in the Gas Demand Report.

2.2.4. Recommendations

On completion of all analysis and evaluations, a work plan for the most feasible scenarios forms an integral part of the final reporting, advising the NPA and Government of Uganda on the likely pathways to follow for the development of the value chain, including estimates of schedules and costs. These are provided in detail after each chapter and in summary in the executive summary and the conclusions chapters.

3. GEOLOGY

3.1. Geological Setting of Uganda

Uganda has a geological history spanning more than 3 billion years (3Ga). It is part of the Congo-Tanzania Craton, lying on the north-eastern part, comprising of Archaean lithospheric fragments which are intersected or surrounded by Proterozoic fold belts related to a series of orogenic cycles including the Eburnian (2.20–1.85 Ga), Grenvillean (1.10–0.95 Ga) and Pan-African (0.75–0.50 Ga) orogenic cycles. Post Pan-African orogenic cycle are sedimentary suites of the Karoo and the Neogene East Africa Rift System comprising mainly of Molasse assemblages including shales, sandstones and conglomerates.

The iron ore deposits of Uganda fall into two systems, the **Karagwe-Ankolean system (1.8-2Ga)** hosting the hematite deposits of South West Uganda and the Alkaline/Carbonatite complexes of the Cenozoic period (~40Ma) hosting magnetite deposits of Eastern Uganda as shown in figure 1.

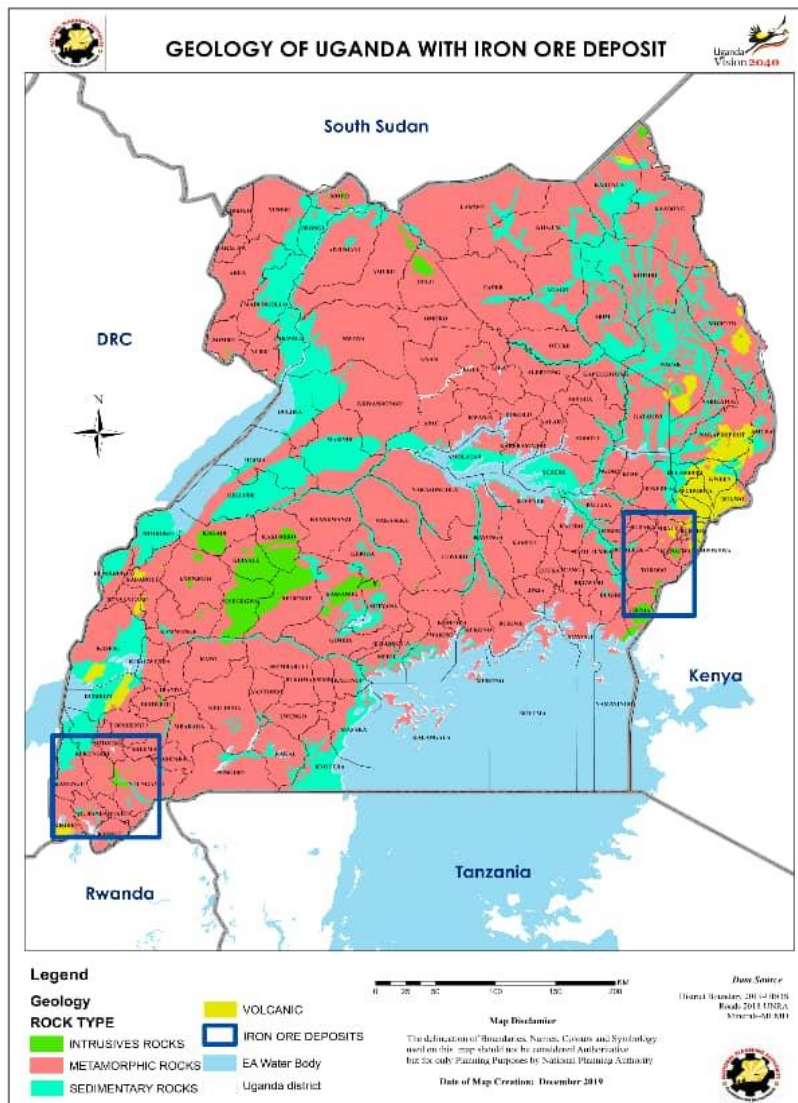


Figure 1: Geological Map of Uganda and the locations of Iron Ore Deposits (Source – NPA)

3.2. Hematite deposits of South West Uganda

Between the two ironstone deposits, the hematite ores of South West Uganda are considered more prospective with potential for huge discoveries. They are also of a superior grade with most of the samples collected in the area posting iron content ranging from 55 to 68 %Fe (Ref: Literature Review, DGSM library literature and SINO Minerals Investments Ltd site visit) as opposed to magnetite deposits of Eastern Uganda with iron content of 40 to 62 %Fe (Ref: Literature Review and 3M Mining Ltd Exploration data from the area). For this reason, this research focuses on the iron ore deposits of South West Uganda as the preferred source of raw material for the proposed integrated iron and steel value chain development.

Reconnaissance survey of hematite deposits in South West Uganda, in and around the Town of Kabale, was conducted with a view to better understand and to assess the potential resource base expected to provide raw material feed to the proposed iron and steel making processes. The iron resource is expected to meet mining targets of iron ore per year as determined by the raw material demand for steel products.

3.3. Hematite Mineralisation and Structural Control

The iron ore deposits are comprised of a series of hematite beds interbedded conformably with metamorphosed sedimentary horizons of pink/grey phyllites that are believed to be of mudstone, shale and sandy origin, which points towards a sedimentary source of the deposits. Banding and vuggy quartz have been observed on the Kihimuro outcrops (figure 2) suggesting a Banded Iron Formation (BIF) type deposit. The deposit is comprised of a series of thin beds ranging from 5m to about a hundred metres wide, with a general Northwest-Southeast (NW-SE) strike direction and dipping at 30°SW to almost vertical.

Geophysical survey conducted prior to this mission demonstrates similar strike trends to the observed structures in the field. Mapped outcrops coincide well with the geophysical anomaly, an indication of the authenticity of the geophysical signature mapping (figure 2).

The geophysical mapping also conforms with the NW-SE strike trends of the hematite beds mapped during the reconnaissance survey. It can therefore be inferred that the strike extent of the geophysical anomaly depicts the lateral extent of the hematite deposits. The widths of the geophysical anomaly may not depict the thickness of the mineralisation considering that magnetic anomalies can be picked from iron dispersion due to weathering and other geomorphological processes that took place post the deposition of the iron oxides. This is where exploration mapping becomes important to ascertain the true geometry of the iron ore mineralisation and therefore an estimate of the quantities of ore available on the ground. Furthermore, expert analysis of the geophysical data can provide indications of thickness and minimum depth of the mineralisation expected on the deposit.

Faulting and zones of breccia have been observed in the hematite outcrops, an affirmation of the literature that the deposits were subjected to intense structural deformation through folding and faulting, which probably occurred during the uplift and thrusting during the Eburnian orogenic cycle. Faulting probably explains the discontinuities and disappearances of the hematite beds from surface. The parallel hematite beds with striking similarity could probably have been continuous beds of ironstone that were thrust, folded and uplifted, a process that could have triggered surface processes of weathering and erosion to remain with remnant parallel discontinuous lenses of the deposit.

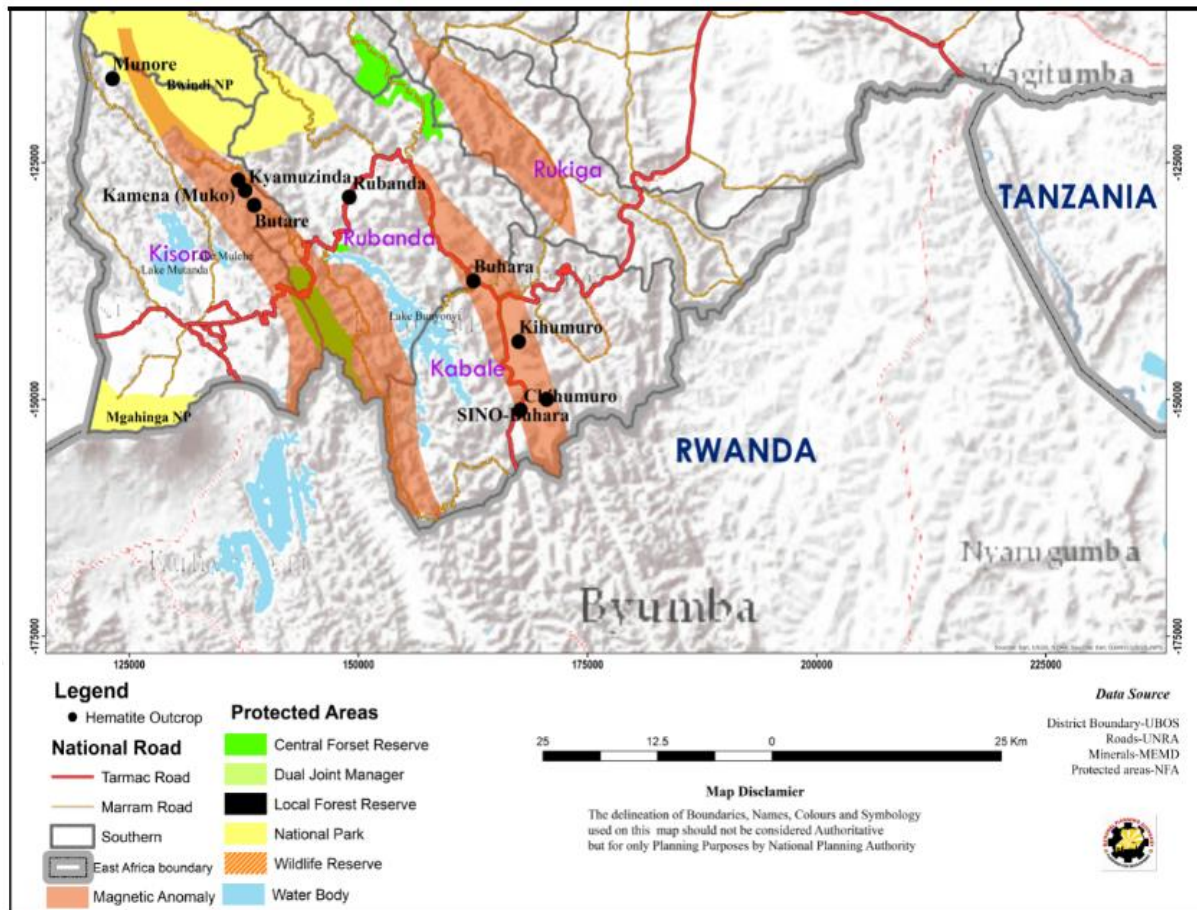


Figure 2: Geophysical map overlain with outcrops mapped in South West Uganda.

Mineralisation is said to be more concentrated at the hinges of the folds' "anticlines (and possibly synclines") depicting post thrusting (re)mobilization of the iron to form the deposits.



Figure 3: Specularite Hematite (Rugando Hill)

Texture of the hematite mineralization ranges from massive to foliated, with most of the foliated hematite being specularite with strong metallic lustre. The platy specularite has silica in-between indicating that the deposit could be a metamorphosed BIF. Vuggy quartz in the silica bands is a clear indication of hydrothermal enrichment of the iron within the deposit, which probably pre-dates the metamorphic process which produced specularite hematite schist, a variety of hematite characterized by aggregates of silvery, metallic, specular ('mirror-like') hematite flakes (Figure 3).

3.4. Geo-Metallurgical Properties of the Hematite Deposits

Texture and structure of the hematite as described above has significant bearing on metallurgical processes of iron making. Whereas massive hematite is likely to produce lump ore, which can be fed directly into iron making processes, the probability of high fines ore content, which would require palletization or sintering for it to be amenable to iron making processes cannot be ignored. On the other hand, friable and flaky specularite may require processing (milling) especially in DRI processes, with subsequent requirement for sintering or palletization after milling for amenability to iron making processes as the platy flakes may not support enough aeration for the flow of reductants. Evidence of these hematite properties are clear from the crushed and screened ore stockpiles on sites of the small-scale miners operating in the region. The massive hematite is likely to spit out high run-off-mine lump ore content on crushing and screening than the specularite ore, implying that if economic deposit(s) of lump ore are available, they are likely to be mined at lower cost considering they do not require further processing after mining and crushing, which, unlike fine ores or specularite, can be fed directly into iron making processes without the need for processing, sintering or palletisation.

3.5. Geological Resource/Reserve

Several companies with operations, either exploration or mining, on the iron ore deposits in the country have come up with mineral resource figures which have been used to determine the total known iron ore resource in the country, including results of the mapping undertaken by the Directorate of the Geological Survey and Mines (DGSM). It is however emphasised that from the information in the available literature, and the reconnaissance mapping undertaken by the team on the ground, the resource figures quoted are crude estimates. There are no known reporting standards followed in defining the resource as is normally required in the profession of mineral exploration and ore evaluation. One company, SINO Minerals Investments Limited, with a 30Mt resource that remained open ended at depth, is the only company to have carried out reasonable exploration, with results that can be used to define a 3D geometry of the deposit in its license area. As such, caution should be taken when using these figures for the purposes of planning the development of the I&S industry. The resource has been therefore classified as inferred mineral resource, which emphasises the need for further exploration to fully define the resource if meaningful planning for the development of the I&S value chain is to take place. Table 1 shows the identified mineral resources from both the southwest and the eastern part of Uganda, with the southwestern hematite deposits having higher grades than the magnetite deposits of the eastern part of Uganda.

Based on the reconnaissance survey carried out in southwestern Uganda, it is obvious that high quality hematite deposits exist in the area, and with proper exploration, the potential for a measured mineral resource that is more than double the 583 Mt in Table 1 can be a reality. A full exploration program for the entire deposit in the southwest should be a priority for the government if it is to unlock the value hidden in those iron deposits. It would be prudent for the government to consider the development process as a national agenda whereby government investment into developing these resources can create accretive growth to the government asset base. Value of that investment can always be clawed back in several ways including PPP, JVs or through tender, with a benchmark price that reflects the amount of investment the government committed during the mineral resource development stage.

Table 1: Known Iron Ore Mineral Resource in Uganda.

Type of iron ore	Location	Region in Uganda	Inferred Qty (million tonnes)	Fe Content (%)
Haematite (Fe₂O₃)	Buhara (include SINO's 30Mt indicated Resource)	South Western Uganda	205	68
	Rubuguri		55	69
	Rugando		65	67
	Muko		155	68
	Mugabuzi Hill		12	60
	Nyaituma		5	60
Magnetite (Fe₃O₄)	Bakusu, Nakhupa, Nangawale, Surumbusa	Eastern Uganda	41	50
	Tororo		45	60
TOTAL			583	65.87

Source: Directorate of Geological Survey and Mines

3.6. Conclusions and Recommendations

1. High grade hematite ore exists in South West Uganda with known grades ranging from 55 to 68%Fe. These are typical grades required for production of quality iron from DRI processes, which therefore leads to production of quality steel products at lower costs.
2. The government is encouraged to consider the exploration of the deposits as a national agenda and therefore invest in exploration to fully evaluate the deposits and delineate a mineral resource that can be used to plan the path the I&S value chain development should take. Exploration should be carried out by a professional organization that can produce results that conform to the ongoing international reporting standards like the NI43-101 of Canada or the JORC code of Australia. There is significant value addition that is gained by using these standards especially when seeking investment from private/listed corporations, be it PPP, JVs or open tender.
3. Based on outcrops observed in the iron formations, there is high variability in hematite ore geometry, with discontinuous beds of varying thicknesses. Geological mapping of the entire ironstone region is highly recommended as a first step ahead of any other exploration activity in order to fully understand the extent and structural controls of the mineralization. Geological mapping and the geophysical survey information available will then pave the way for targeting and subsequent exploration strategy.
4. The available geophysical data should be further analysed by an expert geophysicist who can provide leads on the likely geometry and the minimum expected down-dip extension of the deposit. This information can provide information on the minimum required exploration activity in the early stages of the program.
5. The different hematite types (massive and friable specularite) should be sampled for metallurgical tests in the early stages of development of the iron and steel value chain. The results will inform

the stakeholders on the likely scenarios required to prepare the ore for the reduction processes – Hematite fines from mining activity and the platy specularite ores may require Sintering or pelletisation before feeding into the iron reducing processes.

6. Aggregate size analysis of ROM ore carried out by SINO Minerals Investments Company Limited from its massive hematite ore retained screens of 80% lump and 20% fines, with a lump aggregate size cut-off of 5mm, a clear indication that there would be a significant amount of resource that would report to the fines during exploitation of deposits similar to SINO's. In the case of specularite ore, the fines content is expected to be more than 50% on screening, which further emphasises the need for screens and metallurgical tests, which then informs the project developers about the strategy to beneficiate the ore in readiness for iron making.
7. The fact that the ore is outcropping points to the possibility of open pit mining method as opposed to the more costly underground mining. However, the variations in deposit thicknesses may have a bearing on the amount of overburden to be stripped when mining, a scenario which further emphasises the need for mapping to ascertain lateral and width extent of the deposit lenses to be in a position to select the more prospective areas of the deposits that can be developed at lower costs.
8. Iron ore, being a bulk commodity requires well planned sustainable exploitation strategy, which can only be delivered by well experienced corporations with enough capital to invest in the industry. It is common practice for a government investment arm to take position either as a standalone investor or as a partner in the business. The numerous small-scale miners that have taken position in the area, with either exploration or mining licenses issued to them, are a potential hindrance to large scale mining investment unless the government re-aligns its policies on the criteria for investment in bulk commodities like iron ore.

4. MINING AND MINERAL PROCESSING

Mining and mineral processing forms a major component of the I&S value chain after mineral exploration. Whilst mining concentrate on the extraction of the mineral resource from the ground, mineral processing deals with the beneficiation of the mined iron ore, a process of value addition ahead of metallurgical processes of iron making and steel production. By virtue of the equipment used and the expert personnel required, huge capital investment is required for this part of the value chain. It therefore calls for careful planning and design by experts if meaningful returns are to be realised from the investment.

4.1. Iron Ore Mining Overview

4.1.1. Defining Parameters

In any mining development, there must be a mineral resource defined for mining to take place. It is from the mineral resource that a mineral reserve can be defined taking into consideration the economic factors influencing the profitability of the investment. Figure 4 shows the relationship and interaction between exploration, mining and metallurgical processes.

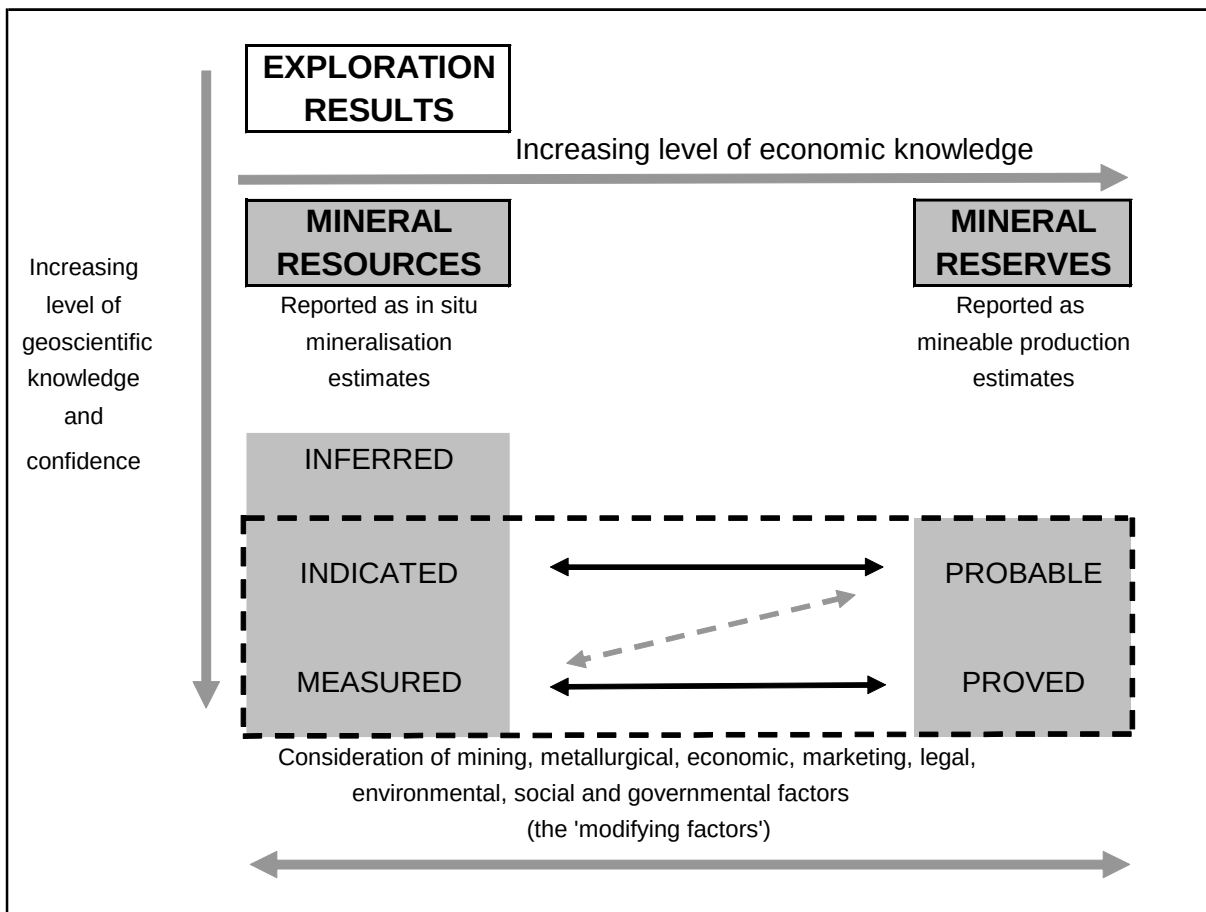


Figure 4: Relationship between Exploration Results, Mineral Resources and Mineral Reserves

As indicated in figure 4, there is a certain level of exploration required to define a mineral resource which in turn will pave the way for defining a mineral reserve, which is required to be in a position to perform mine planning and design and the subsequent mining of the mineral resource. As a minimum,

the mineral resource should be explored to indicated level to be able to define a probable mineral reserve, and a proved mineral reserve can only be derived from a measured mineral resource, both of which have to be defined using credible international reporting standards (NI43-101, JORC, SAMREC). This further emphasises the need for systematic exploration that is credible and acceptable to investors as emphasised in the earlier chapter on geology and exploration. There is a number of modelling techniques that can be used to fully predict/estimate the resource/reserve figures of the deposit to allow proper mine planning and therefore profitability of the operation, with examples like: geological modelling at exploration level to define the resource, mine modelling to define the reserve and NPV scheduler to define/produce a range of profitable pit shells based on defined input parameters. Several valuation protocols like SAMVAL, CIMVAL etc. are available for credible estimates of profitability and therefore the value to be created by the business unit as part of the I&S value chain.

Due to limited information, this chapter is therefore mainly based on conceptualisation as no exploration has been done to delineate the mineral reserve, which is required to effectively analyse the constraints that determines the mining method, production figures, bill of quantities on equipment, manhours, costs, etcetera. It is therefore virtually impossible to model any data, be it technical or financial, without a basis for benchmarking. Detailed and accurate information can only be provided when a definitive feasibility study (DFS) is in place for implementation.

4.1.2. Selection of Mining Method

For a mine design, a choice of mining method to be used must be made. Also, the type of access to the mine must be considered. After material is mined, a choice for material handling must be made. This, however, is all influenced and determined by the characteristics of the ore body being mined. Geological factors play an important part in arriving at these conclusions and among them include deposit shape, deposit size, grade of ore, distribution of ore, depth of ore beneath the surface, ore thickness, ore strength and the strength of the host rock. The need for proper exploration and laboratory tests of an ore deposit to determine these characteristics cannot be over-emphasised. Visual assessments during the reconnaissance visit provide some leads on what to expect from the iron ore deposits of South West Uganda (Table 2).

Table 2: Observed Geological Characteristics and Geometry of the iron ore Deposits

General ore characteristics	Prevailing conditions
Rock Strength	Generally Hard
Ore Strength	Hematite – Hard; Specularite - Soft & Friable
Deposit Shape	Elongate, steep and shallow dips, 5 to 100m wide
Deposit Size	Unknown but potential for huge deposit
Ore Grade	55 to 68%Fe
Ore Distribution	Generally continuous in discrete lenses
Ore Depth	Unknown (intersected at 140m depth by SINO Minerals)
Ore thickness	5m to 100m
Total reserves	Unknown (only ~500Mt inferred Resource)

In addition to the geologic characteristics, the following, among others, have to be considered when evaluating the mining method: required production, ore recovery, life of operation, mining and capital costs, support infrastructure, availability and cost of (skilled) labour, investment risk, remoteness of operation, topography of the area, climate and altitude, groundwater properties, slope stability, ground conditions, blasting restrictions, environmental restrictions, waste rock disposal, social and community constraints etcetera.

From a visual point of view, the ore deposit is expected to meet some of the defining characteristics for an open pit mine.

4.1.3. Rationale for Open Pit Mining

Most iron ore deposits, by virtue of their bulkiness and outcropping, are exploited using surface mining system dominated by open pit mining method. Drawing analogies to the hematite deposits of South West Uganda, open pit mining is considered appropriate based on the following ore characteristics.

- Outcrops of the deposit have been observed, which points to surface mining as a possible system of exploitation of the mineral resource.
- The geometry of the ore bodies is observed to have orientations ranging from steeply dipping ($\sim 90^\circ$) to shallow dipping (30°). Deposits of this type of geometry are best optimised using open pit mining.
- The fact that the ore body has variable down-dip orientations implies there is significant amount of waste material that must be removed, particularly on the hanging wall side of the deposit, to allow access to the underlying ore. Waste has to be transported to a waste disposal area to make it possible for further access of the deposit beneath, hence the preference for open pit as opposed to open cast mining where the waste-cast tends to block access to ore resource at depth.
- Open pit mining also allows room for manoeuvre of the heavy equipment used during excavation.

It being a hard rock, there is need for drilling and blasting, which can then pave way for material handling.

4.1.4. Implications of physical properties of the deposit on mining

With two ore types in the area, trial mining at feasibility is called for in order to ascertain material and equipment requirements for extracting the ore. Generally, the following scenarios are expected following field characterisation of the rocks constituting the ore and waste material.

- The massive hematite ore would require drilling and blasting to break the rock for haulage to metallurgical plants for processing.
- The country rock, which constitute waste material when mining, is predominantly hard phylitic rock which also require drilling and blasting to break it for haulage to waste dumps.
- There is a possibility that the soft friable specularite ore can be “free dig” material, or may require minimal blasting, in some areas. However, as in hard rock, excavation would require strong equipment as variations in hardness/softness is expected within the in-situ ore. The free dig approach can only be ascertained after trial mining at feasibility.

4.2. The Mining Process

When ore parameters are defined, production will then follow. Production can either be owner operated or outsourced depending on the best option that maximises profits as determined in the feasibility. There are some risk factors to take into consideration, which needs to be mitigated if proper production that adheres to the mine plan is to be realised.

Depending on the planned production parameters to meet the required tonnages, the following must be taken into consideration for the type of deposits found in South West Uganda.

4.2.1. Equipment Requirements

Considering that there is both hard rock ore and friable ore, the following equipment is suggested for the mining operation.

- Drill rigs – required for blast hole drilling. The types and quantities of explosives can only be determined when the rock characteristics are tested and determined, and the aggregate size, as required by the selected iron ore reduction process, is known.
- Grade Control Drill rigs – the requirement for grade control drilling can only be determined when the ore characteristics of grade and continuity have been determined.
- Excavators – these are required for the process of “mine, haul and dump” and they include mechanical shovels, back-hoe diggers, and front-end loaders. Type and bucket size are determined by the production rates required as provided in the mine plan.
- Dump trucks – capacities range from 40 to 200t; based on the current demand of steel in the country, small sized dump trucks would suffice for the operation.
- Dozers and graders for road preparation and resurfacing.
- Water trucks – required for suppressing dust in the pit and the haulage system to the process plant, if not tarmac.
- Vehicles – for day to day running of the operation.

Being heavy equipment, they constitute significant amount of the cost required for the operation; both capital expenditure (capex) and operating expenditure (opex) are high. Capex and opex estimates are based on Original Equipment Manufacturer (OEM) and service provider quotations, DFS, as well as estimates from consultants’ data bases. A backup engineering service for the repair and servicing of these mining equipment can significantly add-on to the operating costs.

4.2.2. Grade control

Grade control procedures ensure that the ore and waste tonnage are separated from each other to minimise ore losses and dilution. The Project relies on good grade control procedures, measurement and supervision. Good drilling and blasting techniques and operations provide the basis for efficient downstream loading and hauling activities and are linked to the grade control initiatives. The ability of the grade control function to distinguish waste from ore and to direct the loading and hauling operation accordingly is a key success factor. Good skills, expertise, training and knowledge must be included in the grade control program, together with operating procedures and strict controls. Good grade control can enhance the head grade of the operation, thereby improving revenue.

Reconciliation forms part and parcel of grade control as it is used to account for the mass and quality balances of the mined ore. After an ore block is mined, ore quality and quantity are expected to reconcile between the estimated block grade and tonnage and the realised mined ore grade and tonnage. Deviations can be due to poor grade control during the mining process, hence the need for well trained professionals for this part of the operation.

4.2.3. Mining Risk

Mining is a high-risk operation hence the need for careful planning and implementation of the mining operation as determined by the mine plan and design. The ability to identify and mine the ore

separately from the waste portions is a key success factor for a mining operation. The drilling and blasting process will require strict control and supervision to ensure ore and waste are blasted correctly. This will minimise dilution and improve head grade feed to the process plant or reduction furnaces. Grade control in the open pits will require highly skilled technical personnel (geologists supported by grade controllers) to constantly monitor loading activities and to direct hauling activities to the concentrator for ore or waste dumps and/or backfill areas for waste material. The head grades take account of dilution and ore losses. The operation should aim to minimise ore loss and dilution at ore-waste fronts to no more than 5%.

Table 3: Major Risks in a Mining Operation

Events	Causes	Reason	Likelihood	Consequence	Mitigation
Technical Risk Factors					
Inadequate ore/waste identification	Incorrect identification/poor grade control	Inadequate skills, training, supervision	Possible	Major	Training, correct skills, monitor drill chips, grade control
	Faulting/Folding/geology		Likely	Major	
Poor Drill Techniques	Poor rig operation, inadequate planning	Inadequate training, supervision, monitor control	Possible	Major	Employ suitable equipment, operators, training, planning, supervision, monitoring and control
Slope Instability	Inadequate Slope design	Lack of in-situ rock strength and Geotechnical test work	Possible	Significant	Drill geotechnical holes, test work, modelling
Generation of dust, fumes, noise pollution	Noise, air pollution, blast vibration	Hauling activities, blasting	Almost certain	Significant	Water trucks, correct explosives, timing of blast
Cost Risk Factors					
High Capex	Mining equipment	High costs obtained from OEMs and service providers	Likely	Major	Check designs and equipment specifications, use mining contractor

Slope stability should be ensured by drilling geotechnical holes and performing test work followed by appropriate modelling. This will confirm slope angles and minimise any negative risk associated with slope failure. Mine designs are often optimised with one or two haul road, each entering and leaving the mine. Any slope failure which affects the haul road will compromise production capabilities for a period resulting in revenue loss. Specifically, for the iron ore deposits of South West Uganda, the fact that the ore beds are thin implies there is not enough room to provide several in/out haul roads, making it susceptible to slope failure risk which can easily block pit access roads in the event of the

risk happening. Also, the foliated specularite hematite is vulnerable to slips and slope failure. It therefore calls for professional geotechnical engineers to map, test and model the deposit to the required standards that minimises the risk of slope failure. This also applies to waste dumps and tailings dams, which can slip away if not properly designed.

Climate in Uganda is predominantly characterised by cyclical wet seasons which could significantly impact production. Stockpiling a portion of the mined ore during the dry season and treating this material during the wet season is recommended and should be included in production planning. Maintenance of roadways and precautions during the wet season should be implemented to minimise the risk of accidents and lost production.

Major risks, also termed High Level Risks, can seriously affect revenues of a mining operation. They need to be identified and measures taken to mitigate them. They should be classified according to the level of risk, their likelihood to happen, and the damage they can cause when they happen so proper mitigation can be put in place as demonstrated in table 3. In the event of a risk occurring, proper investigation should be done, and remedial action taken to lower the risks in future, and the whole process properly documented including predicted residual risk, and lessons learned for future references.

4.2.4. Risk arising from Artisanal and Small-Scale Miners (ASMs)

The visit to Kabale revealed significant presence of both artisanal and small-scale miners who are in the practice of mining iron ore for sale to steel companies. In the envisioned scheme of value chain development, this group of miners can potentially pose significant risk to the goals of this vision.

Artisanal miners, by nature of their operations, use poor informal and individual mining practices seeking to eke out or supplement a subsistence livelihood. Due to lack of resources, their mining methods are detrimental to the environment, communities in the areas they operate and to economic development of the region and the country:

Risk to the Environment

ASMs use manual equipment to scrounge for the iron ore bearing rocks among other country rocks. Due to lack of equipment, they leave behind a legacy of dangerous pits/openings that can be a hazard to animals and people plying in the areas. Waste material is disposed randomly within the vicinity of the operations without any waste management planning, thus degrading the environment for any possible future land use. By exposing rocks to the surface without any mitigatory measures, chances of triggering weathering and erosion into the drainage basins of the area are high, with catastrophic pollution impact to water features of the area rippling into the fauna and flora livelihoods of the area. With the bulk of the region comprising of protected areas (natural forests (Bwindi National Park), Water features (Lake Bunyonyi) and Kabale (Urban settlements), ASM operations must be checked and gagged to ensure proper protection of the environment.

Risk to the Communities

ASM operations are awash in the area within metres from people's dwellings, a situation that can cause catastrophic hazards to communities if the operations are allowed to continue unregulated. The operations have potential to trigger landslides on both the insitu rocks weakened by mining or the waste material casted onto hill slopes, which can cause direct threat to people's lives and damage to their households, belongings and farmlands.

Risk to economic Development

ASMs can only mine a few metres from the surface due to the risks associated with digging deeper into the ground without proper mining equipment. As such, their operations have potential to make

ore bodies difficult to mine professionally due to several risk issues, among them being increased capex due to high stripping ratio, rock stability issues, mitigating ASM-induced environmental degradation etc. There is also the possibility of resentment by the ASMs in the event of professional miner investing in the area, a situation that can trigger conflict between the warring groups, with serious impact of scaring away possible real investment in the area.

Risk Mitigation

In order to create an enabling environment, the government must first understand the mode of iron mineralisation in the area. With knowledge of the deposit, it is the responsibility of the government to parcel out the deposit to interested operators based on ability and experience. Large experienced miners with the financial muscle to invest in modern, efficient and sustainable mining operations and the right skills can be offered the opportunity to develop the bulk of the deposits, whereas ASMs can co-exist alongside, mining the tail ends of the deposits that are normally difficult to optimise economically and therefore not so attractive to big mining operators. By so doing, the ASMs can learn from the bigger players and therefore gradually improve their operations, thus allowing a win-win situation to prevail.

4.3. Definitive Feasibility Study

Due to capital intensive nature of a mining operation, a Definitive Feasibility Study (DFS) is a requirement, also referred to as Bankable Feasibility Study (BFS) if funds for the operation are to be sourced on the open market like debt, equity and/or crowdfunding markets. As such, the following, in addition to the factors raised above, should be taken into consideration when preparing the DFS:

- Complete detailed open pit mine designs and production schedules for selected pit shell.
- Define associated services, engineering and infrastructure requirements.
- Prepare production schedules for mining activities.
- Assist in preparation of mining contractor tender documents if outsourcing of some of the operations is considered economic or more efficient.

The deliverables for the DFS should include (but are not limited to) the following:

- Strategic life of mine plan.
- Detailed mining equipment and infrastructure requirements.
- Mineral resource to reserve statement (JORC/NI43-101 compliant).
- Production (ore and waste) profiles.
- Grade and tonnage profiles.
- Tender documents for contract mining.
- Assistance with adjudication of tenders
- Progress reports.
- Environmental and Social impact assessments.
- Tailings dam design and operation if mineral processing involves production of tailings.
- Negotiations with government and local authorities.
- Surface infrastructure.
- Metallurgical test work, pilot plant, and plant design.
- Techno-economic modelling and financial analysis.
- Obtaining permits, licences, authorisations and permissions for mining activities.

The Government, being a major player in promoting the industry, should take a position and invest in feasibility studies. Other than determining the value of the deposit/s, this could form a basis for attracting investors when selling/marketing the project to potential clients. The government

investment can always be factored in (or recouped) when agreeing on investment deals with the selected investor/s.

4.4. Mineral Processing

Mineral processing allows the process of beneficiation to commence, thereby adding value to the mined ore. The DFS process informs whether the ore requires this process or not. From a visual assessment during the field visit and information gathered from stakeholders, both types of iron ore in the South West requires some sort of beneficiation.

4.4.1. Massive Hematite Ore

The bulk of the massive hematite ore (~80% lump as quoted from SINO) may not require mineral processing but the ~20% fines, which needs to be prepared for the iron making process through sintering or palletisation does. The exact percentages of lump ore to report directly to the iron making processes and that of fine ore for palletisation are determined by the iron making plant's grain size specification requirements. Trial mining and screening tests, as mention in the DFS above, should be carried out with the aim to meet the specific ore requirements of the iron making plant.

Common practice in iron production is to have the iron making part of the value chain to take charge of the process of palletisation due to the availability of expertise with knowledge of the pellet size requirements for the iron making plant. Only in few cases are miners responsible for this process. Should this process be part of the iron making process in the Uganda scenario, the cost associated with fines beneficiation should not be part of the mining and processing business unit but the iron making part of the value chain.

4.4.2. Specularite Hematite Ore

From visual assessments during the field visits, it is obvious that specularite hematite ore requires beneficiation due to its high friability characteristics. Whilst it could be less costly to mine as compared to massive hematite; the amount of capital investment required in beneficiating the ore can be so huge that it can offset that cost-saving benefit realised during mining.

The first process in the beneficiation is comminution, which is the reduction of solid ore materials from one average particle size to a smaller average particle size, by crushing, grinding, vibrating, or other processes. In so doing the hematite ore can be partially freed from the associated gangue minerals like silica, alumina and other deleterious material embedded in the matrix of the ore minerals. This process employs the use of crushers and mills to crush and grind the ore thereby liberating the ore from the gangue minerals. Sag mills are normally employed for milling the ore. Both the crushers and the mills are high capital-intensive equipment, which calls for the right type of crusher and mill for maximum benefit.

In order to zero-in on a particular mineral processing method, a series of tests have to be done by professional bodies with the right equipment for such tests. Only when the tests are completed should the best option for beneficiation be employed. Options for processing include but not limited to the following methods:

4.4.2.1 Gravity Spiral Concentrator

The gravity spiral circuit is designed to extract and concentrate heavy minerals of which iron oxide is one. As in any gravity concentrators the separation of minerals in a spiral is based on particle size

and specific gravity. This is a high-tech operation that calls for experienced professionals who can interpret data generated by the process like particle size recovery curves. A significant amount of capital is called for in the development of this process plant.

4.4.2.2. Wet High Intensity Magnetic Separation (WHIMS)

Platy metamorphosed hematite like specularite, can be subjected to magnetic separation, which is achieved by induced magnetism in an aqua media. Prior to magnetic separation, ore normally passes through a preliminary concentration process in a vortex cyclone, which separates heavy iron oxide particles through an underpass and the lighter material as an overflow. The hematite will then be transported in slurry form to the WHIMS for final concentration to the required grade.

As is the case with spirals, WHIMS technology requires highly experienced professionals with the skills to optimise production using this technology. Similarly, WHIMS are capital intensive to build like spiral plants.

4.4.2.3. Dense Medium Separation (Gravity Separation)

The process uses a dense liquid which is typically a suspension of fine magnetite or ferrosilicon particles suspended in water. Particles are sorted primarily on the basis of their densities. The iron ore comprising of particles with a wide range of densities is introduced into a medium suspension of a given density. Hematite particles of absolute density of 5.2t/m^3 are heavier than the medium density which allows them to sink and collected as an underflow, commonly referred to as sinks, while silica particles (density 2.65t/m^3) and other lighter materials like alumina, which are lighter than the medium density, rise to the top and are collected as an overflow, commonly referred to as floats. The magnetic properties of the fine magnetite particles allow them to be recovered using magnets for reuse. This kind of plant also requires high capital investment.

In all the processes, the final product is fine grained hematite, which would then require sintering or palletisation using the same process as explained in earlier sections.

4.5. Conclusions and Recommendations

- Mining is a capital-intensive operation, more so with iron ore mining which requires huge capex and opex to get the operation running. In order to minimise the risk on capital investment, there is need to follow the defined procedures in developing this part of the value chain, starting with defining the mineral reserve (hence the requirement for exploration) followed by DFS, which then captures all the aspects of mining including mine planning, mine design, cost etc.
- The current ASM operations in the area has potential to pose big risk to mining investment, communities and the environment. With the envisioned iron and steel value chain, the SME's are not capable of implementing sustainable mining operations that can economically feed into the value chain. Government intervention is necessary, and measures should be taken to ensure that concessions for such a potentially huge deposit are issued to capable investors with the relevant experience and expertise to develop sustainable mining with corporate and social responsibility for the environment and communities in the area.
- With proper exploration strategy to define an indicated resource in place, the government can be in a position to allocate zones of higher mineral intersections to experienced players with enough capital for sustainable mining while also incorporating SMEs to exploit the tail ends of the mineral deposits which are rather difficult to exploit using mechanised mining methods. In so doing, the government creates synergy between the bigger and SME players in feeding the iron making processes with iron ore.

- In addition to being capital intensive, the industry requires highly skilled personnel to deliver projects of this magnitude. Whilst the use of expat experts is inevitable, there is need for local skills development in order to speed up local content participation in the value chain.

5. IRON AND STEEL PRODUCTION - THE PRESENT AND THE FUTURE

5.1. Introduction

Understanding the present steel market dynamics forms an important part of the study as it helps in the understanding of supply and demand scenarios in the country, which are important for national planning and for future development of the industry. Field visits were conducted for data collection through visiting the main players and collecting data on production rates, products range and applied technologies in production. This was done to facilitate an understanding of the present industry situation of the existing steel plants. The exercise allowed interaction with the players themselves and therefore facilitated a good understanding of their constraints, thus creating a platform to pinpoint problems associated with the industry, which in turn facilitated provision of informed advice regarding the direction the industry should take to fully meet the steel requirements of the country. It was considered the most efficient method of studying the industry to enable provision of advice for the sustainable growth of the Iron & Steel (I&S) supply chain in Uganda.

Collection of data was through reconnaissance mapping of the iron ore deposits in Kabale, and face-to-face interactions with company officials and observations of operations where the team was allowed access to the plants. The main constraint likely to make the analysis biased was the lack of verification means for the data collected verbally from stakeholders. Meetings with companies' officials were often followed by plant visits though in some occasions the plants were shut down due to various reasons (power shortage, lack of purchase orders, etc.), and in other cases the team was not allowed access to the plants. Nevertheless, the mapped data, to a large extent, conformed with available data from the Ugandan steel production and consumption indicating that the collected data was authentic.

A cross-check and an integration of the retrieved data from I&S players was done with the support of the UBOS and online databases. The Bureau made available Country statistics on population growth and supplied import & export steel data, which was used to properly set the actual iron and steel parameters including the amount of produced and consumed steel per person each year. Similarly, online databases provided invaluable data and information for the subsequent economic evaluations, and therefore enabling the possibility to compare Uganda's steel situation with other steel players in the world for proper evaluation and analysis.

The growth rate of the above-mentioned parameters was evaluated by means of two different models: (i) a custom built economic model using historical data projected into the future and (ii) a capability model developed on the basis of the team's experience given the availability of resources (commodities and energy), infrastructure (goods and energy transport systems) and data from similar situations in other emerging countries.

The analysis and results of the two models provided complete and exhaustive panorama of the industry growth scenario in the next five years. Over the next five years, only basic indications are given about the expected parameter trends.

As a parallel exercise during the visits, an assessment of the iron and steel technologies in key players' plants was conducted. Installed equipment were evaluated in comparison with the most efficient and innovative technologies worldwide. This was done as a basis for determining opportunity areas for quality and efficiency improvements and therefore provide advice on how to improve those operations.

The research also looked at regulatory issues in order to facilitate and bring about technology upgrade in the I&S sector. The possibility of facilitating enforcement of regulations promoting technology upgrading was considered, as well as an overall environmental control (pollutant emissions and specific consumption of energy) to enable the Ugandan iron & steel system to ultimately achieve the target of high quality and sustainable production.

The innovative promising technologies were not neglected but were not considered as vital for the Country's development now, since both Country and new technologies are in a very preliminary stage of development. An example is the Hydrogen based DRI that has not yet started any industrial trial and that requires a lot of investment in research on equipment, material and innovative solutions for prototyping and the subsequent scaling up so as to reach a reliable industrial production stage. Such a technology will therefore not be available as an operation in the next few years. Its availability will be dependent on trials in pilot plant stage and the required capex being proven viable.

5.2. Iron and Steel Value Chain – Uganda Scenario

The simplified iron and steel value chain (figure 5) showing the current scenario in Uganda clearly demonstrates the gap the Ugandan Government would like to develop. There is the iron ore deposits that remain untapped on the upstream side of the value chain while the industry is thriving on the downstream side with activities ranging from steel making and shaping, fabrication and production of final products. The grey area constituting the non-existent components of the value chain of iron ore mining and iron/steel making is the missing link in the value chain, which, if developed, can unlock the much-needed economic growth in the country.

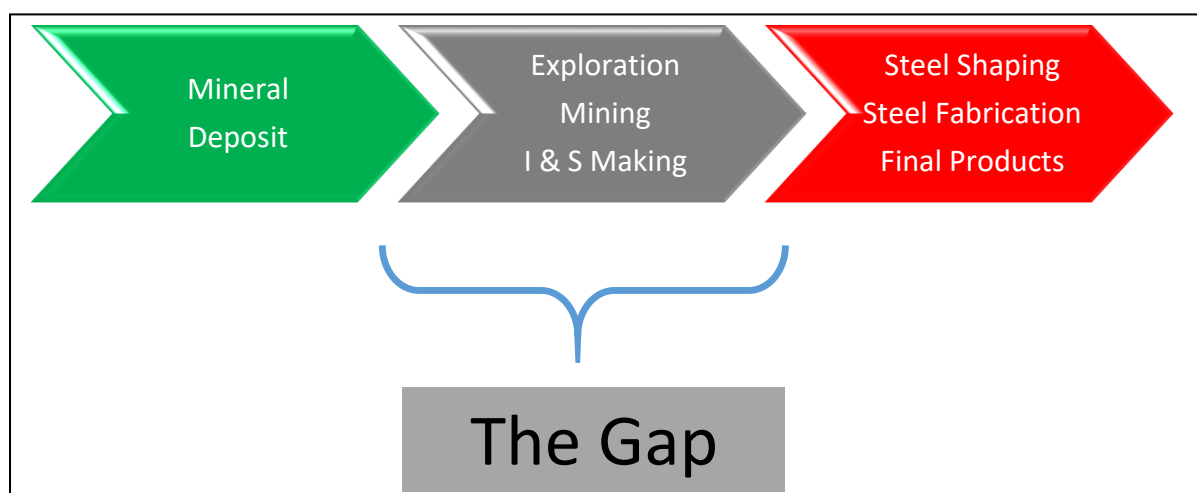


Figure 5: I&S value chain showing opportunity areas (The Gap)

It is clear from the current status of iron and steel industry in Uganda that there is need to establish a viable and sustainable iron and steel industry starting from mining of iron ore up to the marketing and profitable selling of the finished product.

Some steelmaking is on-going using induction furnaces based on steel scrap as the raw material; but with the objective of making steel from iron ore, this process does not represent closure of the value chain gap. Moreover, most of the companies are only into cold drawing plus some processing and then marketing of the products.

This section analyses and discusses the current iron and steel companies in Uganda, and recommendations on how the companies can establish iron ore-based steel production. It briefly discusses the markets that can take up additional production, and therefore the country realising its

goal of uplifting the industrialisation agenda for the rest of Uganda. It then goes on to discuss details of options to be considered in the future based on economic modelling results of the Uganda scenario.

5.2.1. Current Scenario

The current iron and steel value chain in Uganda is as illustrated in Table 4. The potential for developing the complete value chain is there given that there are industry players at each stage of the value chain. What is required is to identify means to fully support and develop the capacities of those players for a sustainable development of the value chain, especially the upstream and midstream (Phases I to III) industry players who are yet to start production in their respective phases of the value chain.

Table 4: Current positions in the iron and steel value chain for steel companies in Uganda

	VALUE CHAIN PHASE								
	I(1)	I(2)	II	III(1)	III(2)	IV	V(1)	V(2)	VI
COMPANY NAME	Exploration	Mining	Iron Making	Steel making for 100% of requirement	Steel Making to produce small % billet using arising scrap	Casting	Hot Rolling	Cold DRA Wing	Market
IMCU									
DRIOC									
SMI									
Steel Rolling									
Tororo Cement Ltd									
Tembo Steel – Iganga			W.I.P.						
Tembo Steel – Lugazi									
Bavima Steel Ltd									
Madhvani Steel Ltd					XXXXX	XXX			
MMI Steel Ltd									
Pramukh Steel Ltd									
Tian Tang Steel									
Roofings Ltd – Lub									
Roofings Ltd – Nam									
Uganda Baati									
Viva Steel Ltd									
EA Roofings Ltd									
STEEL TUBE (1)									
STEEL TUBE (2)									
Mesha Steel									
Yogi Steel									

KEY:

COLOUR	DENOTION
	100% production of billet and other cast steel intermediates
	69 % production of billet
	Very low, low and close to 50% production of billet
XXXXX	Have installed Induction furnace and CCM but 0 % production
W.I. P.	Refers to physical construction work in progress on the ground

Since the last time (2017) the NPA document titled “Project Document to Inform the Feasibility Study for Establishing a Comprehensive Iron and Steel Industry in Uganda” was written, a few developments have taken place along the value chain of the iron and steel industry in Uganda, with potential for significant value addition if fully supported. The following information has been gathered from field visits to SW Uganda and data compiled during visits to existing industries.

- IMCU closed mining activities at the end of 2013 following the ban on export of iron ore.
- DRIOC took over WCH and are mining, processing and selling the iron ore to Steel Rolling Ltd for DRI, and to steel companies charging DSO at 10% in the scrap melt shop.
- Madhvani Steel Ltd. has integrated backwards from rolling only purchased billets to establishing a melt shop that will be producing billets via a continuous caster, though production is currently on hold.
- Steel Rolling Ltd. is producing DRI by RKF using haematite and briquetted carbon from bagasse and other organics.

5.2.2. Upcoming Scenario in the short term

- Projections are that Tembo Steel will start production of DRI using coal and RKF perhaps by mid year 2020 (from Document “Uganda Steel Potentiality”)
- Further expansion in DRI production is anticipated from a second plant from Tembo Steel.
- The actualisation of construction and commissioning of DRI Plants by Kigezi and Pramukh should further increase the total DRI output in Uganda.
- The construction and commissioning of a plant by Dong Song which is expected to produce up to 300,000 tonnes per annum using iron ore will further increase the tonnage of steel made from iron ore.

5.2.3. Anticipated growth of Installed capacity and markets

A total of 9 companies are starting at the very end of the value chain (Phase V (2) in table 4). There would be better value addition to the industry chain should these companies purchase locally made (rather than using imported) semi steel as this would strengthen the business case for those starting at phase III to buy DRI from phase II investors. This appears to be the basis for the coming in of Kigezi and DongSong i.e. to feed DRI to their downstream clients as well as feeding billets to companies starting at phase V (1) e.g. Roofings Namanve is indicated as active in phase III to phase VI but on the ground 90% to 95% of their billets are imported, and only 5% to 10% is via their SMS/CCM. There is therefore scope for developing the upstream industry, at the least to feed into the existing demand currently at about 500,000 t/year. Further opportunities for development on the upstream side are foreseen with growth in demand as predicted by the economic models described further in this chapter.

5.2.4. Market Forecast from Historical Data

The primary market for the iron and steel value chain is the local Ugandan market. The regional market, especially EAC, is significantly big with high demand given that the raw material in the region is also predominantly semi-steel imported from countries outside of the region. However, Uganda should focus on the local market first in these early stages of industry development. Increase in production and improved performances giving rise to production of steel products at competitive prices would eventually enable the Ugandan steel industry to be competitive in the nearby region. It is therefore important that the government incentivises the industry players to achieve this level of performance given the benefits associated with Uganda being able to compete with other players in the region. Three market segments have been identified as key targets for the locally produced steel:

1 Local Market

Growth expectations are based on substituting current imports and meeting the natural demand from a growing population. This is limited growth and may not drive the steel making industry to its full potential, but it will suffice in the early stages of development.

2 Current EAC Block.

Figure 6 shows year-on-year growth in steel imports in the EAC block from 2016 to 2018, and most importantly is the fact that the growth is leveraged by semi steel raw material imported from other countries, an opportunity area for Uganda to strategize on in its development of the steel industry and markets. Assuming Uganda attains globally competitive production of steel products in the future, when the EAC steel imports are expected to be around 4 million tonnes per year, Uganda stands to benefit from this regional growth in demand by supplying its steel products to the region. Targeting up to 20% of the EAC market (as envisioned by the government) could place Uganda at levels of mid-tier steel producer in the world, with an export market of 800,000 tons/year.

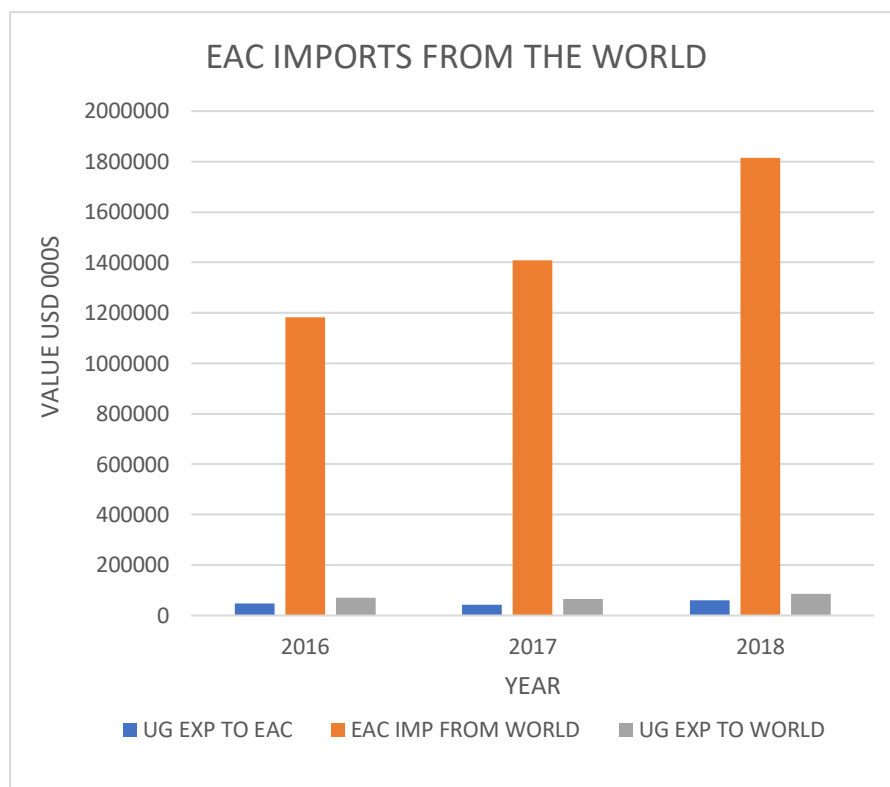


Figure 6: EAC Global Steel Imports and Uganda's Exports to the to EAC and the World (Source: <https://www.trademap.org/Index.aspx>)

3 EAC and part of SADC (DRC, Zambia, and Angola)

The Uganda steel manufacturers association reported that trade is expanding even outside the EAC block. Trading with DRC and Northern Angola is said to be on the increase. This was further confirmed by the forecast data for steel demand for regional countries for the years 2021 to 2025, which is shown in table 5:

Table 5: Regional Steel Demand Forecast (X1000 tonnes)

	2020	2021	2022	2023	2024	2025
Angola	1591	1672	1757	1847	1941	2040
DRC	186	195	205	216	227	238
Kenya	2057	2161	2272	2388	2509	2637
Madagascar	150	158	166	174	183	192
Mozambique	559	588	618	650	683	717
South Africa	8870	9323	9798	10298	10823	11375
Tanzania	1133	1191	1252	1315	1382	1453
Uganda	192	202	212	223	235	246
Zambia	305	320	337	354	372	391
Zimbabwe	184	194	204	214	225	236
Tot: All steel products	15227	16004	16820	17678	18580	19527
Tot: Long steel products	9898	10403	10933	11491	12077	12693

(Source: World Steel Organisation)

It is evident that the market is available in the region. What is required is for Uganda to improve the competitiveness of steel produced from local resources and therefore fulfilling all the requisites for reducing the price of final products as further described in the report.

By backward integration, the upstream side of the value chain can start by mining enough iron ore to feed into the existing industry until such a time that quality is no longer compromised should the mining be boosted to cater for the export market.

5.3. Steel Industry Key-players

The team visited 15 Steel Players (namely: Roofings Ltd, Steel and Tubes Ltd, Tembo Steels Ltd, East African Steel Corporation Ltd, Tian Tang, Pramukh Steel Ltd, MMI, EA Roofings, Yogi Steel, Bawima Steel, Steel Rolling Ltd, Yogi Steel, Tororo Cement Steel, Mesha Steel and Viva Steel) of which Roofings, Tembo Steel and Steel and Tubes have two plants each. Details of the visits are provided in Annex 2.

Two other Companies were visited with plans to set-up new plants in Uganda due to start production in the next 2-3 years. These include Dong Song Group and Kigezi Steel Ltd.

The following section presents findings that came out from the collected data and a consequent series of recommendations and/or suggestions, targeting the sustainable development of the iron and steel value-chain in Uganda:

5.3.1. Present Over-capability of Steel Production

The first immediate outcome emerging from the visits of the steel players was that there is an over-capability of steel production compared to the actual country need.

The total amount of final products produced in 2018 by the 15 steel players interviewed was slightly more than half a million tons while their installed plants capacity is more than 1.5 million tons.

This difference between produced quantities and installed capacity has significant implications on the country's iron and steel industry development in the incoming few years; in order to reach the point where it is necessary to install new steel plants or empower some value chain portions, this gap between capacity and demand has to be fully closed.

5.3.2. Present range of steel products in Uganda

The second outcome that emerged from the visits to Uganda's Steel producers was the huge variety of different sizes of plants; from 1,000 t/year, up to 300,000 t/year, and producing different mix of products.

Only 8 plants have the capability to produce liquid steel but only 7 are actually using this option; the other plants are importing semi-finished products that consequently make the final products. These are not competitive due to the high transport costs.

The actual steel production is limited to long products with low carbon content (barbed wire, nails, meshed wire, hollow sections, etc.) and some cold rolled coils to make roofing material or galvanized sheets.

High alloy steel necessary for high strength products requires dedicated production plants not yet available in Uganda. Therefore, products for construction works (heavy profiles and structural steel) are almost absent.

The major planned projects in the country like the new dams, the upgrade of Entebbe airport and construction of new regional airports and upgrade of aerodromes, the new SGR (railway) mostly require structural or high-quality steel, which at the moment is not locally produced. Only re-bars or small profiles by steel producers can be utilized for these projects. There will thus be a minor direct impact on the local I&S industry, as it is still necessary to import most of the required steel for big infrastructure projects. However, the new infrastructures will accelerate the overall country development and consequently an increase of steel products requirements, helping a lot in the development of the entire industry value chain. There is an opportunity for investment in those steel products not yet locally produced.

5.3.3. Present Liquid steel production in Uganda

Table 6: Steel melting production and finished product production as of 2019

Company	Steel melting production (ton/y)	Finished products production (ton/y)
ROOFINGSTEEL (NAMANVE)	13,200	149,000
ROOFING STEEL (LUBOVA)	0	360
STEEL & TUBE	30,000	86,000
TEMBO STEEL (LUGANZI)	60,000	60,000
TEMBO STEEL (IGANGA)	36,000	24,000
MADHAVANI STEEL	0	24,000
TIAN TANG	18,000	43,000

PRAMUKH STEEL	25,000	36,000
MMI STEEL	0	36,600
EA ROOFING	0	21,600
YOGI STEEL	28,000	28,000
BAWIMA STEEL	0	10,000
STEEL ROLLING	0	0
TORORO CEMENT STEEL	0	8,800
MESHA STEEL	0	1,000
VIVA STEEL	0	1,500
DONG SONG (New Plant)	0	0
KIGEZI STEEL (New Plant)	0	0
	210,200	529,860

Table 6 reports the data of the liquid and finished steel production for each steel player. On the basis of the production data collected from the visited steel players, the production of finished steel products is presently at approx. 530,000 t/year and the liquid steel production is at approx. 210,000 t/year as of 2019, leaving a deficit of approx. 320,000t/year of liquid steel required for Uganda to meet its current production capacity without having to import the required raw material.

This import level confirms that there is an immediate need to increase the internal liquid steel production, to cover at least 70-75% of finished products. Flat steel should still be imported for a while due to unavailability of Plate Hot strip mill in Uganda (it requires a huge financial investment and high production level/plants).

5.4. Uganda's Macroeconomic Data

It was necessary to evaluate the standard parameters for the iron and steel market according to the present and future country situation. These parameters are related to the consumption of steel per person year by year and their sustainable growth is important for the iron and steel value chain. To this effect, it was necessary to know the trend of projected population growth in the country.

In order to correctly calculate the steel consumed in the Country, the imported quantities of iron & steel were taken into consideration and balanced with exports.

The following are the data that was acquired from the Uganda Bureau of Statistics and used in the feasibility study:

- 1) Population growth trends (figure 7), and
- 2) Import/Export of Iron and Steel products through the years (figure 8)

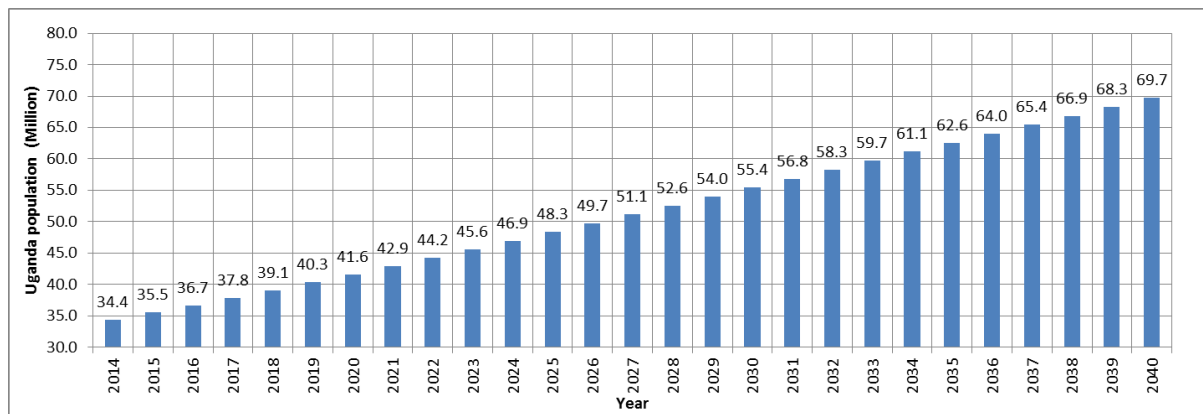


Figure 7: Trend of projected population growth in Uganda (UBOS, 2019)

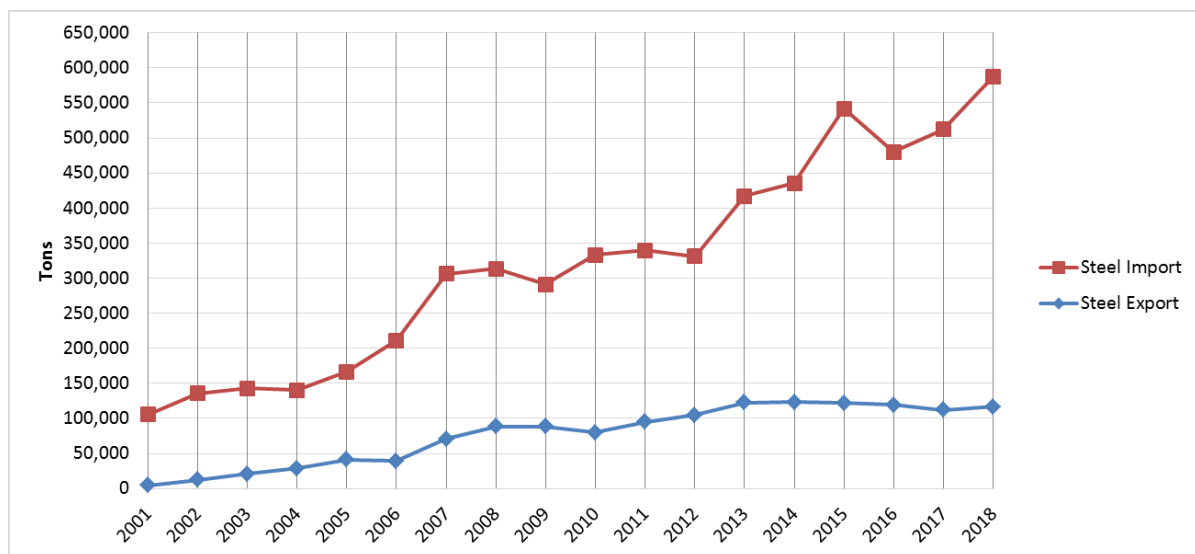


Figure 8: Graph of Steel Import/Export in Uganda (UBOS, 2019)

5.4.1. Modelling of Uganda's Iron and Steel Value Chain

The present situation of the iron and steel value chain is very fragmented. At the beginning of the chain is iron ore, which is available in Uganda but is not being mined. This part of the value chain is almost inactive, and the entire system needs to be structured and organized as explained in chapter 3 and 4 of this study. Considering the actual level of liquid steel production and the projected level in the next years, it is clear that iron ore deposits are more than sufficient to cover the steel production needs for more than 50 years i.e. fully supporting the Uganda's actual and expanding steel industry sector.

Presently, the iron and steel industry is producing liquid steel by melting scrap in induction furnaces, the only suitable technology for the power grid situation in Uganda. There are a few industry players undertaking the process of installation of direct reduction plants with RKF's utilizing coal as a reductant. With the installation of coal based DRI plants, it is possible to generate energy of up to 12-15 MW through recovering heat from waste gases. The energy could be used for powering the induction melting furnaces independent of the main electric power grid and therefore limiting the need of its reinforcement. This approach seems to be the best option in the country's current environment. It allows for gradual increase of mineral utilization, with the only need being the organization of the import of a proper reductant or provision of reductant by biotechnologies.

A capacity model of the iron and steel chain, year by year was developed up to 2024 in order to keep it reliable based on the expected infrastructural developments and foreseen changes of the industry in this timeframe. Forecast beyond 2024 was only based on reasonable and intelligent guessing of the market and technological trends.

The industrial ecosystem is in fact a very complex mixture of activities and surrounding conditions, making models difficult to converge and produce reliable output. The ecosystem depends on many internal and external factors and influences, like, for example, cost of production, consumption units, raw material availability, infrastructure and transport availability, availability of professional staff and highly technological equipment, human factor influence, government support, mentality of investors, among others. Owing to this, the modelled ecosystem can be different to the expectations due to the predominance of the above factors, which will evolve in a way that it is not easy to predict. Table 7 provides a visualization of the model output for a period of one year:

Table 7: Capability model for year 2019 inferred from 2018 data

YEAR			2019																														Transportation from Mombasa
UGANDA POPULATION			40,330,000																														
PRODUCTION			5,2																														
LUL STEEL / PERSON (kg)			8,1																														
LONING PRODUCTS			5,0																														
FLAT PRODUCTS (Tonn)			4,8																														
FLAT PROD. LINES PROD			13,1																														
FINISHED PRODUCT			15,4																														
UGANDA INTERNAL PRODUCTION (LST / TON)			15,4																														
ASSURED INTERNAL PRODUCTION (INCLUDING SHORT QUANTITIES) / KG / TON			15,4																														
FINISHED PRODUCT			15,4																														
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5.4.2. Present Uganda Steel parameters coming out of the Capability model

The typical Iron & Steel market parameters were included in graphs (figure 9) showing the predicted values through the years, for:

- I. Total Country liquid steel production
- II. Total Country I&S products production

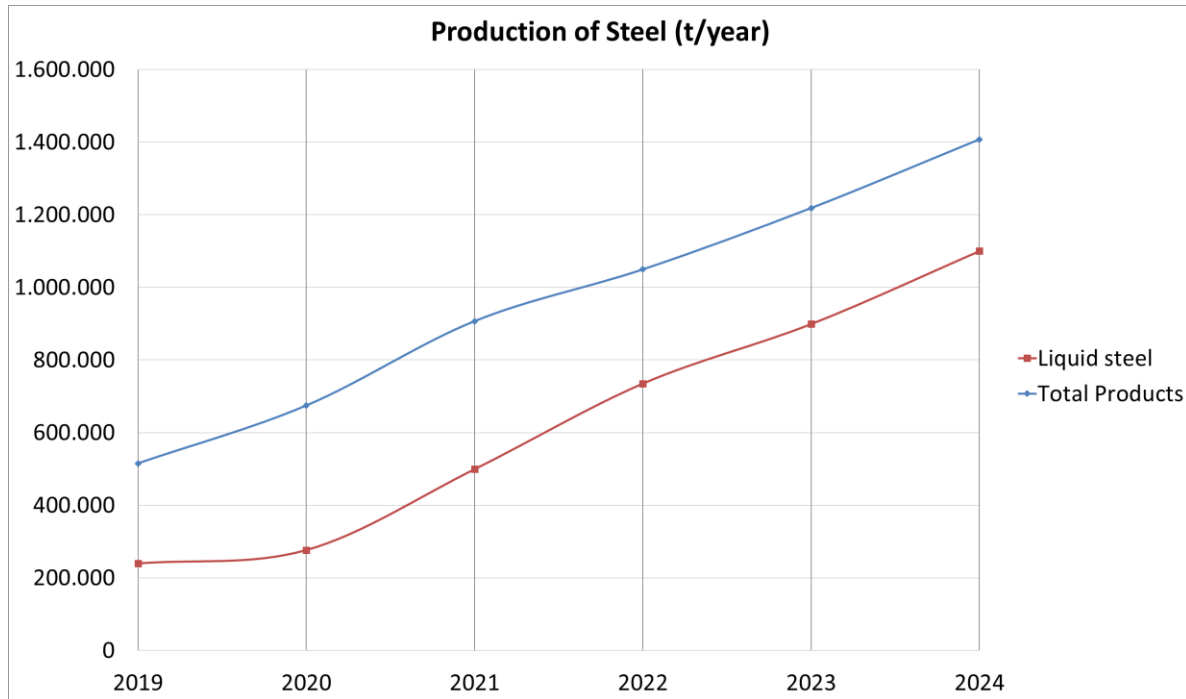


Figure 9: Graph of Projected Steel Production in Uganda

In accordance with the Capability Model (figure 9), liquid steel production can increase up to 1.1 Million t/y by year 2024. This optimistic view is based on the assumption that Uganda will take advantage of the iron ore resources presently available in the country and the natural gas from Tanzania (MoU already signed between the two countries to construct the pipeline) to produce liquid steel. Furthermore, the massive support of political measures and a significant projected growth of the economy has been considered to support the growth-trend of the iron and steel industry and reach the expected ramp up.

This implies that there will be an increase of steel demand in the Country, which will be a consequence of a strong country development based on important measures of industry sustenance and infrastructure investments. The overcapacity presently installed in the existing plants and the presence of the new scheduled plants are promising signals for a rapid increase in steel production to meet with the projected per capita values indicated in figure 10 (referred to total products and liquid steel).

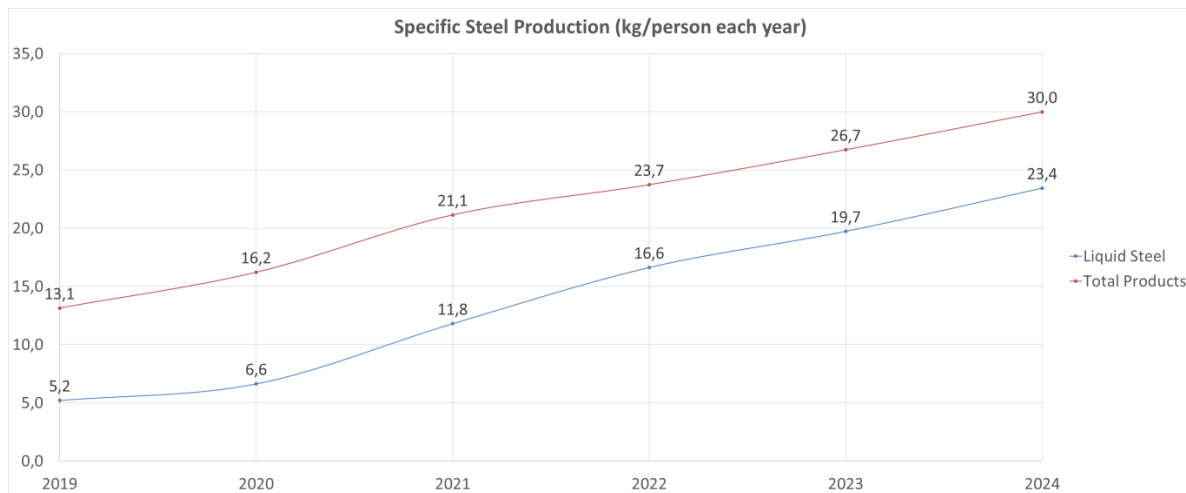


Figure 10: Graphs of Projected Steel Specific Production in Uganda

5.4.3. Economic model of Uganda Production and Consumption of Steel

In order to compare the assumptions included in the Capability model year by year for the Iron and Steel industry growth, an economic model was developed based on historical country data. It assesses the impact of Iron and Steel value chain on GDP without considering specific disruptive measures for a fast development of the industry. Further detailed description of the economic model is provided in Annex 3.

The task was commissioned to Rina Consulting (Italian company well known for the combination of deep technological expertise and rich industry insight – www.rina.org), who utilized a simplified Klein model (See: Greene, 1990¹) including the parameters relative to the Steel Sector impacting on the study, i.e. steel production and steel specific consumption.

The result is the Dynamic General Equilibrium Model (DSGE) for the project. In general, DSGE models are multivariate time-series models that are used in macro-economics, for policy analysis and forecasting. These models are systems of equations that are typically derived from economic theory. As such, the parameters are often directly interpretable based on theory. Therefore, DSGE model calculation allows current values of variables to depend on past values and expectations of future values. All of the parameters used in the DSGE model are to predict the effects of steel sector growth on Uganda economic sectors as well as consumption and future investment based on actual economy.

DSGE is a model for policy analysis that is structured into three groups: demand, supply, and monetary policy. The equations that describe these three groups come from the Micro Foundations group, with clear assumptions about behaviours from main agents in the economy, including households, businesses, and states. These agents interact in a clear market over time, leading to “general equilibrium” of the model.

Input data for Uganda economic model was based on the following historical data:

- 1) Apparent steel consumption obtained based on the relevant data coming from World Bank, World Development Indicators Data Base, World Steel Association and Trade Map Data Base.
- 2) All goods consumed in Uganda except steel.
- 3) Investment in the steel sector as sum of loans and foreign + domestic investments.
- 4) Gross Domestic Product and the Production of Steel in a dynamic way.

¹ Greene, W.H. (1990) Econometric Analysis

The model output includes the steel parameters of interest for the iron and steel industry referred to as apparent data (i.e. it consider in total, the amount of imported and exported steel and also steel quantities included in goods like for example cars, appliances, etc.) integrated with the actual data collected during the visit to steel players.

Table 8 below summarises, the assumptions required to run the economic model. These assumptions are generally validated with similar cases and are keeping into account the specific country data relevant to population, government expenditure and investment.

Table 8: Table of Model assumptions

Parameter used	Source of data
Population trend	Uganda Bureau Of Statistics (UBOS) data
Government Expenditure generally on the economy	Assumption of 1% increase in the annual expenditure starting from 2019
Total Investment in Steel sector	Obtained from the sum of outstanding loans to Metal sector (from UBOS) and Direct Foreign Investment calibrated to the metal sector, considering that half of it goes to the manufacturing sector; metal sector weights 8.3% of manufacturing sector (World Bank Data Base). Assumption of 2% increase in the annual investment starting from 2019
Import and Export of steel	UBOS data

The model uses a system of equations and identities as shown below:

Equation 1: Consumption of steel = function of {GDP, Population, Consumption of steel (time t-1), Net export of steel (Export-import) Government expenditure}.

Equation 2: Consumption of other goods= function of (CPI, GDP per capita, Net export of other goods)

Equation 3: Investment in the steel sector= function of {Real Interest Rate, Total investment (all economic sectors) (time t-1),

Identity 1: $GDP = \{ \text{Consumption of steel} + \text{Consumption of other goods} + \text{Government expenditure} + \text{Total investment (all economic sectors)} + \text{Net export of other goods} + \text{Net export of steel} \}$

Identity 2: Production of steel = Consumption of steel + Net export of steel

As commonly known in this kind of model, there is a problem of endogeneity that can be overcome by applying the 3 Stage Least Squares Method. The goodness of the model with reference to each parameter is given by the value of R^2 (goodness of fit). This was found to be above 0.9 for each output, which means more than 90% of the variability of the specific considered parameters was well represented by the model.

Table 9 gives the independent variables for each considered parameter and its goodness of fit value.

Table 9: Table of 3 Stage Least Square Estimation

Consumption of steel	
Independent variables	3SLS Coefficients
GDP	-.0092425
Population	20.35363*
Consumption of steel (time t-1)	.1051833
Net export of steel	-1.314868***
Government expenditure	-.0118188
Constant	-3.71e+08*
R²	0.9335
Consumption of other goods	
Independent variables	3SLS Coefficients
CPI	8.06e+07***
GDP per capita	1.05e+07***
Net export of other goods	-.2745209*
Constant	-9.04e+08
R²	0.9918
Investments in the steel sector	
Independent variables	3SLS Coefficients
Real Interest Rate	35833.75
Total Investments (time t-1)	.0376584***
Constant	3.03e+07**
R²	0.9415

*** Significance level = 0.01, ** Significance level = 0.05, *Significance level = 0.10

The model was forecasted for 5 years (2020- 2025) and the results are shown in figure 11. The graphs show that the model predictions fit with the production of steel, specific consumption of steel, private investment in the steel sector and GDP.

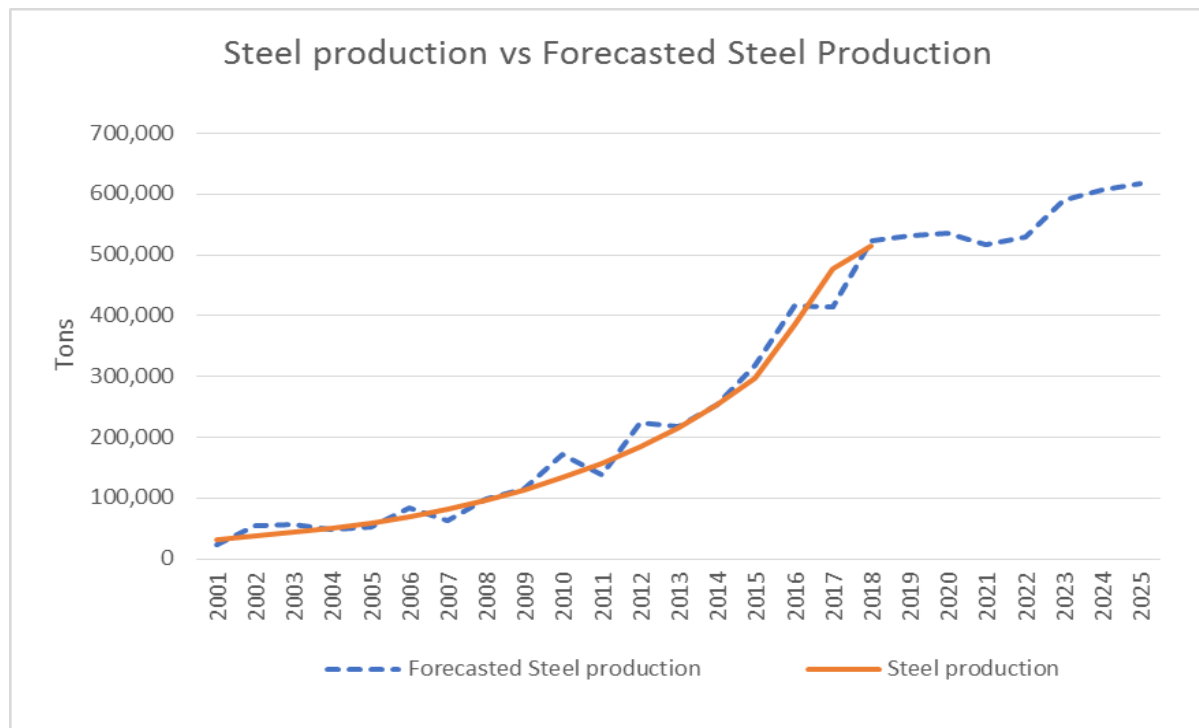


Figure 11: Graph of Uganda Steel production by economic model (Source: RINA-C)

Figure 11 shows that the model results (blue dotted curve) fit well with the actual historical steel production trends from 2001 to 2019 (red curve).

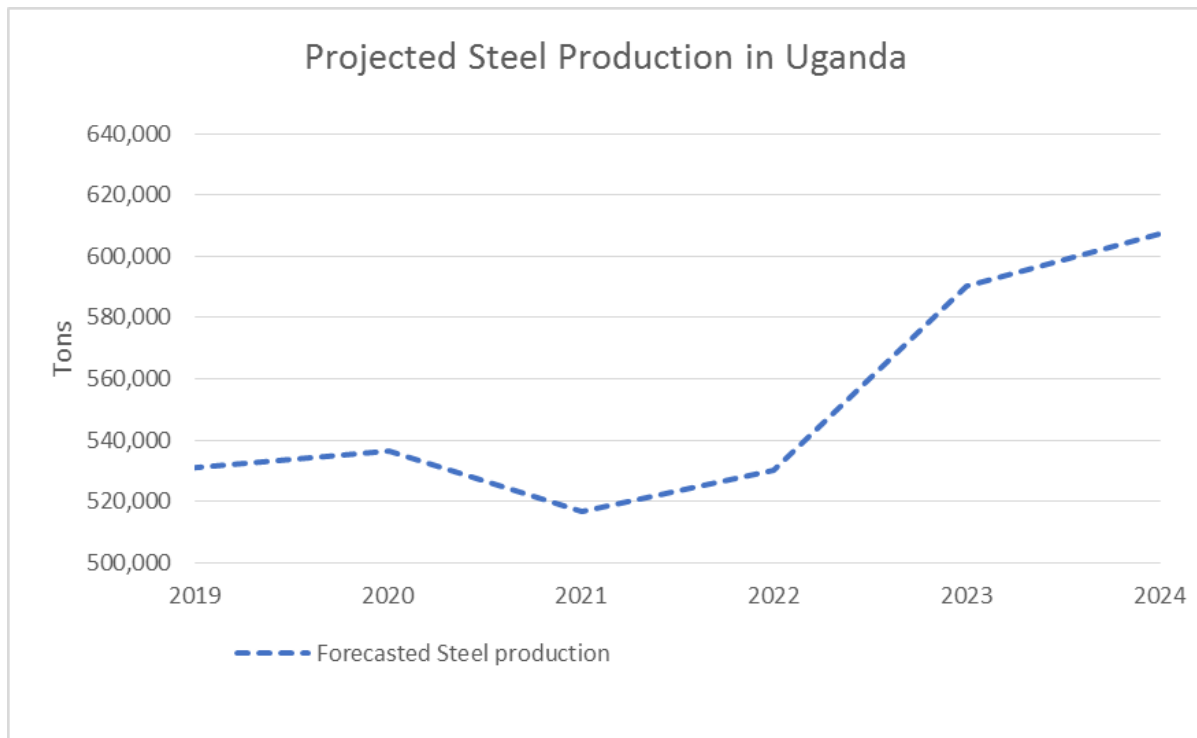


Figure 12: Graph of Uganda Steel production (extracted for the forecasted years only)

Figure 12 shows the enlarged part of figure 11 for the years 2019 to 2024 and the prediction shows a general overall increase over this period. Even though there is an initial fluctuation between 2020 and 2022 there is on average a distinctive steel production growth over the next 5 years. The initial fluctuation is due to both the model, which follows a cyclical economic pattern, and to a possible temporary phase where the industry faces the conversion of its production lines because of the new investments.

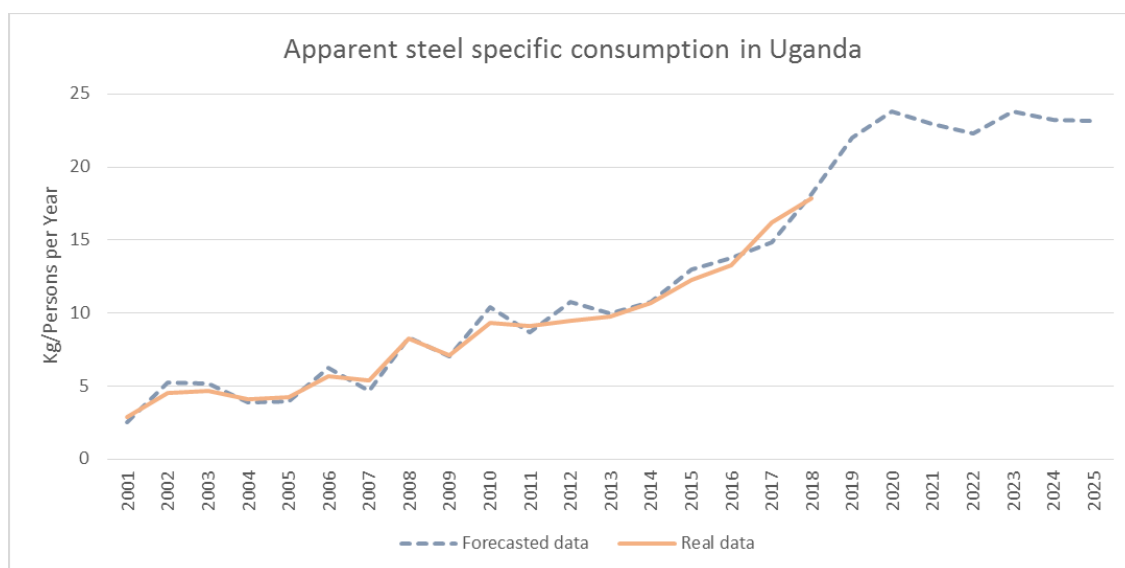


Figure 13: Graph of Uganda apparent steel specific consumption per person each year (Source: RINA-C)

Similar to figure 11, figure 13, shows how the model results (blue dotted curve) fit with the actual historical data (orange curve) as far as per capita steel consumption is concerned. Please note that the specific consumption of steel refers to apparent values of consumption as opposed to the figures reported in the Capability model.

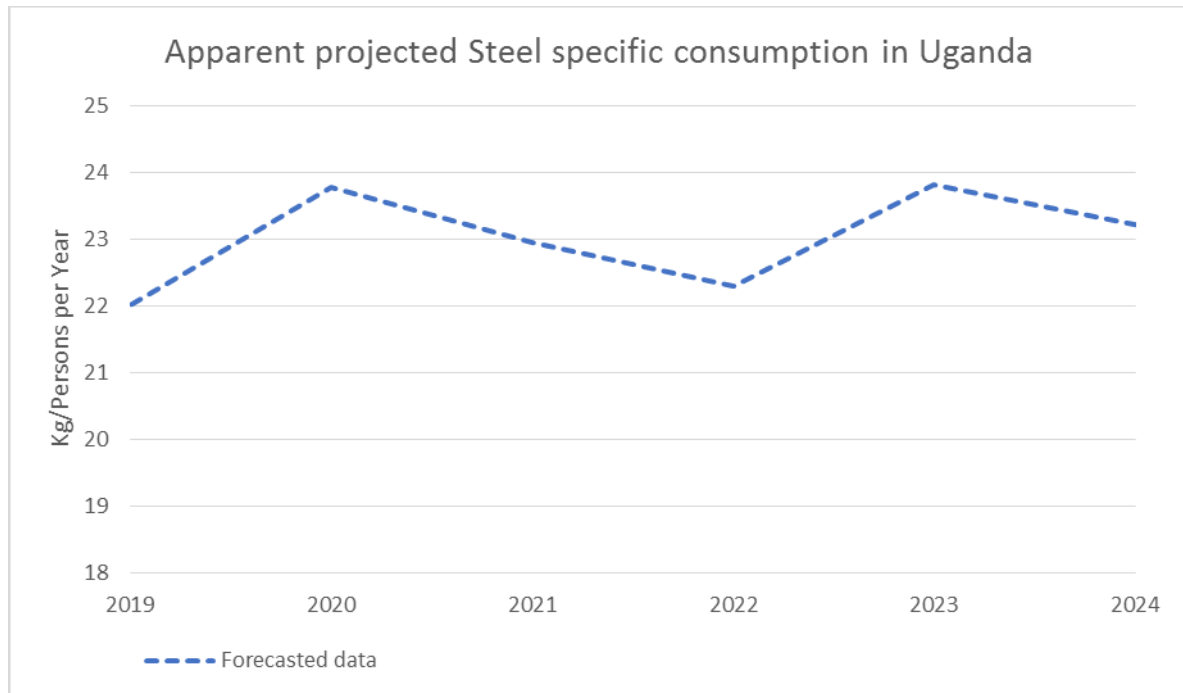


Figure 14: Forecasted specific Uganda steel consumption per person each year (Source: RINA-C)

Figure 14 is the enlarged projected part of the specific steel consumption over the period 2019 to 2024. It shows that consumption is reaching a steady value because its increase is balanced by the fast growth of the country's population.

Population data are generally referred to people that consume the steel while available data for Uganda are referred to any age from child to elderly people. For this reason, the curve is not completely representative since children make up the majority of the population compared to the data of other countries where the population growth is low, and the children percentage much lower.

Figure 15 shows the graph of investments. Investments in the past years were provided only with data on bank loans to the metal sector while, in the last years, foreign direct investments and local private investments have been considered too. It can be observed that the model prediction averages around the actual trends of investment over the years, which gives credence to the model prediction over the planning period of 2020-2025.

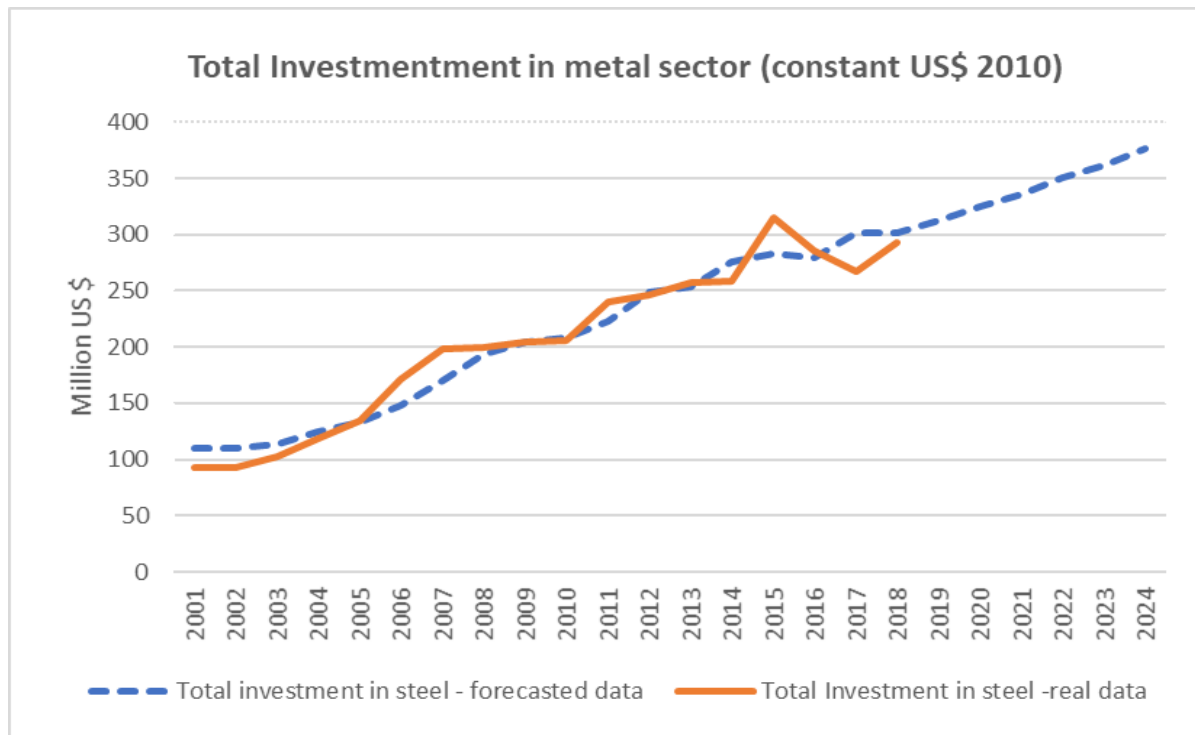


Figure 15: Graph of Uganda Private Investments Steel Sector forecast (Source: RINA-C)

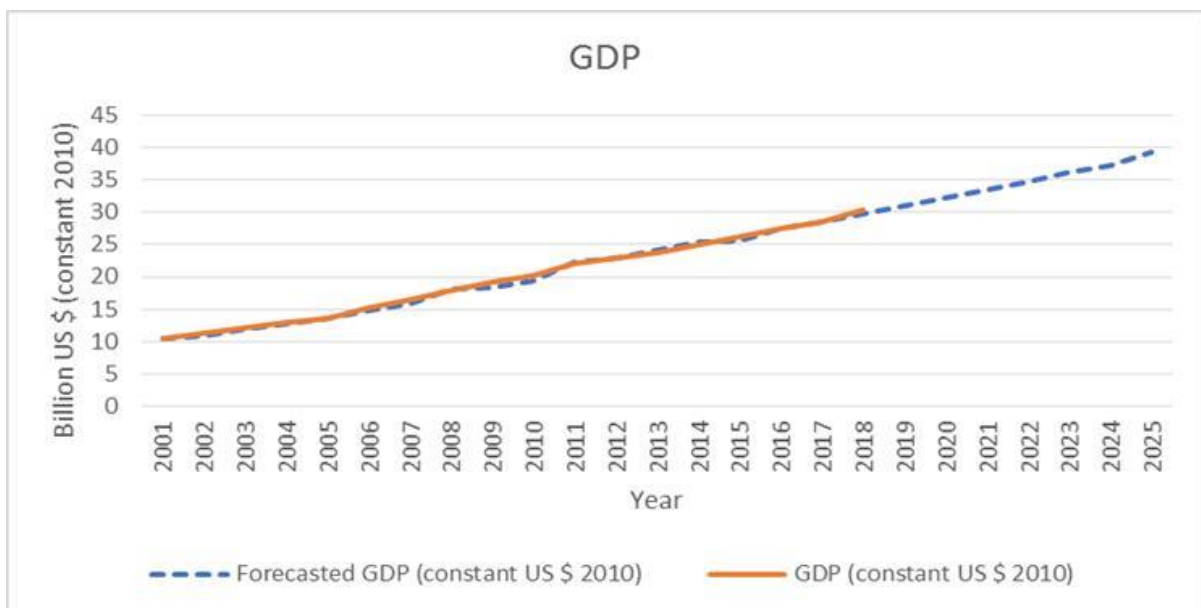


Figure 16: Graph of Uganda GDP data (Source: RINA-C)

Figure 16 shows a steady increase of GDP up to 2025 reaching a value of about US\$40 billion. Again, the model seems particularly fitting the historical figures.

In order to get an overall final evaluation of the future steel market in Uganda the following aspects must be taken into consideration:

- 1) The results of the economic model
- 2) The information collected from the steel players and the related Capability model
- 3) The steel market situation of similar countries worldwide.

5.4.4. Uganda's Steel Industry in comparison with other countries

The typical iron and steel parameters (production and specific consumption) calculated for Uganda were compared to some benchmark situations worldwide (figures 17 & 18). Data from World Steel association (Steel Statistical yearbook 2019) have been considered for the comparison.

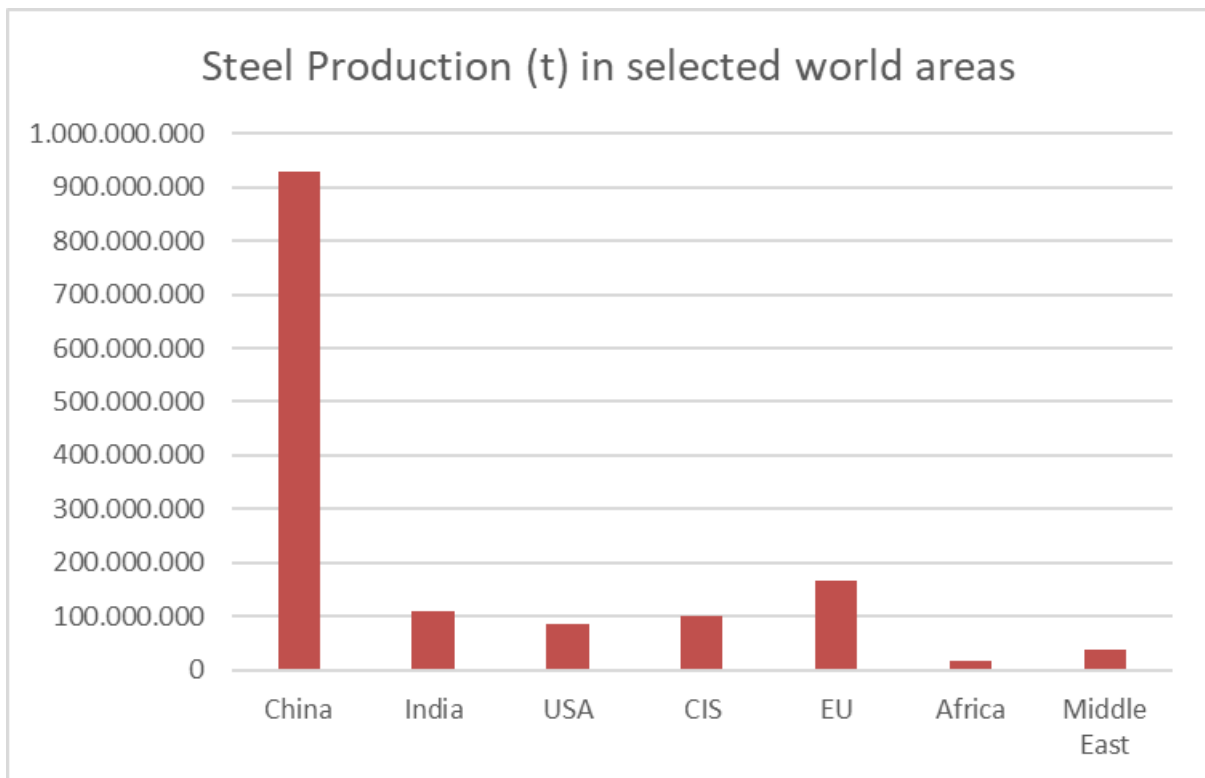


Figure 17: Graph of Steel Production in selected world areas.

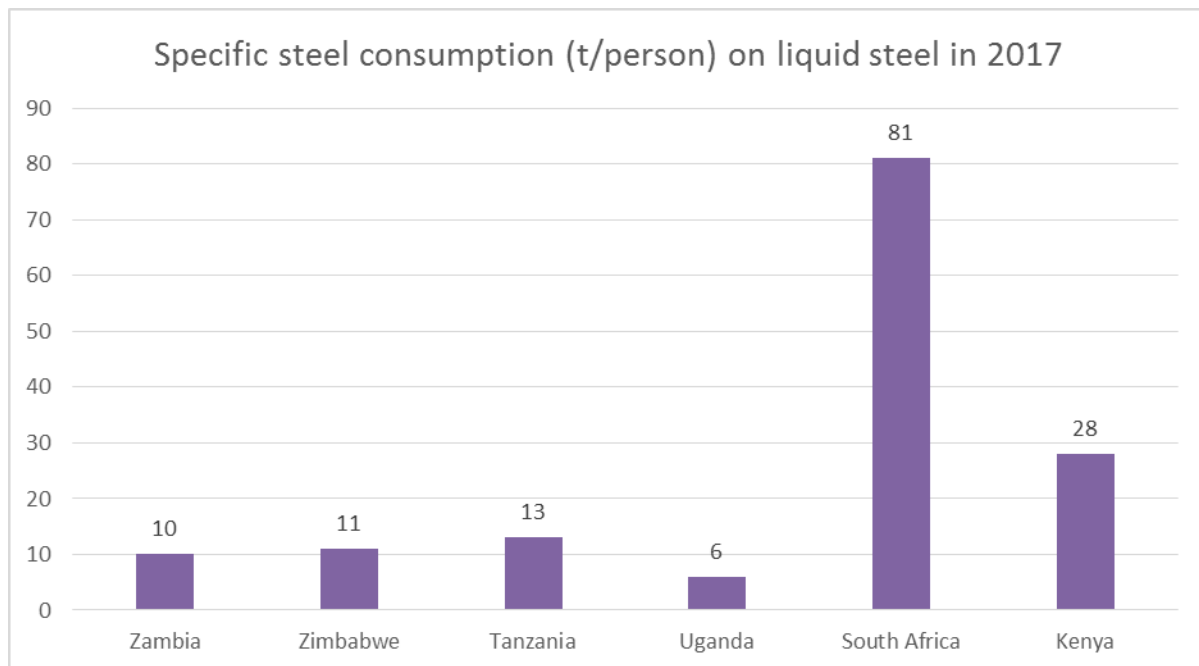


Figure 18: Specific Steel Production in 2017

China produces almost 50% of the world steel while Africa, with a population of almost 1.3 billion (about 17% of the world population), produces just 0.9% of world steel (15 Million ton/y). Among the East African Countries, Uganda with only 6 kg/person as of 2017 statistics, which is the lowest specific steel production in the region, ranks at the lower end of steel production.

5.4.5. Economic model versus Capability model output

The economic model is a precise instrument to forecast the trend of development of the industry. It was set on historical values, country economy parameters and assumptions (refer to Table 8), which can be eventually modified to verify alternative scenarios. The considered variables of the model have the proper goodness of fit, showing accuracy of the model prediction. However, the economic model does not consider some of the country specific situations, like rate of industry development, which may develop at a faster rate when compared to the model provision. Particularly important is the overcapacity of the actual key players in Uganda, which allows for immediate supply of additional steel once the economic situation calls for more steel product requirements without the need for waiting for further investments in new equipment.

Another important factor is the predicted increase of liquid steel production in the short term once some key players set-up the planned new DRI plants (some of which are already under construction); it surely will rapidly lower the cost of final products, which in turn means the possibility to increase steel demand in Uganda and probably also for the export market.

Finally, government involvement in the development of the industry will accelerate the provision of measures necessary and required to develop liquid steel production, and this can be achieved by:

- 1) implementing proper policy on import of semi-finished products in order to favour and protect local players investing in DRI technologies (e.g. Imposition of taxes on imports of semi-finished products can promote sustainable local production of liquid steel).
- 2) Developing the exploitation of iron ore deposits in the different parts of the country (mainly Kabale).
- 3) improving the electric grid system.

- 4) facilitating investments in the sector.
- 5) making available natural gas in the country.

If these measures are backed up by general development of the country's economy and infrastructure (particularly transport), the optimistic scenario of steel production trend showed in figure 9 could become a reality. Section 5.5 lists the measures required to reach the required targets.

Figure 19 gives the comparison of steel production given by the Economic model (i.e the situation of the sector's natural growth without any targeted interventions) versus the Capability model taking into consideration the new plants start-up (as anticipated by the key players' visits) and the Country's boosted economic growth.

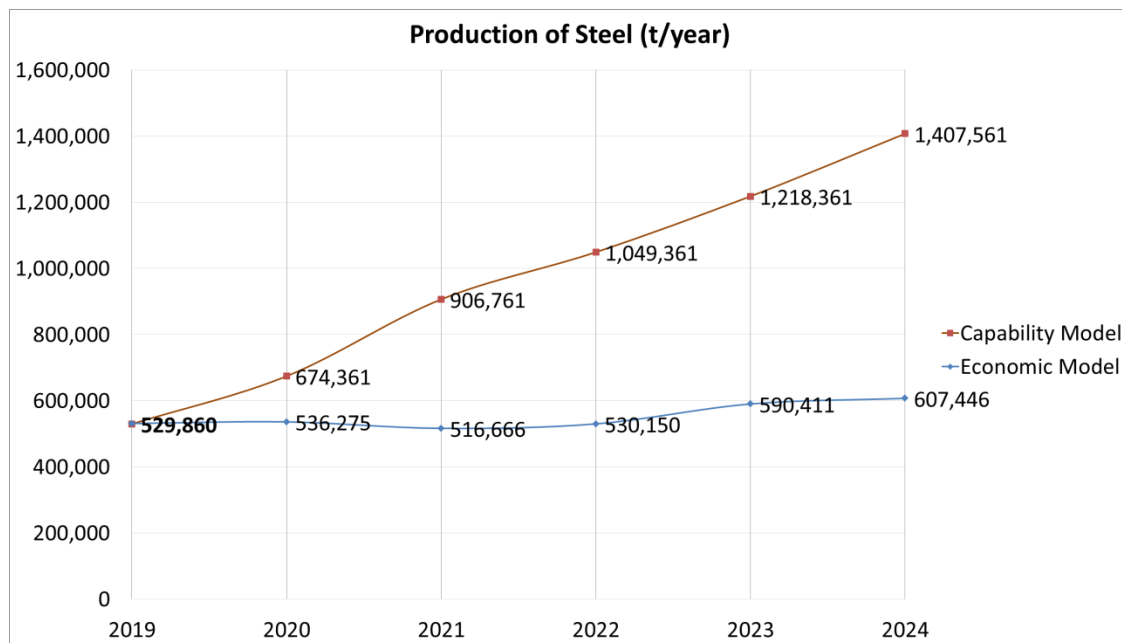


Figure 19: Graph of Steel Production comparison (Economic model vs Capability model)

It can be observed from figure 19 that if the country is to achieve its development targets for the I&S industry as per the Vision targets, there is need for the proposed interventions to be made to boost its growth.

5.4.6. Uganda steel consumption data and impact on liquid steel

The steel consumption per capita in 2019 is estimated at 13.1 kg/person, while the liquid steel production per capita is at a level of about 5.2 kg/person. Out of the 13.1 kg/person of steel consumption, long products are approximately 8.1 kg/person (62 %), while the rest 5 kg/person (38%) is Flat products. Products for construction works are almost absent in Uganda, the strong majority consumption being roofing sheets, barbed wire, nails, meshed wire, hollow sections etc. This consumption distribution and products type confirms that there is no adequate steel demand coming from construction.

Construction type of products, such as profiles, pipes, rails, etc should be considered gradually in the country (starting from a production from imported semi-finished material) considering the difficulty and cost of local production of these high-alloy / high-strength steel qualities. The installation of a rail mill is considered urgent and necessary to produce rails locally and support railways development in the country and region.

Liquid steel production of 5.2 kg/person as of 2019 (46% of total steel consumption) might increase to a level of 10.9 kg/person once the Dong Song Plant starts liquid steel production. However, from the study visits, there are some concerns as to whether the plant will be constructed and whether the process of charging iron ore in electric arc furnace will be really an industrially viable process. Nonetheless, for the projection of the study, the team considered it real in the model calculations - the installation and start-up of this plant.

Liquid steel in Uganda is estimated to reach 52% of finished steel products in 2021 but it should be further increased in order to close the gap and to reduce the imported semi-finished materials (billets and wire rods). The increase in liquid steel production in turn will decrease the cost of the finished products and make the steel industry more competitive for export in the surrounding countries. Presently, the cost of finished products is heavily affected by transport costs (25-30% of the total cost) of raw materials of mostly South African origin (coming from Mombasa via rail or trucks).

Considering that there is no proper and organized availability of scrap in the country, many of the steel producers are not in a position to increase the level of liquid steel production by electric induction furnaces (no raw material) and also because the country is experiencing a diffused unavailability of stable and good electrical energy from the grid, as reported by many interviewed steel makers.

With the anticipated increase in liquid steel, export of steel may slightly rise from the actual estimated 120,000 t in 2019 to almost 168,000t in 2024. These estimated figures will allow for a potential increase in consumption in the next few years and thereby facilitating the realisation of basic assumptions made on the steel industry development. With this increase, the consumption of steel in Uganda will thus be about 30 kg/person in 2024, though this level is still lower than major developing countries in steel consumption in Africa. It also needs to be pointed out that there is rather consistent production of only low added value finished products, like Nails, Meshed wire, BRC etc. An increase in the range of products and product quality is necessary for a sustainable development of the steel industry.

5.4.7. Progressive development of Uganda's steel consumption and introduction of new technologies

It is reasonable to foresee that with the gradual increase of steel products demand; steel overproduction will be covered by this demand and the country will be ready to install new plants targeting new technologies. Considering the trend of import and export of Iron and steel products and considering the present overproduction capacity, it is recommended to align to the model predicted figures in the next years.

When the first plants, (Dong Song, Tembo or any other), start their operation, there could be a sudden increase of liquid steel in the country, thus lowering the cost of liquid steel and giving a boost to export. This condition is well reflected in the Capability model.

According to the experience of the researchers that conducted this study and to what is already recorded for other developing countries, the following list of implementations for the industry (divided by single value chain position) is considered appropriate from a technological point of view, the most sustainable development path:

- 1) **Mining:** the initial development of the primary portion of the value chain can be realised through the utilization of direct reducing plants and later on by the use of mini blast furnace technology. Therefore, iron ore is surely an important asset and mining must be properly developed. Uganda absolutely needs to start exploiting its iron ore resources and through this activity support the steel industry growth. It will facilitate liquid steel production and all downstream activities. As the

realisation of the development of this initial stage is projected to be a gradual process, and with a big number of utilizers (clients) spread around the country, it is not foreseen, at least for the next five years, that a centralized beneficiation facility is a viable option (which is costly but considered to be more efficient in the future if a mini blast furnace is to be installed).

- 2) **Melting and refining facilities:** at the present stage of power grid infrastructure and electric power availability, the only possible means to make liquid steel in Uganda is with induction furnaces, though they are very limited in their refining process capability. The required steel chemistry per steel grade can only be achieved with a proper scrap selection. This results in a limited product mix or higher production cost by using more expensive and cleaner scrap. Due to the lack of refining capability, induction furnaces are mainly present in rebar production with carbon content in range of 0.20-0.25% and with phosphorous content of max. 0.035%.

In order to reduce steel production costs and improve the liquid steel quality, it is necessary in the future to develop other technologies like electric arc furnace (EAF) and/or BF/BOF:

- 2.1) The EAF cycle contributes 28% of liquid steel production worldwide. The EAF process is more flexible than the induction furnace due to the possibility to: (i) refining the steel by means of the slag formers addition, (ii) control temperature and (iii) use different charge mixes (low scrap quality). The average EAF size has a capacity of 100-120 ton with a tap to tap time of about 40 minutes and energy consumption of about 630 kWh/t. The utilization of the EAF furnace could bring several advantages to the steel market in a short-term period, but at the moment this technology cannot be installed because of the lack of a reliable electric grid and the peaks of electric energy absorption of this equipment.
- 2.2) The BF/BOF cycle is used to produce almost 1.2 billion t/y of liquid steel (71% of the world liquid steel production). This cycle is more complex because a coke oven plant and a sinter plant are required to produce the blast furnace raw materials. Moreover, the energy required to produce liquid steel is in the range of 2,900-3,200 kWh/t. The utilization of iron ore as a raw material allows the realisation of liquid steel with high grade of cleanliness. The blast furnace technology will be an option in the future but since it requires a stable liquid steel demand of at least 1.5-2 Million tons/year and a proper infrastructure able to both provide the high quantities of raw material to the steel-shop and to deliver the final products to the market, both of which are not presently available in Uganda, it can be planned for future introduction (about 10-15 years).

The above described technologies can be adopted in a phased approach with the increased demand of the market and with the country becoming ready for their use in terms of energy and infrastructure. This will allow production of better steel qualities and, aside from long product plants, will also allow gradual introduction of flat products producing plants. With the steel qualities in place, the needs of increased mechanical properties will move the industry towards alloyed steels. This evolution requires, beside electric arc furnaces, ladle furnaces and connected technologies to make environmentally free and efficient processes.

- 3) **Rolling mills:** the actual process for long products is, in general, very primitive in Uganda mostly because quality is not an issue on the final products. The technologies being used are generally 50-60 years old and very much relies on manual operation, missing almost entirely higher levels of automation.

With an empowered market demand on new steel qualities and new products, it will be possible to initiate the development of the industry exactly from this stage of the value chain. Rolling bigger profiles and flat products will be a good option for growth; new plants will gradually appear in the market. The process can start with selected products, for example, rolling facilities for

producing rails and seamless pipes, in particular, installation in the country of a rail mill could facilitate and speed up the railways network development all over Uganda. The cost of transportation is presently negatively affecting the entire industry and the most efficient and low-cost transportation of raw material and steel products is by rail through a diffuse network.

With market increase, subsequent demands for better quality will become an issue, which may result in campaigns to refurbish or renew equipment as anticipated, together with the adoption of new plants with modern technologies in the industry.

- 4) **Finishing and heat treating:** only a few facilities in the country presently have this kind of process and these are limited to roofing material and galvanizing. This kind of equipment can be installed in the value chain at any time and this should follow the trend of the demand in the market. It is in fact necessary for the steel players to properly feed other industries of the country that start up and develop. It is impossible to predict at this stage which finishing and heat-treating facilities will be required in Uganda without the knowledge of the development of the industries requiring those products, for example, if Kiira Motors starts operations, probably all technologies applicable to automotive will be installed (annealing and superficial heat treatment). If other industries develop, for example production of appliances and other products, then different approaches apply for heat treatment technologies. Anyhow it will be surely possible for the steel players to monitor the demand of quality steel with high mechanical properties, which will subsequently result in specific heat treatment plants with the necessary finishing equipment to be set up in Uganda.

On the side of infrastructure and electric power needs, there are no specific constraints for the finishing equipment and this characteristic will also facilitate installation of new plants if this will be economically feasible and required by the market.

5.5. Considerations for improvement of the Iron and Steel value chain

Presently, the downstream companies are encountering several obstacles and difficulties in almost every field of the supply chain in developing their business. The challenges obtained from the field visits are given in the proceeding section together with suggested possible remedies.

Iron and steel are produced from raw materials through a series of processes that can be clustered into 4 groups:

- 1) Exploration and mining
- 2) Iron/steel making
- 3) Iron/steel refining and casting
- 4) Steel rolling

The set-up of these phases represents the iron/steel value chain and for each phase some actions are needed to make the whole production competitive. The actions can be divided into two groups for each step: general actions (which are the pre-conditions to facilitate growth of the industry) and specific actions.

General Actions or Pre-conditions

Considering the actual situation in Uganda, the following general actions are required:

- 1) **Improve the availability and the stability of electric power.** Steelmaking is an energy intensive industry (EII) and electric power at competitive price represents a key point for the development of the whole steel industrial sector. Presently, it is necessary to invest in the distribution line more

than in energy production. New Power plants are in the pipeline, but energy urgently requires a proper distribution network to be reliable and, most of all, stable for the steel industry players.

- 2) **Improve the infrastructure:** Uganda does not border with the sea and so it is important to improve the land transport systems (railways and roads) for raw materials and final products movement. Steel making processes require big amounts of raw materials that in order to improve the whole value chain must be delivered on schedule. The delivery of the raw materials on delay can cause an unexpected stoppage of the whole production line, which, for some plants can have very negative impact. Transport should have the lowest possible cost in order to avoid affecting the final product prices. Presently, it is incredibly high. In the immediate term, the government must, on its own or through PPP, build roads and railways to facilitate transportation of iron ore to the smelting plants and to the ports for export if the ban is lifted. Another option is to use freight by ships/barges on Lake Victoria to enable movement of iron ore and coal from indicated sources to private sector industries in Jinja and those proposed for Tororo and Kampala. The infrastructure and other constraints within the mining subsector require immediate attention to enable private sector entities that are already mining in South West Uganda and delivering the ores to Jinja to transact their business. This key pre-condition is of utmost importance and as such, is further described in detail in the logistics section of the study (Chapter 7) and in Annex 4 on Supplier Development Program.
- 3) **Adopting trade policies that promote import replacement:** Trade policies can play a key role for the development of the steel sector in Uganda. The measures to discourage the importation of the semi-finished products could boost the production of the domestic liquid steel, which in turn could significantly lower the costs and promote export of the steel products.
- 4) **Additional government policies:** The development of internal steel market (construction, steel components, bridges, highways, etc.) can be boosted by means of tax deduction for investments in steel production sector and reduced tariffs for utilities. State involvement in the steel industry can contribute to the development of steelmaking industry.
- 5) **Labour skill:** The modern steelmaking can be considered a high technology industry that requires skilled labour with degrees in engineering and chemistry. The improvement of the knowledge of steel production by means of specific University courses in metallurgy through international scholarships and other avenues, at all stages of steelworks, is a prerequisite to train the leadership for the steelmaking industry of the future. Moreover, knowledge sharing and engagement of strategic networks involving international partners, must be carried out. Artificial intelligence to analyse data collected by IOT of the machines, online modelling of the plants, automatic blockchain systems / contracts are just some of the skills required by the modern steelmaking plants. A Skills Development Plan (Annex 5), which forms part of this report further elaborates the importance of skills development for sustainable growth of the industry.
- 6) **Environmental aspects:** Iron and steel industry is highly intensive both in material and energy. For the development of the industry sector, the environmental aspects must be taken into consideration. The utilization of state-of-art-technologies is mandatory to minimize the environmental impact of the steel industry. For the two most important routes of (i) integrated cycle (sintering plant, coke oven plant, blast furnace and BOF) and (ii) EAF cycle, the main environmental emissions and the filtering systems to minimize emissions are reported below.

Integrated cycle:

Sintering plant: The preparation of the blast furnace burden with proper physical and chemical characteristics requires agglomerating of fine ores, additive and recycled iron-

bearing materials that are ignited with the coke breeze. The main stack emissions of sinter plants account for up to 50% of the total dust emission from integrated steelworks. The off gas contains dust, heavy metals, SO₂, NO_x, HCl, HF, CO, org C, PAH, PCDD/F, PCB. The most common off-gas filtering system in the sinter plant is composed by dry electrostatic precipitators (ESP) and bag filters. Moreover, the injection of adsorbents (active carbon, lime etc.) in the upstream of the bag filter is installed to reduce the emission of the other hazardous compounds.

Coke oven plant: the coke is produced by means of the heating of coal in an oxygen-free atmosphere. The coking process produces off-gases (direct, diffuse and fugitive emission) composed by dust, H₂, CH₄, H₂S, SO₂, NH₃, HCN, CO and hydrocarbons. The off-gases coke oven gas is fully collected and further processed by the cleaning and recovery of by-products and used by other steelworks consumers as a fuel. The wastewater is treated by biological technique by means of the aerobic biological system with activated sludge.

Blast Furnace: It is a counter current reactor into which iron-bearing materials (iron ore lump, sinter and/or pellets), additives (slag formers such as limestone) and reducing agents (i.e. coke) are continuously fed from the top of the furnace and the pre-heated air is injected from the lower part of the reactor. The blast furnace is primarily a source of dust and gaseous emission to air. Raw blast furnace gas contains dust (including heavy metal and carbon), carbon monoxide, carbon dioxide, sulphur, ammonia, hydrocarbons etc. The gas is purified to remove coarse dust by means of dry cyclone or dust catchers and subsequently wet scrubbing is used to remove fine dust, heavy metal, SO₂ and cyanide compounds etc. In some plants, an electrostatic precipitation is applied. The cleaned blast furnace top gas is then stored in gasholders for subsequent use as a fuel. The dust content after the cleaning treatment is less than 10 mg/Nm³. The wet scrubbing generates a contaminated wastewater flow that is usually treated in circular settling tanks. The sedimentation properties of the sludge are very often improved by dosage of flocculation agents (anionic polyelectrolytes, mixed polymers or activated silicic acids) or by sludge contact plants.

Basic Oxygen Furnace (BOF): The removal of the impurities occurs by means of the injection of oxygen in dedicated reactor. This process is a source of off gas contained dust, organic C, SO₂, NO_x, CO, heavy metals etc., solid wastes/by-products and wastewater. Dust and off-gas emitted during the blowing period are combusted in the flue-gas duct and the sensible heat is recovered. The off-gas dedusting is usually performed by venturi-type scrubbers or dry and wet ESP. The dust concentration after the treatment depends on the plant configuration but however usually is in the range of 10-50 mg/Nm³.

Electric cycle:

EAF: The direct melting of iron-based materials such as scrap is usually performed in an electric arc furnace which plays an increasingly important role in modern steelworks concepts. The electric arc furnace steelmaking process is a source of primarily dust and solid wastes/by-products. Primary off gases represent approximately 95% of total emissions from an EAF. Most of the existing plants extract the primary emissions by the 4th hole. The EAF off-gas contains dust, metals, nitrogen and sulphur oxides and organic matter. Generally, the treatment of off-gas flows (both primary and secondary emissions) is performed in the same device, mostly in bag filters. Only in a few cases are ESPs and wet scrubbers applied. The combination of direct fume extraction and a hood system is often used. This combination achieves a collection of about 98% of the primary emissions. This measure has also improved the working environment significantly and reduced the diffuse dust emissions via roof fans up to 2.5 mg/m³.

5.5.1. Specific Obstacles, difficulties encountered by Key-players and possible remedies

Downstream steel producing companies are encountering specific obstacles and difficulties according to information collected during the site visits of the key players. Specific visits reports are further detailed in Annex 2 - Technical Competitiveness Report.

Here below are listed the most important factors that are impacting the industry development, and which are not permitting Uganda steel producers to improve and expand their finished production levels, hereby referred both in quality (with only few exceptions) and in quantity (for almost all producers):

1) Quality

The quality of steel products in Uganda is at a rather low level as evidenced by the reports of the visits done to the steel players. This is mostly due to the following reasons:

- a) Finished product demand and consequent production is concentrated in low rebars, nails, wire mesh, BRC and similar products made of low alloy steel rolled from long products, which are generally classified as low quality since they are not made of high alloy or/and not further processed, and heat treated.
- b) There is no production of medium high-quality finished products except for minor exceptions and the reason highlighted by the steel players during the visits was the low demand.
- c) No useful Quality Certification is required for common market sale in Uganda according to information received from some of the players. They stated that all of Ugandan production is quality certified but final end users cannot differentiate, and therefore appreciate, the differences in quality depicted by different certification levels. If the statement is confirmed, then it means that the current quality level of certification is not capable of distinguishing between materials quality and is thus not adequate for the country.

2) Cost Competitiveness

The competitiveness can be improved through:

- a) Increase of liquid steel production, which therefore limit/eliminate the import of semi-finished products, of which the transport cost is, in some cases, around 25-30% of the total cost
- a) Investment in high quality Rolling and Process Equipment with better tolerances
- b) Increase in the high-quality finished product demand, so only steel producers with better quality products can take more market share.

3) Energy Efficiency

Energy efficiency can be improved by:

- a) Improving the stability and power of the national electric grid (one of the primary reasons for low production and lack of investment in steel industry)
- b) Installing more technologically efficient equipment (for example EAF/LF)
- c) Increasing the production rate (that will decrease energy specific consumption per ton)

4) Scrap quality and availability

Scrap quality and quantity (both presently at very low level) can be increased by progressing construction and investment in Uganda, which is a time dependant issue. More investment and progress in infrastructure development will certainly generate more scrap. The scrap market must be organized as it happens worldwide as a commodity, sold at fixed price and collected as per its quality; organizing the market of scrap can also facilitate its use. The system should generally be handled by the state or by dedicated traders in order to avoid uncontrolled collection of steel materials (often stolen or not regularly collected). Below is an example of Scrap handling procedures in Europe:

EU Countries Scrap yards and handling Procedures

The European steel scrap specifications have been agreed upon by most commercial firms in the EU-27 member states that are involved in the recycling and processing of ferrous scrap. The specifications classify scrap into six categories: old scrap, new uncoated scrap with low residuals, shredded scrap, steel turnings, high residual scrap and fragmented scrap from incineration. These specifications also contain environmental, health and safety related considerations such as the content of dangerous, flammable and explosive substances, Cu, Sn, Cr, Ni, Mo, S and P and radioactive materials [53, BDSV 2009]. The definitions of the items in this list of specifications apply only to non-alloy carbon steel scrap as raw material for the steel industry.

Scrap metal is normally stored outside on large, uncovered and often unpaved ground. The ferrous scrap is loaded into baskets by magnets or by grabs. The handling minimizes any rogue, non-magnetic material like stones, wood or non-ferrous metals from entering the process.

Some types and quality of scrap may lead to the deposition of residual materials on the ground, as well as to the deposition of heavy metals and hydrocarbons during metal storage and handling operations.

Depending upon the types and qualities of the scrap being processed, handling operations (i.e. charging, discharging) may also lead to inorganic (dust) and organic emissions (e.g. oily splinters), under certain weather conditions.

5) Material, Production & Logistics planning

Almost every steel producer informed the team during site visits that their material, production and logistics planning are dependent on the local steel demand level, import price of raw material and transportation cost. Planning can be made more stable once the Uganda steel demand increases and eventually also the transport system improves (by railways, by motor-roads, by barge over the lake, etc.). Please refer to precondition and logistic section (Chapter 7) of the study for details.

6) Process efficiency

Main considerations are evidenced in above points 1,2,3,4 and 5.

7) Plant Lay-out

This point has important impact on the potential increase of production rate and possible process utilization. For Example, the Melt Shop requires a lot of area for: scrap, slag, material movement, bag filters etc. and if this requirement is not foreseen in setting up the Rolling Mill, there can never be a cost efficiency improvement, production improvement, quality increase etc.

It is necessary to create favourable conditions that can attract investors to put money in new steel industry equipment. The investment positive ground for steel industry can be assured only by a proper market demand of products and this ground is generated by the development of the country's economy, starting with new infrastructure and challenging major projects. This is the real development key for the iron and steel value chain and any measure (like for example those already described to help the steel players in terms of preconditions and general state policies) will start the process and accelerate it.

The empowered economy needs to boost the civil sector with housing, new industrial and service buildings, roads, etc. This is the approach which was adopted by most of the developing countries to

kick-start and move forward steel demand since the construction industry consumes a lot of steel products and is one of the most powerful economy drives.

With the availability of good economic conditions, financing schemes are possible: which may involve the private and/or public sector; both foreseeing a good revenue in investing in the steel market. Additionally, foreign investors will be attracted by the growing economy.

The I&S industry requires huge resources for financing new plants and/or improving or revamping the existing ones. For this reason, the government, in parallel to the direct help given to the industry players with market protection and general economic advantages, should also promote investment by private entities with regulations that can ensure return on investment (ROI) for the private investors.

Once interest in investing is generated, bridging the technology gap would not be difficult because the market itself will widen the demand for production of new quality steel products from long to flat steel production.

It will be necessary to have rolling imported semi-finished materials as a first step and then gradually produce better liquid steel to produce the entire value chain in Uganda. The path is to proceed with the general actions listed above.

5.6. General technical notes on visited plants

This section gives the general overview of the steel players plants visited with the specific details for each plant given in the attached Technical Competitiveness report (Annex 2). Unfortunately, in many occasions the plants were shut down due to various reasons (power shortage, lack of orders, etc.) and in some other cases the visit was not allowed. For this reason, the technical assessment of existing equipment must be considered partial even if it was possible to evidence a general technical overview for the country with the following considerations.

5.6.1. Plant technology and process

The majority of the visited plants are utilizing obsolete technologies, mostly of Indian origin, with very low-quality control, very low-quality output and very high manual involvement.

In some cases, it was also noticed that there is no proper continuous feeding line from the CCM to the Rolling Mill and billet Reheat Furnace was not installed (thus limiting or even interrupting the production process and generating waste scrap).

Most of the plants are importing semi-finished materials (Hot Strip Mill and/or Cold Rolled Flat Coils & Wire Coils), on which they need to pay high transportation fees (in the range of 25-30 % of the value of imported goods) that are subsequently transformed into finished products. This certainly is not contributing to the competitiveness of the final products.

5.6.2. Process technology improvement

Most of the equipment installed in Uganda's steel plants is of very old technological concepts with very low manufacturing tolerances (few producers excluded as per details of the plant surveys, Annex 2). Many of the mill plants are not environmentally friendly and safety of the employees is also at a very low level, since many of the operations in rolling mills are still done manually (very frequent use of "repeaters" for rolling process). Automation is only done in a few plants.

Considering that Uganda's final product is presently made of only nails, barbed wire, low carbon rebars, wire mesh, binding wire etc, because players are in line with the market demand, it is acceptable to leave and continue production with existing rolling mills for those kinds of quality and quantity. In fact, there is no justification for any steel player to invest in modernization of their plants since they will never payback the investment without a market boost.

In order to reach the condition in which some revamping/refurbishing jobs in the plants will be justified, it is necessary to close the huge gap between the installed plants capacity of the steel players and steel demand. Once the local demand reaches a higher level, considering both quality and quantity demand, then there will be a need for new additional plants, which will subsequently raise output of finished product and quality.

Considering that Uganda's actual finished products installed capacity as of 2019 is about 1.5 million tons and actual production is approx. 530,000 tons (210,000 tons only coming from liquid steel), the need to make interventions in plants for the existing productions is not foreseen in the next five years.

By means of the Capability model, analysis was done considering the maximum growing trend and estimations were made that the steel industry in Uganda can increase capacity by installing new plants starting by the year 2028 as follows:

- Installed Finished Product Rolling Capacity = about 2,850,000 tons (out of which a new technologically advanced plant should be available for at least 1.3 million tons), from the actual capacity of over 1,500,000 tons
- Finished product produced in Uganda = 2,500,000 tons (from actual 530,000 tons!)
- DRI Production = over 950,000 tons (from the actual level of 30,000 tons). The installation of more DRP is essential to raise the actual level of local production and simultaneously reduce the cost of production, since local scrap is available in very low quantities (and quality); importing scrap is too costly.
- Liquid Steel production level = 1,850,000 tons, from the actual level of 210,000 tons. The technological process to be used should be by installation of bigger, modern Induction Furnaces and Medium size EAF/LF, with most technological CCM, for the plant Integrated size in the range of 200–300,000 tons/years.

In the meantime, as the steel demand grows, it is a technological priority, to install in Uganda plants that are presently not available which can produce specific new products that could give a very big advantage in the Country's development. Two of them could be for example:

- (i) a Rail Mill Rolling plant that will support and sustain Railway lines development, which contributes to transport cost reduction, in addition to faster transport capability for Industry needs.
- (ii) a Pipe mill for seamless pipes with high resistance that could help infrastructure with mechanical pipes and facilitating installation of pipelines at high pressure.

5.6.3. Process top technologies worldwide and possible application in Uganda

Modern technology in the iron and steel industry worldwide follows several guidelines in terms of overall process and equipment.

However, considering the actual Ugandan iron and steel value chain, there is need to evaluate the proper approach to take towards the adoption of these new technologies in order to achieve a reasonable and sustainable development of the above-mentioned industry. Any investment should be justified with a proper payback of the investment by the steel player or by a subsidization of investment by the government.

Listed below are the targets of the latest developed technologies in the steel industry:

- 1) Achieve top quality final products particularly for special steels, flat products, etc. Presently these are not produced in Uganda, and for this reason, it is not to be considered a key issue in the next few years.
- 2) Achieve energy efficiency in the steel industry. This target is much closer to the actual Ugandan environment in steel plants especially for the potential to auto produce electric energy and lower the dependence to the public grid. For example, the new DRI plants scheduled to be installed should consider the possibility of utilizing the energy in off gasses to drive mini steam powered turbines to generate electricity.
- 3) Achieve higher level of automation. This technology enables the production of high quality and repeatable steel products in the plant and improves flexibility in the operations, which in turn improves profitability as a result of cost reduction and final quality improvement. Presently, plants are totally manual except for a few exceptions, and the introduction of higher automation should be limited to the new installed plants in the country since the present low level of products does not justify retrofitting better automation equipment.
- 4) Achieve better environmental levels for lowering emissions and production of pollutant elements. This is important everywhere, but it is important to highlight that this could generate an increase of Capex in new equipment and in Opex for the additional handling of these environment free equipment and the subsequent disposal of collected pollutants. Proper regulations should be enforced in a phased approach in the Country by the government in order to align the industry to the best available technologies with reference to this subject.
- 5) Achieve fully automatic Industry interconnection. This is related to the tendency of worldwide steel industry to align to the new paradigm of Industry 4.0 (like they call it in Europe) or MIC 2025 (Chinese diction) or equivalent names in other areas.

It is important for Uganda Steel industry to attain automatic connection between suppliers of products, producers (steel key players) and customers in order to achieve seamless unmanned performance of the entire value chain, which in turn cuts costs and increases profitability. With the full implementation of this new concept, contracts are stipulated by the automation system itself when needed on the basis of pre-arranged costs and conditions and the automation control of each plant is able to predict when goods & materials are needed through signals elaborated by sophisticated series of IOT equipment and through an intelligent supervising system, controlling the entire plant. Depending on the key players production levels for their clients, all the internal processes of production are handled by advanced control logics. Obviously, it is not expected that such a level of top technology will be reached immediately in Uganda, not even by the best plants in operation in the country. The implementation of this new paradigm will be necessary in Uganda only when a market demand, driven by economic conditions brings the industry to a high production rate for most of the steel products, and when the industry reaches capability to boost export. Additionally, these kinds of investments must be supported by a state system which gives incentives to companies involved in this type of evolution/innovation.

Presently, the country's priority should be initially focused on a general development of the industry, and later align technology to such high levels.

- 6) Achieve fully automatic safety systems. Apart from the obvious value of human life and health, the safe operation of man labour is economically important for a company and should be improved as much as possible using all the available technologies. This is also related to the

necessity to protect workers who are the best asset in production due to their know-how, and difficult to acquire/source them for specific high technology equipment/plants.

5.6.4. Suggested development path in technology for I & S Players

Owing to the analysis of Uganda's steel industry given above, it is suggested to develop the level of technology in Uganda step by step.

The first step is to increase liquid steel production by means of new carbon based DRI plants installed inside the existing premises of the key players. There is already a DRI plant in advanced stages of construction (Tembo Steel) plus others already planned that will follow if the first one will give positive results. The major concern is the availability of reductant required for the direct reduction of iron ore, which can potentially be sourced from Tanzania and/or from other sources like biomass. The government should assist these players to take off in the immediate term up until gas is made available through the projected pipeline from Tanzania, which would then allow the construction of new and bigger plants based on natural gas reduction. The tonnages of these new plants, as captured in the Capability model, may not have an immediate impact on the economy, but are an important driver of the fast growth assumed in the model in comparison with the situation pictured in the economic model. The start-up of this and other plants will surely force other developmental projects into play like upgrading the transportation system to get iron ore from Kabale to the iron making plants, assuming that coal could reach plant sites by barges from Tanzania. There is also a need for the gradual reduction of the final product cost, of which the entire country system will benefit from. One particular way to create an investor attractive environment for the steel players that are already interested to invest in this kind of technology is to apply, as soon as possible, trade policies that stifle imports of semi-finished products, thus giving a competitive advantage to players investing in liquid steel production with DRP process.

Whilst eight companies have plants capable of producing liquid steel, only seven of them are using this option mainly through scrap melting to produce about 210,000 t/year of liquid steel, the bulk of the steel (320,000 t/year) is imported as semi-finished products that are consequently processed into the final products, a rather expensive and not competitive process due to the high transport costs. The Government of Uganda should therefore extend specific assistance to the companies that are going beyond melting scrap in Induction furnaces but are investing in iron ore mining, processing & smelting, and steel making. Table 10 shows the companies that have taken a step further in adding value to the industry by using local resources to produce either raw material for the steel makers or to feed into their own steel plant.

Table 10: Companies Investing in Value Addition to the Iron and Steel Industry

VALUE CHAIN PHASE	I(1)	I(2)	II	III	IV		
COMPANY NAME	Exploration	Mining	Iron Making	Steel Making	Casting	Tonnes DRI in 2024	Comments
IMCU							Reported to have done exploration. Mining stopped after ban
DRIOC							Collecting from artisanal miners. Sizing the ore & selling to Steel Rolling
SMI							Did some exploration. Setting up ore processing plant after having fulfilled other procedures.
STEEL ROLLING						30000 DRI	Currently Producing DRI
Tembo Steel – Iganga						130000 DRI	Constructing DRI plant. Already have SMS/CCM based on scrap+ 10% ore
PRAMUKH STEEL LTD						35000 DRI	
KIGEZI		XX	XX	XX	XX	20000 DRI	Licenced to establish DRP/SMS/Billet Caster
DONG SONG				XX	XX	300000 Iron ore	Licenced to establish Ironmaking/SMS/CCM no progress yet.
	In literature, not cross checked with core samples						
	Mothballed with security details in place; no operational activity						
	Sourcing ores from artisanal miners, and running the crushing/screening and loading for Steel Rolling and takers of small iron ore quantities						
	Professional exploration done						
	Physical construction on the ground						
	Operations in progress year after year						
xx	Plans approved. Physical construction on the ground not yet started						

In order to sustain these investments, the government can provide specific assistance to the companies' needs in line with fair industry practice that allows competitive investment. Specific assistance echoed by some of the players are provided on the list below (Source – NPA²)

Tembo Steel Ltd.

Most of the required machinery for the DRI plant is already procured and the construction phase of the plant has commenced in Iganga. Tembo Steel considers the following assistance as enablers of competitiveness for its investment:

1. 10-year corporate tax exemption so as to re-invest and expand the plant.

² Developing the Iron and Steel Industry in Uganda – Harnessing the Low Hanging Fruits.

2. Concessional power tariff.
3. Subsidy in the form of a grant for operating capital amounting to USD 5million
4. Waiver of tax on importation of coal from Tanzania
5. In the on-going Common External Tariff (CET) review, zero tariff on wire rods importation should be reviewed as this is produced locally by Tembo Steel Ltd. This will save the country about USD 40 million in importation of wire rods and USD 300 million with the DRI plant established.
6. Assistance in getting a mining license for iron ore (this though would not hinder operations as two companies have already invested in mining equipment and have land and licenses, which would supply Tembo's DRI plant once operations start).

Pramukh Steel Ltd

Fifty acres of Land is already acquired in Njeru for a DRI plant with proposed production capacity of 35,000 TPA. Pramukh seeks the following assistance to start the DRI plant by 2024:

1. Ten-year corporate tax exemption so as to re-invest and expand the plant.
2. Concessional power tariff.
3. Low interest loan amounting to USD 20 million, at an interest rate of 4-5% per annum.
4. Tax exemption on importation of plant machinery.
5. Waiver of tax on importation of coal from Tanzania.

Madhvani Steel Ltd

The company is in phase V of the iron and steel production value chain; it imports billets and makes iron bars, which is its only product. It has an installed annual capacity of 40,000 tonnes but annual production is at 15,000 tonnes at present. The Company has a mining license located in Manafwa and plans to set up a pig iron processing plant at a total cost of USD 80 million. The required assistance from government comprises of the following:

1. Tax holiday for a minimum of 20 years
2. Exemption of duty on importation of raw materials.
3. Exemption of WHT/VAT/duties on imported plant and machinery and all other project related activities including consultancy services.
4. Soft loans with government support.
5. Assistance from the government on acquiring the required 100 acres of land.

Kigezi Steel Ltd

This is a company that is yet to start operations, but the directors indicated that they have plans to establish a sponge iron plant. Assistance required from government to ensure the success of this venture include:

1. A loan amounting to USD 40 million.
2. Acquisition of 100 acres of land, either in Namanve industrial park or in Masaka.
3. Activation of MV Kaawa to provide transport services across Lake Victoria from Tanzania to facilitate transportation of coal to plant site.

Should the government approve some or all of the requested assistance, it is advised that the incentives packages be tied to specific deliverables over a specified period of time as will be agreed between the government and the respective investors. A mechanism to monitor progress made by any selected entrepreneur(s) should be put in place to ensure adherence to the agreed terms and measures taken in case of default.

In the event of these new plants being developed, the processes should be fully automated, with environmental control equipment in place (coal is never a clean material to be processed). Production of electric energy utilizing the latent heat in the exhaust system of the kilns should be taken into consideration. Some path towards high-level automated control system should be considered for all

new equipment installations or for all completely new-built plants. This is another area of the investment the government can take a position to incentivise the players as it facilitates efficiency, safety and sustainability. Available technologies are further elaborated in the next chapter with suitable options suggested for the Uganda scenario according to the estimated evolution of the industry.

6. IRON ORE REDUCTION OPTIONS

The present chapter has been introduced to give the readers a full panorama of the available technologies in iron making (with basic information and data) and the justification for their selection as far as advising Uganda's path in developing the iron and steel industry is concerned.

At the end of the chapter are reported some consideration of costs necessary to produce the steel with the different technologies.

6.1. Introduction

There are two main processes for producing steel:

1. Starting from iron ore mineral (various processes, of which the most common is by BF in combination with BOFs, or alternative technologies like direct reduction processes, normally referred to as "Integrated steel plant"), or
2. Starting from collected scrap or steel-based materials (various processes re-melting iron & steel, generally by Electric furnaces (Induction furnaces and EAFs), called "Electric steel plant").

In the integrated steel plant, the production route generally consists of sintering or palletisation plants, coke oven systems, blast furnaces and BOFs. Such arrangement requires high capital expenses and raw materials with stringent specifications. Coking coal is needed to make coke strong enough to support the burden in the blast furnace.

Integrated steel plants of less than one million tons of annual capacity are generally not economic/viable through this process. The coke oven, sintering and blast furnace plants are expensive units, and, for proper handling, it is necessary for coking coal to be of good quality.

In the electric steel plants, ferrous scrap is melted and partially refined in the EAF. The production with this plant is more flexible but requires scrap metal supply of proper quality and huge amount of electricity that is stable (mostly from the grid).

Direct reduced plants (DRP), an alternative route of iron making starting from iron ore mineral, has been developed to overcome some of the difficulties presented by the conventional blast furnace route. DRPs are successfully manufactured in various parts of the world and they use either natural gas-based or coal-based technology. Iron ore is reduced in solid state by means of Direct Reduction (DR) at temperatures of 800–1050°C and once reduced it is called DRI (direct reduced iron)

The need for development of DRP and commercialization of DRI arose in late 50's when EAFs started facing problems of scrap availability of the desired quality through the traditional source channels.

DRP represents a feasible starting point in Uganda to the development of the steel industry

The advantages of this technology in the Ugandan scenario are the following:

1. DRI production does not require coking coal that is not available in Uganda as is the case with other parts of the world.
2. Hot-briquetted iron (HBI) is a compacted form of DRI designed for ease of shipping, handling, and storage and could be applied in the Ugandan environment.
3. Hot Direct Reduced Iron (HDRI-gas based) is iron which is not cooled before its discharge from the reduction furnace and immediately transported to a waiting electric arc furnace, thereby saving energy. This concept can be applied in plants opportunely designed for the purpose.

4. The direct reduction processes use palletized iron ore or natural "lump" ore. One exception is the fluidized bed process which operates with sized iron ore particles.
5. The direct reduction process can use natural gas contaminated with inert gases, avoiding the need to remove these gases prior to DRI process.

The utilization of direct reduced iron in electric arc furnace, partially or completely substituting scrap, brings several potential advantages that can be summarized as follows:

1. Known uniform output composition.
2. Low levels of residuals/tramp elements compared to scrap.
3. Capability to maintain phosphorus level in steel within 0.002%
4. Low content of dissolved gases, except for coal based DRI, which is typically high in nitrogen (600 ppm).
5. Uniform size and higher bulk density compared to scrap.
6. Capability of forming protective cover of foamy slag in the bath by virtue of availability of oxygen and carbon in the process.
7. Lower refining requirement of produced steel.
8. Possibility of producing a huge variety of high-quality steels as it contains very low tramp elements.

However, there are some disadvantages and conduction problems in using direct reduced iron in the Electric Arc Furnace, which are:

- Lower liquid metal yield, lower metallization, lower carbon content (in sponge iron produced in coal-based plants), high gangue content resulting in additional requirement of electrical energy in its melting and finally higher meltdown time.
- Sponge Iron is highly susceptible to oxidation and rusting if left unprotected, necessitating the need to process it immediately after formation.
- The bulk iron should be handled with care since it can catch fire because it is pyrophoric.

To overcome these disadvantages, processes using a combination of DRI and Hot Metal in Electric Arc Furnaces are being developed.

The substitution of scrap by DRI has limitations depending on the furnace operating conditions. Only 20% to 30% of DRI can be used in the furnace charge when feeding is done in batches. The proportion of sponge iron in the charge can however be increased up to 100% by providing continuous feeding arrangements and by use of high-power input rate in furnaces with higher capacity transformers.

The growth of the world DRI production is shown in figure 20.

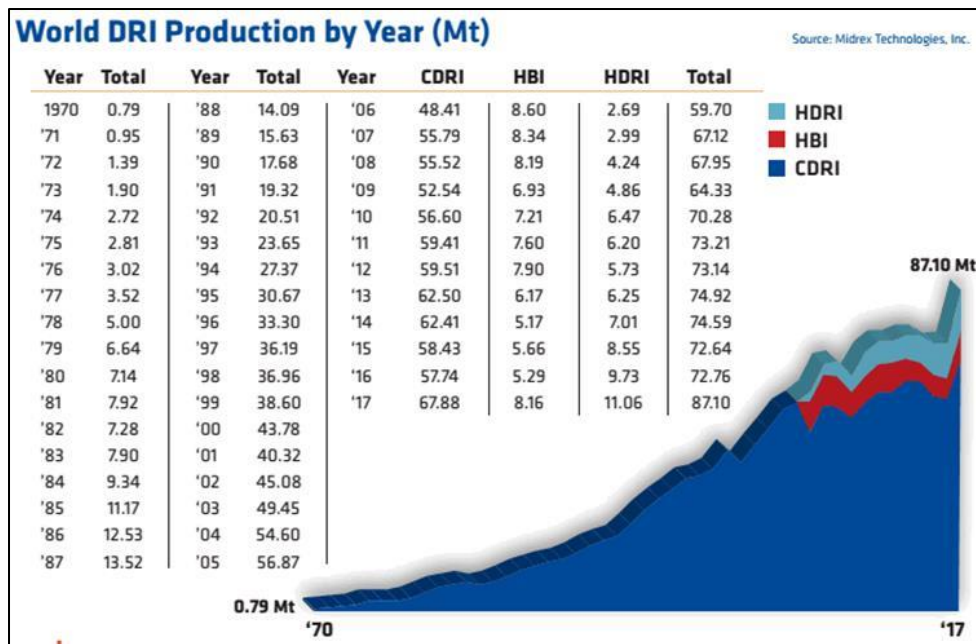


Figure 20: World DRI production (Source: Midrex Technologies, Inc. 2019)

Most of the increase in DRI production in the world relied primarily on the Midrex process till 2005, when competitors reached the same level of technology (Hylsa process). World's production of DRI crossed 70 Mt, for the first time in 2010.

6.2. Direct Reduced iron making routes

The processes available for direct reduction can be broadly categorized into two groups, according to the type of reductant used:

- 1) Coal based DRI Processes.
- 2) Gas based DRI Processes.

The recent trend of DRP is towards rotary furnaces (RKF and/or rotary hearth furnaces) using coal as a reductant. For bigger plants the most efficient and economical way to produce DRI is with shaft furnaces/fluidized bed reactors for gas-based processes.

Presently, many different direct reduction processes based on coal or natural gas are available for iron production from iron ore.

The principal raw materials used in all these processes are iron oxide (lump ore, pellets or fines), non-coking coal/natural gas, and a small amount of flux (dolomite or limestone).

Coal based DRI

In coal based DRI processes, the reduction of iron ore takes place in solid state in a reducing atmosphere. During reduction, the oxygen component of the iron oxide comes out in form of CO or CO₂ leaving free pore space in the iron enriched part of the lump ore or pellet. This makes the final product porous in nature and it becomes spongy. Due to this physical property it is also called Sponge Iron.

In coal-based sponge iron process either an RKF or a rotary hearth furnace is used.

Gas based DRI

In gas-based processes, the reduction of iron oxide is carried out by a mixture of CO and H₂ at a temperature of about 750-950°C. The reducing gas is produced by reformation of natural gas. The reformation process is a partial oxidation of hydrocarbons, which is normally enhanced by the use of a catalyst like nickel.

A vertical reactor is used for gas-based reduction of iron ore, whereby gas is injected from the bottom of the reactor and the iron ore charge from the top. On dispersion upwards, the gas interacts with the iron ore charge moving downwards by gravity, thereby allowing the reduction reaction to take place to produce DRI, which then settles at the bottom of the reactor.

Gas-based reduced iron can be cooled indirectly or briquetted in hot condition to get hot briquetted iron (HBI).

In accordance with the current Uganda steel industry, the following three options for DRI production (proven industrial technologies) have been considered:

1. RKF (coal-based reduction process): easy to be realized and implemented. Installation in Uganda has already started for some of the key players in the steel production.
2. RHF (coal-based reduction process): more efficient as compared to kilns and capable of processing big quantities of high quality DRI because of its big rotary furnace diameter.
3. VFR (gas-based reduction process): most efficient of the three processes and top quality DRI but high capex and necessity to install big plants with high production rate (over 300,000 t/year).

6.2.1. Rotary kiln furnace

DR plants using RKF have relatively limited capacity, which is dependent on the size and diameter of the furnace.

The minimum capacity per unit is about 50,000 t/year, which can reach up to 200,000 t/year per unit. However, it is possible to optimize the charging system, installing 2 units on one common charging system, thus having the yearly production capacity reaching up to 400,000 t/y. This would be a suitable size for the Ugandan environment considering that the steel industry is young and growing requiring limited amount of steel products for its market.

In coal-based DR processes the main raw materials fed into RKF are: 4-18 mm lump hematite ore, or 9-30 mm hematite pellets (both 1,550–1,600 kg/t DRI) and a relatively coarse fraction of non-coking coal (6–20 mm).

The coal added at the feed end (also referred to as co-current coal) not only acts as a reducing agent, but also supplies most of the heat required to increase the charge temperature to the desired reduction temperature.

A finer fraction of coal (<6 mm) is also normally injected (or blown-in) from the discharge end of the kiln (counter current coal) using air as the carrier gas helping to complete the reduction and to supply heat (as a result of combustion) in the region close to the discharge end of the kiln.

The temperature of the charge bed inside the kiln is limited to a maximum of around 950–1,050°C in order to ensure that the ash in coal does not fuse. As a result, the entire reduction occurs in solid state. There are specific iron ore characteristics required for the RKF process as shown in table 11.

Table 11: Characteristics of iron ore for RKF process

Characteristics	Constituent		Value
Chemical	Fe (%)		65 min
	SiO ₂ +Al ₂ O ₃ (%)		3.5 max
	CaO+ MgO (%)		2 max
	S (%)		0.02 max
	P (%)		0.035 max
	Pb+Zn+Cu+Sn+Cr+As (%)		0.02 max
Physical	Size range (mm)		5-30
	Tumbler index (>5mm) (%)		80-92
	Reducibility (dR/dt) 40%; (%per minute)		0.5-0.6
	Decrepitation [%]	<5 mm	15-20
		<0.5 mm	3-5

RKF does not require quality coal, which makes Tanzanian coal appropriate for this process for the Ugandan steel industry.

A flux (like limestone or dolomite) in fine form (below 3 mm, or sometimes 4 mm) must be added along with coal (particularly at the kiln feed end) in order to control the sulphur pick-up by the reduced material from coal ash.

The important factors that affect sulphur levels in the final DRI are the actual bed temperature profile, the amount of coal recirculated into the kiln, the granulometry of the limestone/dolomite used, the extent of recycling of the used limestone or dolomite in the charge, the fines content of the ore feed, the sulphur level in the ore feed, the sulphur level in the coal feed, and the control of temperatures for the feed end bed and gas.

Table 12 reports the limestone and dolomite characteristics for the RKF process.

Table 12: Table of Fluxes characteristics

	SiO ₂ (%)	CaO (%)	MgO (%)
Limestone	8 max	46 min	8-10
Dolomite	5 max	28 min	20

The product discharged from the kiln (reduced iron oxide plus some coal) at a temperature of 950–1,000°C is cooled to below 120°C in an indirectly water-cooled rotary cooler.

No water is normally allowed to come in direct contact with the reduced product since it is prone to re-oxidation.

After cooling the material below 120°C (sometimes 150°C), the reduced iron oxide (being magnetic in nature) is separated from the non-magnetic portion of the kiln discharged material (consisting mainly of charcoal), using magnetic separators at a stable temperature of 70–90°C.

The rotary horizontal drum is the main component of the whole plant; it is a simple erection, supported by concrete pillars (low highs for erection), which does not require heavy lifting cranes. In addition to easy erection, it is easy to maintain.

This type of plant does not need an oxygen plant, reformer or compressors; it does not require specialized staff, making it easy to manage.

Figure 21 shows a typical mass balance for RKF process to produce 1 ton of DRI.

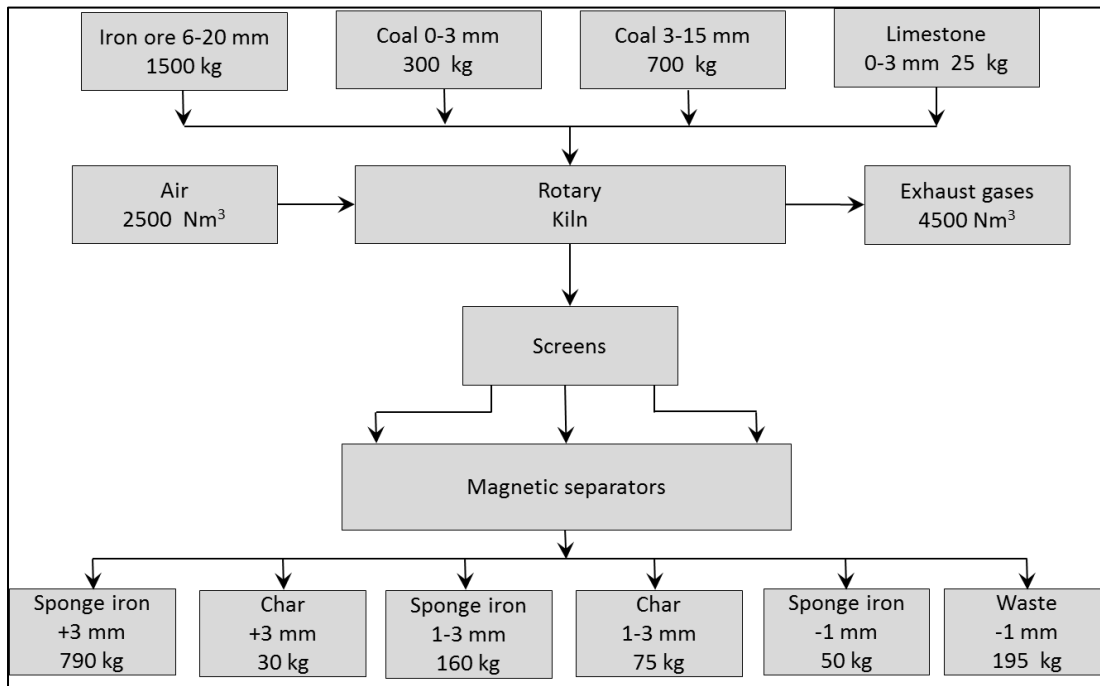


Figure 21: Mass balance per ton DRI in RKF

The total amount of solid wastes generated in an RKF plant can be as high as 195 kg per ton of DRI produced.

The average figures of DRI composition for an RKF plant are summarized in table 13.

Table 13: Table of RKF average DRI analysis

	Fe (tot.) (%)	Fe (met.) (%)	FeO (%)	SiO ₂ (%)	Al ₂ O ₃ (%)	CaO (%)	S (%)	P (%)	Metallization (%)
DRI	88-90	78-79	14-15	2-6	1-2	0.3	0.02	0.04	87-89

The volatiles present in the co-current coal are released at temperatures of up to approximately 600°C, i.e. before the reduction reactions start.

Some of these volatiles are utilized to pre-heat the kiln charge, but the majority tends to leave the kiln with the waste gases without contributing to the actual reduction process.

For this reason, recently constructed RKF plants are equipped with waste heat recovery systems to make use of this energy, which is otherwise wasted away. Generally, it is used to produce electric energy to feed other plant equipment and therefore lower dependence on the main electricity grid.

With the flue gas from the kiln, it is possible to generate steam and run a turbine to produce electric energy. It can be used, for example, to feed an induction furnace as a standalone power source (of 5-7 or more MW capacity, depending on the size of the RKF), without the need for additional energy from other sources of power like the national grid.

Table 14 reports the co-generation of power from kiln waste gases and the amount of power consumed within the plant.

Table 14: Power co-generation from RKF waste gases

Module size (ton/year)	Power Generation (kW)	Own consumption (kW)	Surplus power (kW)
50,000	3-4	1.6-1.9	2.4-2.7
100,000	7-8	2.4-2.6	5.4-5.8
200,000	13-15	2.8-3	8.5-8.9

Nowadays approximately 17-18 % of the total worldwide DRI (about 87 Mt in 2017) is produced with this process.

There are different processes based on utilization of RKF in reducing iron ore. Only a few of them have attained commercial significance, for example: SL/RN, KRUPP-CODIR, Accar, TDR, and DRC SL/RN technology of Lurgi GmbH, West Germany, are among the most successful ones. Table 8 reports the main features of RKF reduction processes.

The approximate cost for a 100,000 t/y RKF plant (including the iron ore sizing plant) is in the range of EUR 18-22 million.

The major strengths of RKF processes are:

1. They can efficiently mix the solid charge which simultaneously undergoes the process of heating and reduction.
2. Since large freeboard space is available above the solid charge in any kiln, the gas phase can tolerate the presence of heavily dust-laden gases.
3. RKF's are commercially proven processes in rugged reactors. In addition to their use in iron and steel, they have also been extensively used in the cement industry. Their low productivity per unit volume is compensated for by the inherent versatility of the reactor.
4. The reduction temperature of iron oxide in RKF's is much lower compared with blast furnaces (about 1,000 °C against 1,300–1,600 °C in the latter). As a result, less energy is required to bring the reactants to the reaction temperature.

At the same time, it needs to be emphasized that RKF technology has several drawbacks/weaknesses. These are:

- 1) Low productivity.
- 2) The reactor rotation at 0.4–0.5 rpm makes it difficult to incorporate process control and quality control systems.
- 3) Low energy efficiency — the energy lost through hot sponge iron (which has to be cooled to separate coal in magnetic separators) is about 1.8–1.9 Gcal per ton and the energy lost

through the waste gases is approximately 2 Gcal per ton (if not recuperated with creation of electric energy).

Table 15 below summarises the RKF reduction processes main features.

Table 15: RKF reduction processes main features

Parameters		SL/RN	CORDIS	Accar	TDR	DRC
Type of reactor		Rotary kiln				
Oxide feed		Lump ore and pellets				
Desired analysis of oxide feed	Fe tot (%)	64	66	65	64	66
	Gangue $\text{SiO}_2 + \text{Al}_2\text{O}_3$ (%)	4	4	4.5	4	4
	S (%)	-	-	0.05	-	-
	P (%)	As low of possible		0.05	0.08 (max)	As low of possible
Size range of iron ore (mm)		10-30	5-25	10-15	5-20	3-20
Reductants	Type	No-coking coal/another fossil fuel		No-coking coal		No-coking coal lignite/anthracite
	Size (mm)	0-16 mm	<35mm	6-50	<15	<32
	Desirable properties	Low ash, high ash fusion temperature, low sulphur, high reactivity				
Energy requirement (GJ/tDRI)		20	15	13.5-15	185-21	16.5-185
Flux (kg/t DRI)		50-70 (limestone)	50-90 (limestone)		60 (dolomite)	40 (limestone)
Electricity (kWh/tDRI)		50-75	70	35	100-120	65-110
Water (m ³ /t DRI)		3	1-1.5	4	3-5	1.5-3.8

Conclusions

- 1) The capacity of RKF plant is limited but adequate for the Uganda steel market: minimum capacity for one unit can be 50,000 t/year, and it may go up to 200,000 tons/year.
- 2) It can be fed with no exceptional quality of iron ore and it can process even friable Hematite. Moreover, the process does not require good quality coal, which makes Tanzanian coal suitable for the process.

- 3) The quality of DRI is suitable for EAF furnaces ($\text{Fe}_{\text{tot}} = 88\text{-}90\%$ and metallization $88\text{-}90\%$), and fines can be re-utilized (by briquetting).
- 4) The approximate cost for a 100,000 t/y RKF plant (including iron ore sizing plant) is in the range of EUR 18-22 million. For these reasons the RKF plant represents the most suitable solution for Uganda's present situation.

6.2.2. Rotary hearth furnace

In recent years, the rotary hearth furnaces (RHF) have been extensively used in many new DR process developments, essentially because they offer some distinct advantages. Though the basic reduction process remains more or less the same involving solid–solid reactions (as in the case of RKF DR), the genesis of these processes is to make the reduction step much faster, by increasing the area of contact between the oxide and the reductant. This basic objective is achieved by using finer materials, and since RKF's are not suitable for processing oxides generally below 3 mm, the rotary hearth furnace provided a solution to this bottleneck.

At the present, worldwide production of DRI with this technology is very limited (in the range of 1-3%) since the cost/benefit range of these equipment is presently not as attractive as RKF's. However, some distinct feature and characteristic of this process may well fit Uganda's environment at a certain stage of its industry development.

Rotary hearth furnaces have a wide range of production capacity (20,000 t/year up to 700,000 t/year) depending on size of the rotary hearth furnace (RHF). The RHF can be charged with any type of hematite or magnetite without quality restriction; moreover, any coal, even with high volatiles, is suitable (for example the low-quality coal that can be imported into Uganda from Tanzania).

In RHF's the iron oxide to be reduced forms a thin layer(s) on the furnace hearth. The layer(s) comprises iron oxide either in the form of pellets, or, sometimes, directly as fines, without any agglomeration. The bed moves slowly as the reduction proceeds using carbon monoxide and hydrogen as the reductant. To this extent, the RHF can be described as a counter-current reactor. Since no motion of the bed except forward movement on account of hearth rotation takes place in RHF, radiative heat transfer permits the bed to reach reduction temperatures of at least 800 °C.

Iron ores fines, iron bearing wastes or other iron bearing materials are fed into the plant where they are mixed with coal and agglomerated. The agglomerates are then fed into the hearth of a Rotary Hearth Furnace to form a layer of material (thickness depending upon the type of process technology), and then heated to 1,200-1,250 °C as the hearth rotates. The reduction process takes only a few minutes. Agglomerates are heated rapidly in the RHF by radiation heat and the oxides in the agglomerates are reduced to metallic iron by the carbon present within the agglomerates. For this reason, efficiency is very high.

Since heat transfer on account of radiation can become a major rate limiting step in any RHF, attention is paid to enhance this process.

Carbon monoxide generated from the agglomerates is used as a major fuel source in the RHF, covering 85%-100% of the total energy need for the process, while the energy balance required is generated by means of fossil gases or other low-cost fuels. Carbon monoxide is completely burned in the RHF by a post combustion system utilizing air injections. The process can therefore attain high-carbon-utilization rate resulting in reduced/less CO₂ emissions.

The end product can be either hot briquetted iron (HBI) or DRI agglomerates, and is a highly metallized iron bearing material which can be used as a pre-reduced feed to the Blast Furnace (to increase its productivity) or as a low residuals, quality and economical scrap substitute for BOF, EAF or SAF feed.

These kinds of plants can also process iron bearing steelmaking and mill wastes, such as BF, BOF and EAF's filter dust and mill scale and sludge (including oily sludge). Zinc possibly contained in dust is

recovered into a sellable zinc oxide powder, while the iron units are reduced to metallic form of quality DRI.

The rotary hearth furnace is the main component of the whole plant; it is made of simple erection supported by concrete pillars and calls for simple maintenance.

This type of plant does not need oxygen plant, reformer or compressors but requires specialized staff.

Table 16 presents typical specific consumption for 1 ton of DRI production of the Fastmet process in a 500,000 t/y plant processing virgin iron ore.

Table 16: Consumption for the production of one-ton DRI in RHF process (Fastmet)

Iron ore (t)	Coal (t)	Binder (kg)	Fuel (Gcal)	Electricity (kWh)	Water (m³)
1.2-1.35	0.3-0.4	30-40	0.5-0.7	80-100	1-2

Table 17 shows the average composition of DRI produced from RHF furnace.

Table 17: RHF average DRI analysis

	Fe (tot.) (%)	Metallization (%)	Gangue (%)
DRI	88-92	92-95	2-6

In the RHF process, the thermal energy is provided by coal or coal plus fossil fuels. The heat remaining in waste gases can be recovered by proper systems from the RHF off-gas stream similarly to what is done for kilns.

However, there are several other coal-based direct reduction processes similar to the RHF available on the market that can be alternatives to RHF for specific reasons. Here below some of the alternatives to the RHF:

- 1) Vertical reactor processes - Kinglor–Metor
- 2) Multiple hearth furnace-based processes - Primius
- 3) Tunnel kilns - Hoganas
- 4) Fluidized bed-based processes – Circofer

Table 18 reports a comparison between the main features of RHF processes.

Table 18: RHF processes main features

Parameters	Inmetco / Redsmelt	Fastmet	Comet	DryIron	IT Mk3
Process status	Started on 1978, it is considered a proven technology				
Type of reactor	Rotary Hearth	Rotary Hearth	Rotary Hearth	Rotary Hearth	Rotary Hearth
Oxide feed	Pellet made out of ore fine, waste iron-bearing materials and pulverized coal	Pellet made out of ore fine, waste iron-bearing materials and pulverized coal	fine coal mixed with limestone and iron ore in separated layers	Briquette made out of fine iron ore and pulverized coal	Pellet made out of ore fine, waste iron-bearing materials and pulverized coal (also anthracite)
Desire analysis of oxide feed	No particular restriction	No particular restriction	No particular restriction	No particular restriction	No particular restriction
Pellet size (mm)	8-12	8-12	No pellet		7-19
Layers	3	1-2	1-3	1	1-2
Bed thickness (mm)	20-30		10		-
Reduction temperature (°C)	1,250-1,300	1,200-1,400	1,200-1,500	1,250	1,400-1,500
ReductantsType	Coal/natural gas	Coal	Coal	Coal	Coal
Electricity (kWh/tDRI)	450	100	380	-	-
Auxiliary fuel (Gcal)		0.7 (Coal)	-	-	-
Coal (ton/t DRI)	0.7	0.5	0.42	-	-
Water (m3/t DRI)	-	3		-	-

The cost of the rotary hearth furnace process depends on process technology and related equipment. The Best production range is 500,000-700,000 t/year. Plant cost is about EUR 50 - 100 million (possible furnace diameter from 10 to 50 m).

The main economic potential advantages of the direct reduction plants with coal in rotary furnaces for production of DRI from iron ores are the following:

- 1) A wide variety of iron oxides and high volatile coal can be used—low-cost iron ore fines and inexpensive steam coal is normally chosen.
- 2) Steel plant wastes and coke breeze and/or carbon bearing wastes can also be used. Any RHF facility in an integrated steel plant provides means of reaching the goal of becoming a zero-waste production facility.
- 3) A wide variety of energy sources can be utilized in the burners, such as natural gas, LPG, coke oven gas, heavy oil, waste oil or pulverized coal.
- 4) The short reduction time enables easy plant start-up and shutdown resulting in flexible operations and quick adjustment of production rates.
- 5) The final product is highly metallized DRI/HBI, free of Zn and Pb which makes it low residual replacement for scrap.

Conclusion

- 1) RHF has a wide production capacity from 20,000 t/year up to maxi module of 700,000 t/year. Lower production plants are technically possible, but ROI is difficult to achieve; for this reason, the optimum plant size is 300,000-500,000 t/year.
- 2) The RHF can be charged with any type of hematite or magnetite without particular quality restriction; moreover, coal even with high volatile (Tanzania's coal quality) is suitable.
- 3) The rotary hearth furnace is the main component of the whole plant; it is of simple erection (it is supported by concrete pillars, low highs for erection) and easy maintenance.
- 4) The cost of the rotary hearth furnace process depends on process technology and related equipment. Best production range is 500,000-700,000 t/year. Plant cost is about EUR 50-100 million (possible furnace diameter from 10 to 50m)

The utilization of rotary hearth furnace can be suitable in Uganda but not in the present environment due to the above-mentioned characteristics; it might however be considered a viable technology for its advantages after a market demand of steel becomes mature (over 1 million of tons). From a schedule point of view surely not before the next five years as detailed in the analysis provided in the two models (Economic and Capability)

6.2.3. Vertical reactor

The reductant normally employed in gas-based DR processes is natural gas.

Direct reduction plants require large volumes of natural gas: specific consumption is around 300–350 Nm³ per ton of DRI.

In the gas-based processes, the reduction of iron oxide is carried out by a mixture of CO and H₂ at a temperature of about 750-950°C. The reducing gas is produced by reformation of natural gas. The reformation is partial oxidation of hydrocarbons. To enhance the reformation process, normally a catalyst (nickel) is used.

Gas-based DR processes can be broadly classified into:

- 1) Retort processes.
- 2) Fluidized bed processes.
- 3) Shaft furnace processes.

Gas-based DR plant consists basically of two circuits: one for reducing iron oxide to make iron, and the other for reforming the reducing gas.

Iron oxide is charged into the reducing furnace, generally from the top, mixing and interacting with the reformed natural gas.

The typical production capability for vertical furnaces is in the range of 300,000 t/year (mini module) up to 1,000,000 t/year (maxi module); but a more optimized plant in terms of cost of installation and production should be in the order of 500,000 t/year.

Nowadays approximately 82.4% of the total world DRI is produced by using vertical reactor processes with the following distribution:

- 1) Shaft furnace processes: 81.7 % (64.8 % Midrex and 16.9 % HYL/Energiron).
- 2) Other gas processes: 0.7 %.

These processes require higher quality hematite than RKF or RHF. Laboratory tests are necessary to characterize the hematite in terms of friability, impurity etc. to assure their possible use in the reactor.

Table 19 reports the iron ore characteristics required for vertical furnace processes.

Table 19: Iron ore characteristics for vertical furnace processes

	Fe (%)	SiO ₂ (%)	Al ₂ O ₃ (%)	S (%)	P (%)	CaO (%)	MgO (%)	Comp. test (kg/pellets)	ISO tumbler test (% wt)
Pellet	66-68	2-3	0.2-0.4	0.01	0.02	2-3	0.08	270 (min)	>6.3 mm 95%min
Lump iron ore	66-68	0.5-0.8	1.6-1.7	0.005	0.02	-	-		<0.5 mm 4% max

Pellets and lump iron ore can both be used, but pelletized ore is the only option if the iron ore is friable. If iron ore quality is not adequate, then a beneficiation plant is required.

In the vertical furnace processes, the iron oxide, as lump ore or pellets, is introduced through a hopper at the top of the reduction furnace. The charge descends by gravity aided by mechanical movements at the discharge end of the shaft.

The counter-flowing gas (reducing gas), injected from the bottom, has two functions: heating the ore charge and removing oxygen from iron oxide.

The reducing gas, containing CO and H₂, reacts with the iron oxide bed and converts it to metallic iron. Water vapor and carbon dioxide thus formed leave the furnace along with the exit gas.

The reducing gas temperature and composition, as well as the time for which the gas is in contact with the charge are essential to manage the process.

Part of the exit gas from the shaft is reprocessed along with fresh natural gas so that it can be reused as reducing gas. The reformed gas, containing 90–92 % hydrogen and carbon monoxide, is then fed directly into the hot condition of the reduction shaft as reducing gas.

The top gas is hot and contains solid particles. Therefore, it needs to be scrubbed before it can be reused as process gas.

These processes require natural gas with adequate quality since it influences the efficiency and other process parameters like the amount of water vapor required in the reformer, etc. The recommended natural gas composition (for the Midrex process) and the effects produced by its constituents, if they are not within the desired specifications, are summarized in Table 20.

Table 20: Recommended natural gas composition in Midrex process

Component	Volume	Effect (%)
CH₄ (%)	75	
C₂H₆ (%)	0-25	
C₃H₈ (%)	0-4	Above 4% the water vapor content must be increased
C₄H₁₀ (%)	0-2	Presence of ethane, propane and butane in different percentages is not an issue, since they are all converted to CO and H ₂ after the reforming process
Liquid hydrocarbons (%)	0-0.5	Presence of liquid hydrocarbon causes problems because it facilitates the formation of carbon which is then deposited on the reformer catalyst surface
CO₂ (%)	20 Max	Above 20% Excess gas is produced and export as fuel
N₂ (%)	20 Max (ideally 10)	Each 10% increases fuel consumption by approx. 2% and reduces shaft productivity proportionately. However, nitrogen does not interfere with the process itself
S (ppm wt)	20	Sulphur above 20 ppm (wt) results in 14 ppm carbon to be deposited on the reformer catalyst

These plants are very high and must be supported by steel structures. The erection is complex due to the need for massive site welding of components of the vertical reactor, and due to a very high level of erection. Probably a crane of 700-800 tons is required to be available on site for erection over 120 meters of heavy weights.

This equipment is not available in every country, and these constraints on erecting the reactor must be carefully evaluated before considering this option.

The quality of the produced DRI in vertical reactors is better than the quality of the DRI produced in RKF's and therefore the most suitable for use in EAF for good steel quality production.

Table 21 reports typical DRI analysis.

Table 21: Vertical reactor average DRI analysis

	Fe (tot.) (%)	Metallization (%)	Gangue (%)
DRI	88-92	90-95	3.5-4

Typical specific consumption of raw materials and services are reported in table 22.

Table 22: Vertical reactor average consumption to produce 1 ton of DRI

Raw material and service	Consumption
Iron oxide (t)	1.55-1.6

Natural gas (Nm³)	300-320
Electric energy (kWh)	90-120

The costs of installation depend on various factors but for a plant size of 300,000 t/y, the approximate cost (including Iron Ore Sizing Plant) is around EUR 90-110 million.

Table 23 compares the two main shaft furnace DR processes: Midrex and HYL highlighting the strengths and weaknesses of each process in terms of the equipment used, type of charge, operating conditions and the inherent flexibility.

Table 23: Comparison: Midrex - HYL III - HYL self-reforming processes

Operation parameters	Midrex	HYL III	HYL with self-reforming (HYL IV/HYL ZR)
Reactor type	Vertical shaft furnace	Vertical shaft furnace	Vertical shaft furnace
Reduction reactor type	Bigger than HYL III and HYL IV with self-reforming	Smaller than Midrex	Smaller than Midrex
Iron oxide feedstock	Mixture of iron ore pellets and lumps	Mixture of iron ore pellets and lumps	Mixture of iron ore pellets and lumps
Reducing gas generation	CO ₂ reforming. No need for steam system, reformed gas quenching, reducing gas heating and CO ₂ removal system	Catalytic gas reforming using steam	Partial oxidation or in situ gas reforming.
Operational pressure	Ambient pressure	5 atm	5 atm
Reducing gas temperature	850-900°C	920-980°C	930-1000°C
Ability to process high Sulphur feedstock	Cannot utilize high Sulphur feedstock	Can utilize high Sulphur feedstock since the off gases are not recycled to the reformer. Hence there is no danger of catalyst poisoning	Can utilize high Sulphur feedstock. Since in situ gas reforming is employed, there is no danger of catalyst poisoning

Conclusions

- 1) In the gas-based processes, the reduction of iron oxide is carried out by a mixture of CO and H₂ at a temperature of about 750-950 °C. The reducing gas is produced by reformation of natural gas.
- 2) The typical production capability for vertical furnaces is in the range of 300,000 t/year (mini module) up to 1,000,000 t/year (maxi module); with optimized size, considering installation costs and Opex, being 500,000 t/year.

- 3) These processes require a higher quality of hematite than kiln. Laboratory tests are mandatory to characterize the hematite characteristics to assure its utilization in the vessel such as: friability, impurity etc.
- 4) These processes require natural gas with adequate quality since it influences the consumption and other process parameters.
- 5) The erection is complex due to the requirement for welding on site all the components of the vertical reactor, and due to the requirement for high level of erection activity (requiring special cranes).
- 6) The costs of installation depend on various factors but for a plant size of 300,000 t/y cost (including Iron ore sizing Plant) is around EUR 90-110 million. The specific investment cost is in the range of EUR 300-370 /ton.

The utilization of a vertical reactor is not suitable for the present needs of Uganda development of iron steel value chain. It is however the best available process in the range of 300,000 t/y DRI production both for efficiency and environmental reasons. It is therefore the best available technology for Uganda once the following pre-conditions are fulfilled: natural gas availability, stable electric energy grid, and increased market for steel products.

6.3. Blast furnace production routes

The typical route for production of iron is by means of an integrated steel plant operated with a blast furnace. The conditions to create such a plant are very complex and at the present stage in Uganda it is not advisable to embark on a project to install a blast furnace.

However, in future if the country reaches some specific preconditions probably it may be possible to adopt a mini module type blast furnace. For this reason, a dedicated component of the chapter on blast furnace production route has been added to “choice of technology” illustrating its basic features.

Iron is produced in blast furnaces by the reduction of iron bearing materials with a hot gas. The large, refractory lined furnace is charged through its top with iron oxide in the form of ROM ore at the appropriate aggregate/lump size, pellets, and/or sinter; fluxed with limestone, and dolomite, with sinter and coke as fuel.

Iron oxides, coke and fluxes react with the blast air to form molten reduced iron, carbon monoxide (CO), and slag. The molten iron and slag collect in the hearth at the lower part of the furnace. The by-product gas is collected through offtakes located at the top of the furnace and is recovered for use as low calorific fuel.

The production of 1 ton of hot metal requires 1,400 kg of ore or other iron bearing material; 400-500 kg of coke; 250 kg of limestone or dolomite; and 1,800-2,000 kg of hot blast. The molten iron and slag are removed, or casted, periodically from the furnace.

A blast furnace is a very huge plant that requires other plants to produce the feedstock materials (coke making, hot stoves and sinter plant or pellets).

Its production capacity is wide (from 300,000 t/year for the mini blast furnace up to 3,500,000 t/y large size blast furnace). Blast furnace plants are usually required where the steel market is large and consolidated due to:

- 1) The blast furnace optimal production capacity is >1 Mt/year.
- 2) These plants are designed to work continuously at the production target; their shut down procedure is complex. The work at low production is not economically convenient.
- 3) They need continuous material feedstocks and a market able to buy final products continuously.

In order to remove the carbon, the hot metal must be refined in Basic Oxygen Furnace (BOF) with scrap and slag formers (lime and dolomite).

The steel completes its refining (alloying desulphurization and temperature adjustment) in ladle furnace or/and vacuum plants (RH or VD/VOD) before reaching the continuous casting machine to produce slabs or billets.

The steel production from BF cycle is not environmentally friendly. However, the utilization of state-of-the-art technologies permits the minimization of steel plant emissions (dust, dioxin, wastewater and hazardous gases).

The CO₂ emission in integrated steel plant is typically in the range of 1.8 - 2.2 ton CO₂/ton steel. The utilization of pellets instead of sinter facilitates reduction of hazardous gas emissions (CO₂, dioxins etc.).

The specific production of BF gas is in the range 1,100-1,300 m³ STP/t steel with a calorific value of about 3MJ/Nm³.

The specific production of COG is 100-120 m³ STP/t steel with a calorific value of about 19.8 (MJ/Nm³). The specific production of BOF gas is 130-180 m³ STP/t steel with a calorific value of about 8.8 (MJ/Nm³).

Taking into account the actual situation in Uganda, the utilization of a mini blast furnace could be suitable only when steel consumption rumps up to over 300,000-500,000 t/y and the following preconditions should be in place:

- 1) Government investment
- 2) A market able to receive the products continuously
- 3) High power electric energy availability
- 4) Adequate infrastructures

The utilization of medium or large blast furnace size cannot be considered feasible even in future in Uganda. In fact, integrated steel plants of this kind, due to their Capex (considering that on top of blast furnace itself also Coke Oven, Sintering plants are expensive units and for proper handling it is necessary coking coal of good quality) are not viable for plants of less than one million tons of annual capacity. This capacity should be summed with the actual steel players production in Uganda and the resulting total is not reasonably reachable for many years.

6.4. Options to produce liquid steel in Uganda

According to Uganda's steel demand, production rate of plants, continuity/availability of electrical power, the best option in the first phase is to consider developing the steel value chain by installing several small sized coal based DRPs (possibly kilns and later RHF's). Tembo steel already initiated this route, others may follow if the venture proves to be economically viable and gain the support of the government. It is important that government backs up this choice by helping Tembo as well as other players that are planning to invest in RKF's such to facilitate their intentions (refer to chapter 6 for government's possible measures to facilitate the industry, and country preconditions).

Also, taking into consideration that natural gas is not yet available, and that Uganda needs to urgently increase the level of liquid steel production using its iron ore deposits, the DRP steel production route with coal as reductant is the only possible option to take. Difficulties are on the logistics side for the requirement to feed to the plants with both iron ore and coal.

Coal amount for DRI Process in RKF is in the range of 800 – 1,000 kg/t of DRI.

Since this is the expected route of producing DRI for years from now, at least up to when natural gas will be available and more efficient gas-based plants will be installed, it must be considered that Uganda needs to import big quantities of coal, which therefore calls for planning at national level on how the coal can be delivered into the country.

For 500,000 t/year of DRI produced by RKF the quantity of Coal (imported) is in the range of 400,000 – 500,000 t/year, which is large enough to call for suitable (dedicated) transport logistics for delivery into the country.

Coal reductant is “less environmentally friendly” than natural gas, but if adequate auxiliary plants are foreseen (for reuse of the OFF gases) and adequate pollution control equipment is installed, the difference can be reduced to insignificant levels.

The second phase is the installation of new additional DRPs. This will be possible only if previous attempts have been proven successful and if the country adopts measures that assure upward demand trajectory as indicated in the Capability model.

Realistically, the rate of steel demand increase will be in between the economic model and the Capability model and it will be as faster as Capability model hypotheses will be realized, sustaining economy growth.

As the steel demand increases, as projected by the population growth, and if pre-conditions will be fulfilled, there won't be problems for the technology to back up the increased demand (all the described technologies will gradually fit into the growth model for implementation).

Availability of gas can accelerate the growth scenario because new DRPs plants with gas as a reductant are more efficient, which will in turn further curtail the prices of final products. The new DRP gas-based plants will coexist with the DRP coal-based plants realized in the first phase up until the DRP coal-based plants become obsolete and get replaced by new technologies in the DRP gas plants. The minimum size of DRI gas-based plants is 300,000 t/y.

A typical consumption of raw materials to produce 1 ton of DRI by gas based DRI plant (vertical reactor) are:

- 1) Iron ore: 1.55-1.60 ton
- 2) Natural gas: 250-300 Nm³/ton (10.4-13 GJ/ton)
- 3) Power: 90-120 kWh/ton

In accordance with the average specific consumption, the required natural gas for steel production for this scenario is shown in the table 24.

Table 24: Natural gas consumption for two steel production scenarios

Steel production (ton/year)	Natural gas consumption (MNm ³ SPT)
500,000	125-150
1,000,000	250-300

When consortium of actors collaborates in generating centralized plants once liquid steel consumption reaches 300,000-500,000 t/y, blast furnace route or big vertical DRI vessels will be possible. This scenario will require big investment, not only for the plant's erection but also for infrastructure. The high cost of investment cannot be absorbed by steel makers alone but must be supported, at least during the development phase, by the government investment institutions and trade markets.

Should the Blast Furnace process be considered as the option, the more reasonable way is to start with mini-blast furnace technology.

6.5. Present production of DRI in Uganda

Presently DRI produced in Uganda using coal is very limited (about 30,000 t/year by Steel Rolling Ltd).

There are only two steelmakers involved in DRI plants in Uganda as of now; one is Steel Rolling Ltd with a plant already in operation while another, Tembo Steel Ltd, has a new plant under construction. Both companies are using RKF technology.

Steel Rolling Ltd is presently using molasse/biomasses as reductant, resulting in low quality DRI with low metallization grade. All the interviewed steel makers utilizing Steel Rolling DRI for their liquid steel production are experiencing high consumption and big quantities of slag production.

Tembo Steel Ltd is in an advanced construction stage of its DRI plant. It is assumed that production will start in 2020. The produced DRI is expected to feed Tembo's own steel plants.

When interviewed during the site visit, Madhavani Steel expressed interest in venturing into DRI production and market. It has at its plant, available space for a potential installation of a coal based DRI plant using coal from Tanzania as a reductant. Madhvani is monitoring developments in DRI production, in particular, has taken a wait and see position on the developments of the DongSong DRI plant. Should the market remain buoyant after the DongSong investment, Madhavani may decide to install its own new DRI plant.

It is reasonable to foresee that other steel players may follow the same route if market indications prove to be positive; for example, Pramukh Steel Ltd is also contemplating joining the DRI production fray. With government support, these players have the potential to partially close the gap between liquid steel production and total steel products production in the next five years.

6.6. Direct reduction technology hydrogen based

This is a new technology under development, with no running production plants anywhere in the world but only experimental prototypes trying to confirm a viable feedback for a future possible industrial exploitation. The technology is promising especially for its huge advantage of being environmentally friendly.

An on-going research is proceeding in Europe and in Uganda where Crane Steel is establishing a DRI pilot plant using hydrogen with the collaboration of the University of Kyambogo's Mechanical Engineering Department.

While everyone agrees on the high efficiency of hydrogen in iron ore reduction, and the green effects of such a process on climate considering the end products are solid iron and water, concerns were raised mainly on the dangers associated with handling of hydrogen and the high costs of hydrogen production using electrolysis method.

The process still requires added carbon, required for liquid steel production, and to properly mix it to get the requested carbon content constantly in the entire reduced mass.

The research being conducted by Crane Steel is in a protective environment where they take advantage of University collaboration, low cost of energy and man-labour; however, it is proceeding with limited funds even with commitment for financial support from the President of the Republic of Uganda through the Ministry for Science, Technology and Innovations Fund.

The development of DRI technologies in the past (Midrex and Hylsa processes) proved that what works in labs most of the time requires a lot of adjustments in real life and what works in small size industrial scale does not always work in bigger industrial scale. A long period (several years) was

required to reach the industrial application level and with huge companies backing up costs of development.

In light of the above, even considering the promising technology for Uganda, it is not expected to play a major role in the value chain of iron and steel in the next decade.

6.7. Estimated costing for production of DRI and Liquid steel in Uganda

In order to give the reader a ballpark cost in Uganda for 1 ton of product (DRI and liquid steel), keeping into consideration the country environment, it was estimated:

- Producing DRI in Uganda (as feed material to Melting Furnaces) has a cost range from 262 to 299 US\$/ton (using natural gas or coal as reductant)
- Producing Liquid Steel has a cost of:
 - i) 456 US\$/ton with use of 100% gas reduced DRI
 - ii) range from 544 to 561 US\$/ton, with use of 100% coal reduced DRI or local scrap (since local scrap quantity is not enough, a blend of DRI and is expected)
 - iii) 655 US\$/ton, with use of imported scrap.

The above rough cost analysis confirms that the difference in final product price depends a lot on the feed material used for Liquid Steel.

The most convenient feed material for liquid Steel production is gas reduced DRI.

Producing liquid steel with DRI coal-reduced and/or local scrap increases cost by 20 – 23%, if instead it is produced by imported scrap cost increases up to 44% (due to additional transportation to Uganda).

Finally cost for producing a ton of material (assumed billet) goes up from 478 US\$/ton by liquid steel from gas reduced DRI to 676 US\$/ton by liquid steel originated by melting imported scrap.

Table 25 illustrated the above reported data

Table 25: Costs in Uganda for 1t of Material

	Cost (in US\$ for 1 ton of product)			
	Material Cost	Energy Cost	Operating Supplies and Staff Cost	TOTAL (Estimated)
DRI (Coal)	\$130	\$160	\$9	\$299
DRI (Gas)	\$104	\$147	\$11	\$262
Liquid Steel (from 100 % charge by DRI Coal processed)	\$358	\$100	\$41	\$499
Liquid Steel (from 100 % charge by DRI Gas processed)	\$318	\$100	\$38	\$456
Liquid Steel (from 100 % charge by Local Scrap)	\$423	\$85	\$35	\$544
Liquid Steel (from 100 % charge by Imported Scrap)	\$534	\$85	\$35	\$655
Billet by Liquid steel from 100 % DRI (Coal Processed)	\$499	\$8	\$13	\$520
Billet by Liquid steel from 100 % DRI (Gas Processed)	\$456	\$8	\$13	\$478
Billet by Liquid steel 100 % Scrap (Local Scrap)	\$544	\$8	\$13	\$565
Billet by Liquid steel 100 % Scrap (Imported Scrap)	\$655	\$8	\$13	\$676

7. INFRASTRUCTURE AND LOGISTICS

7.1. Introduction

This chapter of logistics and when to have which infrastructure in place is informed by the modelled results i.e. how much iron and steel will be demanded at which stage of Uganda's development path (also taking into consideration the regional demand), which technology will be needed at the particular time (as given in the previous chapter). These factors then determine which infrastructure should be set-up and when bearing in mind that infrastructure development has to start earlier than the projects of the steel industry.

In order to make the iron and steel industry viable, there is need to put in place the required infrastructure such as rails, road, over-the-lake facilities, electricity and water. These activities need to be sequenced in the correct priorities and linked to developments in steel demand, the response of production to demand and the technologies that will need to be put in place. The results from the curve plotted from the model predictions has small increases which may only require improvement of rail to reduce Mombasa to Jinja costs. The commissioning of small DRI which are part of the increments on this curve call for the refurbishment of rail to SW Uganda and stabilisation of the grid as more smelting replaces imports of semi-finished steel inputs to rolling mills. Such infrastructure projects cannot be completed overnight. Only through a calculated prioritised approach can the final infrastructure layout be done also paying back by the phased economic activities in the various economic clusters/ecosystems. The proposed Malaba to Ryanika Rail should come online and link to Kigali dry port in Rwanda. This rail track allows goods moved over Lake Victoria to get to rail transportation.

When Uganda eventually invests in a medium to large scale single iron and steel works operation, the same facility will be used for movement of steel from SW Uganda to inland markets, also to Kenya and to the northern areas of South Sudan and beyond. This scenario is informed by the optimistic curve of the model whereby production will grow from 519,400 tonnes in 2019 up to 1,407,561.00 in 2024. The vision is to have a 1.5 million tonne capacity plant in phases in Kabale and this may require investment partners.

The first requirement to interest the private sector is to have an indicative feasibility study which demonstrates viability and sustainability of iron ore mines and an integrated Iron and Steel company in Uganda. One key factor being good infrastructure (road, rail, electricity, water high and speed internet) and logistics. The private sector can carry out due diligence and if satisfied, can go ahead and invest. At this point the investment procedures must be investor friendly. It is for these reasons that an analysis of the current infrastructure and logistics in Uganda was carried out and recommendations provided to ensure that the country is investor ready for the development of this key industry sector.

Adequate infrastructure is required for iron ore mining and iron and steel manufacturing. Actual and not imagined availability of natural gas or a cheap reductant such as coal must be guaranteed including security of uninterrupted supply of the chosen reductant.

Uganda has a trade deficit on steel. Over 60% of inputs for steel manufacturing are derived from imports, yet Uganda has high grade hematite ore in the south west. One reason for non-exploitation among others is the absence of logistical infrastructure as well as the non-existent of a suitable reductant for the iron making processes in the country.

This section presents an analysis of the country's infrastructure, highlights deficiency areas and provides a possible road map (schedule) of projects that must be undertaken to support the private sector that has shown interest in substituting steel imports by exploiting the ores that Uganda is endowed with. Additional infrastructure that must be in place in order to put up a bigger steelworks that can meet steel needs of Uganda, the EAC and possibly the rest of Uganda market share in the global market is also highlighted.

In managing a developed iron and steel industry an enabling environment plus some protection are key elements to be deployed in return for the bigger benefits of national development goals; thus in order to compete with low steel prices from other sources (China in particular) both in homeland and international markets, steel companies in Uganda must design competitively, be located competitively and have efficient and cheap logistics systems. Any disadvantageous structural and logistical issues would make operations uncompetitive for the lifetime of the steelworks. It is important to have affordable electricity and rail tariffs serving the South Western Uganda where currently there is premium grade of iron ore being mined at prices well below world prices. Other cost drivers like water and finance must be made affordable.

7.2. Mining Infrastructure

The infrastructure and other constraints within the mining subsector require immediate redress to enable private sector entities that are already mining in South West Uganda and delivering iron ore to Jinja. The issues requiring attention are:

- The need for a variety of heavy equipment during exploration stage, equipment service centres, fuel stations for fuelling heavy-duty vehicles and electricity. Also required are internal access roads. These issues are covered by the miner, but the government can facilitate by providing enabling policies.
- The need for the government on its own or through PPP to build good external roads and railways for transporting iron ore to the smelting plants and to the ports for export in the event of the ban on export of raw minerals being lifted. This becomes particularly important at the point the mine starts production.
- Electricity is also needed to run laboratories and operate iron ore processing plants, usually established close to the mining sites.
- Processing operations require large amounts of electricity to run the equipment and water for other uses.

7.3. National Intercity Infrastructure

Infrastructure such as roads, rail and freight ships/barges on Lake Victoria are required to enable movement of raw materials from indicated sources to private sector industries in Jinja and those proposed for Tororo and Kampala. Supporting private sector initiatives in the immediate short term is the earliest that Uganda can realise the objective of utilising local iron ore to make steel and substitute the high steel tonnages being imported, which imports are causing a large trade deficit and the attendant undesirable effects on the macro-economics.

The needs assessments are premised on tables 26 and 27 which show the design capacities or levels of utilising iron ore as the input raw materials.

Table 26: Design/Utilization Capacities of Industrial Consumers of Iron Ore, Material Requirements for Iron Ore, Limestone and Coal

Company	DRI Capacity	Iron Ore kg/t	Iron Ore tonnes	Coal kg/t	Coal Tonnes	limestone kg/t	limestone tonnes
Tembo Steels Lugazi	140000	1500	210000	1000	140000	25	3500
Kigezi	20,000	1500	30000	1000	20000	25	500
Steel Rolling	36,000	1500	54000	1000	36000	25	900
Pramukh Steel	35,000	1500	52500	1000	35000	25	875
Total Material by 2024	231,000		346500		231000		5775
DONG SONG Ore			300000				
GRAND TOTALS			646500				

NOTE: On the DongSong flow chart, there is no DRI Plant suggesting Electrical Iron Furnace technology

Table 27: Iron Ore Movement Presented Year by Year as Production Ramps Up

	I	II	III	IV	V	VI	VII	VIII
Year	Tembo	Steel Roll	Kigezi	Pramukh	Dong Song	Total DRI	DSO (Direct Supplied Ore)	DSO +Dons
2020	20000	30000	0	0	0	50000	75000	75000
2021	60000	30000	0	0	130000	90000	135000	265000
2022	60000	30000	20000	0	200000	110000	165000	365000
2023	60000	30000	20000	0	250000	110000	165000	415000
2024	130000	30000	20000	35000	300000	215000	322500	622500

The total annual materials to be handled for real and anticipated current private sector investments require provision of logistical infrastructure and this infrastructure shall also be available for users who will shift to cheaper logistics once they are in place. In the case of rail, provisions can be made for passenger trains to also use the same facilities. The weekly traffic for the annual material flows will be as shown in tables 28 and 29:

Table 28: Projected Weekly Ore Train Schedules for 40 Tonnes per Wagon/ Unit Train of 45 Wagons

Iron Ore	Annual tonnes	Weekly tonnes	Load at	Offload at	# of trains /wk
2020	75000	1470.59	KABALE	JINJA/KPL/TOR	0.82
2021	265000	5196.08	KABALE	JINJA/KPL/TOR	2.89
2022	365000	7156.86	KABALE	JINJA/KPL/TOR	3.98
2023	415000	8137.25	KABALE	JINJA/KPL/TOR	4.52
2024	622500	12205.88	KABALE	JINJA/KPL/TOR	6.78

Table 29: Projected Weekly Coal Train Schedules for 40 Tonnes per Wagon/ Unit Train of 40 Wagons

Coal	Annual tonnes	Weekly tonnes	Load at	Offload at	# of trains /wk
2020	50000	980.39	MWANZA/ then PORT BELL	KAMPALA/ JINJA	0.61
2021	90000	1764.71	MWANZA/ then PORT BELL	KAMPALA/ JINJA	1.10
2022	110000	2156.86	MWANZA/ then PORT BELL	KAMPALA/ JINJA	1.35
2023	110000	2156.86	MWANZA/ then PORT BELL	KAMPALA/ JINJA	1.35
2024	215000	4215.69	MWANZA/ then PORT BELL	KAMPALA/ JINJA	2.63

The above arguments will remain theoretical unless Uganda undertakes infrastructure projects development some of which are planned for in the NDPs. The revival of the meter gauge rail is one such project and this needs to be undertaken immediately. Besides rail rehabilitation and extension to iron ore mining areas, the use of ferries/barges on Lake Victoria must be revived urgently for both ore and coal as well as for products /commodities of trade.

7.4. Restructure Wagon Ferries to Carry Mafi Trailers

The paragraph quoted below is in the report (Corridor Diagnostic Study of the Northern and Central Corridors of East Africa) April 2011.

“Principal cargo transport services on Lake Victoria were designed as part of a railway system, with wagon ferries carrying wagons across the Lake. Link spans were built at all major ports Mwanza, Kemono Bay and Musoma in Tanzania, Kisumu in Kenya and Jinja and Port Bell in Uganda to facilitate rolling wagons on/off the ferries. When the railways were performing well the wagon ferries had an important role to provide an important transport link for both Northern and Central corridors. However, with the near collapse of the railways in recent years the importance and use of wagon ferries declined and the ferries got no proper maintenance.”

This is still true today as it was in 2011. Interventions are required to get rail working again. Road & Rail transport and Ferries at Lake Victoria are all required to provide for all transportation requirements for inward and outward-bound logistics. Such massive infrastructure projects cannot be completed overnight. Only through a calculated prioritized approach can the final infrastructure layout be done. Infrastructure projects have long paying back periods but do payback anyhow through the phased economic activities in the various economic clusters/ecosystems.

With existing logistical problem where there is no rail to talk about, Port Bell needs to be reopened to make use of ships or ferries in Lake Victoria and resuscitation of the metre gauge rail to Kampala.

7.5. Infrastructure and Logistics for a 1.5 million TPA Plant in Kabale

7.5.1. General and Diverse Handling logistics

This function is mainly outsourced to established logistics companies in the country. A database or further licensing of Input Suppliers, Agents and Distributors is required so that investors in Iron and Steel industry can outsource work to other sectors established within the country while the investors concentrate on their core businesses. This is part of the Local Content which is embedded as part of a Supplier Development Program. Such services are required prior, during and after construction and commissioning of the proposed plant.

7.5.2. Utilities: Energy and Water and Natural Gas.

There is need extend the high voltage transmission lines to Kabale in order to have dedicated power supply and sub stations to major industries so as to guarantee supply of uninterrupted high-quality electricity supplies. The proposed electricity distribution network is shown on figs. 22 to 24. Consideration must be made of concessionary rates on the power tariffs at least at levels adequate for the power utilities to recover basic costs. The other critical requirement is to have water reservoirs for adequate availability for all the operations of the steelworks as well as growth of the urban settlement. The completion of the natural gas pipeline and eventual delivery of the gas cannot be overemphasised. This is a strategic requirement otherwise the 1.5 million tpa plant cannot be established. One way or the other, there must be guarantees of security of supply.

7.5.3. Construction of a One Million to 1.5 Million Tonnes/Year Steelworks

The strategy for Uganda is to attain the goal of reducing the import bill where forex is spent on purchasing semi-finished steel instead of making it. This shall be achieved by the private sector who have decided to add value to Uganda's own iron ore and creating employment and other spin off benefits. After Phase 1 of establishing an iron and steel value chain, starting from mining, it is necessary to use natural gas once available to produce all local requirements within Uganda's capacity and export the excess.

This is where the construction of a medium sized steel works comes in. This can be a state-owned enterprise or a JV in which the state holds shares. The infrastructures advocated for immediate private sector players must be revamped for higher volumes of production at every stage to accommodate the proposed new steelworks. Access roads for iron ore exploration must be in place by 2020 to allow schedules in table 27 to happen. There may be need for confirmatory exploration of the limestone deposit as well. Infrastructure will be required from the limestone deposit if confirmed suitable to the steelworks site. Table 30 below shows a wished-for schedule of activities in order to bring in the integrated iron and steel works at the earliest possible time.

Table 30: Proposed Shortest Duration Activities Schedule

			2020		2021		2022		2023		2024	
	Task	CURRENT STATUS	H1	H2	H1	H2	H1	H2	H1	H2	H1	H2
1	Incremental Growth of DRI											
	Rolling Mills	in production										
	TEMBO Steels construction	in construction										
	PRAMUK	planning to start										
	DONG SONG	licence and plans in place awaiting implementation										
	KIGEZI	licence and plans in place awaiting implementation										
2	Exploration	planned for										
3	Meter Gauge Rail	should rehab existing sections & extend to Kabale										
4	HV Transmission Lines	should extend HV transmission to Kabale										
5	Fast Internet Data											
6	Water Resources											
7	Gas line likely sequence											
	Feasibility study	MOU EXISTS										
	Land rights											
	EIA											
	Engineering design											
	Survey/GIS etc											
	Excavation landscaping/fill											
	Civil works and pipe installations											
	Final equipment installation											
8	Choice of site											
	Geotech and other assessments of sites	There are two proposed sites, new ones can come up										
	EIA of proposed sites											
	Link roads											
	Site preparation											
	Construction of DRP, Steel plant and mills											

7.6. Infrastructure Maps and Proposed Sites

In conjunction with activities planned for eventual realisation of a medium sized all in one iron and steel works, an opportunity was taken during the visit to the SW Uganda to find probable sites. These sites have been plotted on a map relative to various infrastructures and logistics as shown in figures 22 to 24. The maps have been included to illustrate logistics that will be required for the proposed sites. If other sites are preferred, the logistical requirements will also change.

- **Relative Locations of Ore Bodies, Steelworks Site & Infrastructure including MGR (meter gauge rail).** This is depicted in figure 22. It is important that plans to rehabilitate the old meter gauge rail are expedited and that an extension be made to Kabale to service the iron ore mining industry and later the iron and steel works.
- **Natural Gas Pipeline & Location of Free Zones Including Possible Gas Line Routes.** This is illustrated in figure 23: There is a planned Kable Free Zone and the natural gas lines are expected to service these areas. This puts the proposed steelworks close to natural gas lines and therefore can be synchronously aligned with the setting up of a DRP based on natural gas.
- **Planned SGR Relative to Proposed Steel Works Sites.** There are plans to later on put SGR linking Kabale to DRC via Mbarara and Kasese, see figure 24. This will be a welcome project that will result in further logistics improvement as there will be a stronger link with countries on the proposed SGR network.

Infrastructure will be required to a possible limestone mine in the Kabale area. The grade of the limestone assays as in table 31 below.

Table 31: Grades of a calcareous rock taken from Kaku River

Compound	Unburnt samples (Kaku-F.25) %	Burnt samples (Kaku-F.25) %
CaO	51.77	64.19
Fe ₂ O ₃	2.08	2.68
MgO	2.03	2.58

7.7. Choice of Site for the Proposed New Iron and Steel Works- Table 32

Table 32 shows reasons for choosing the sites in the positions shown on the maps. The advantages for locating the steelworks in Kabale include proximity to the ore bodies so raw material supply security is more guaranteed as even with interruptions on supply by rail, emergency supplies can be delivered by road to the steelworks. There is no need to transport some materials that may be screened off. The steelmaking processes generate waste and this material should not be part of freight tonnage as embedded in the raw materials. There is a possibility of mining some of the limestone from around Kabale. A geological report indicated that there is an estimated 1Mt of calcareous rock occurring in at least 4 separate deposits along the Kaku River showing promising grades (table 31).

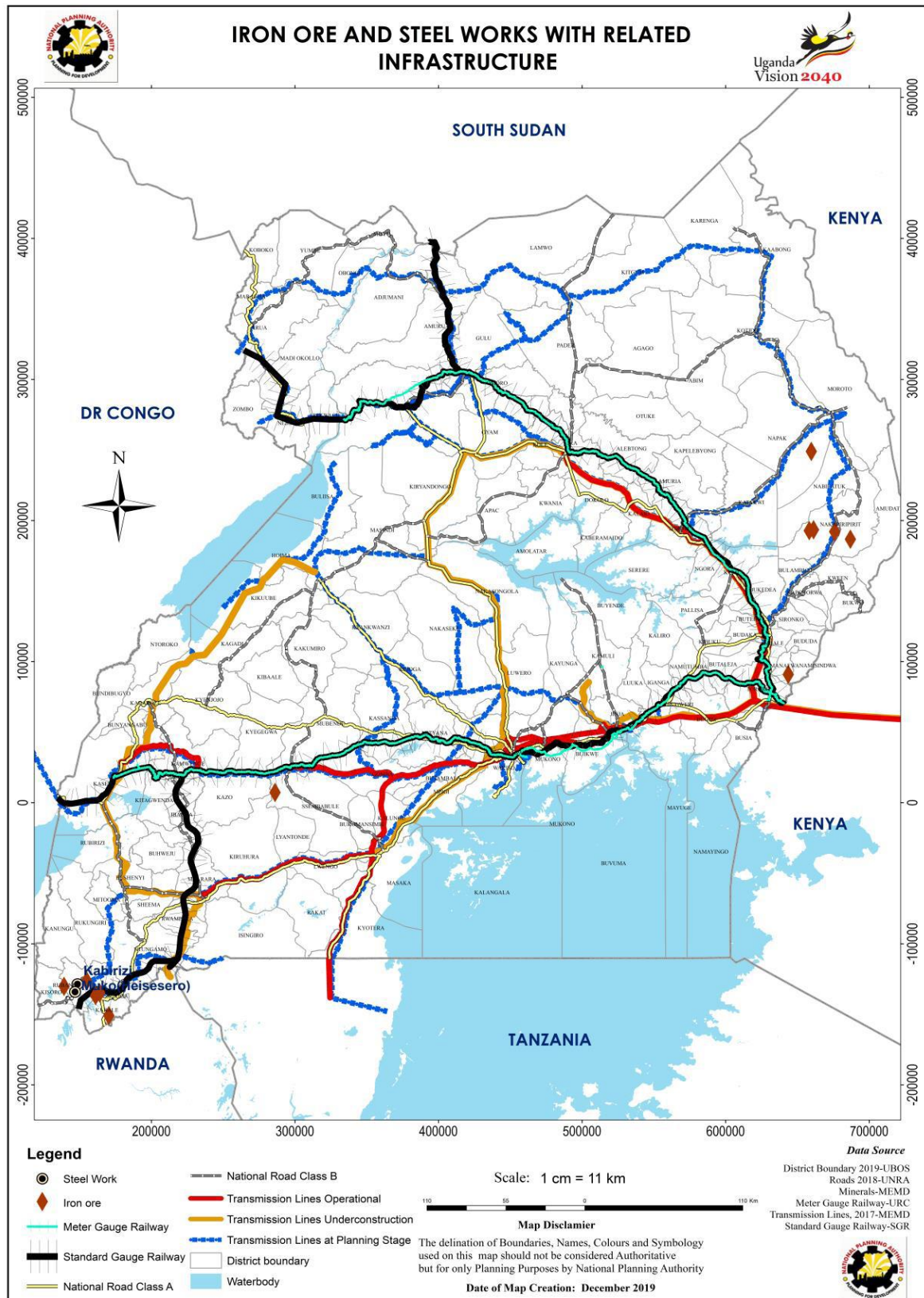


Figure 22: Relative Locations of Ore Bodies, Proposed Steelworks site and Infrastructure.

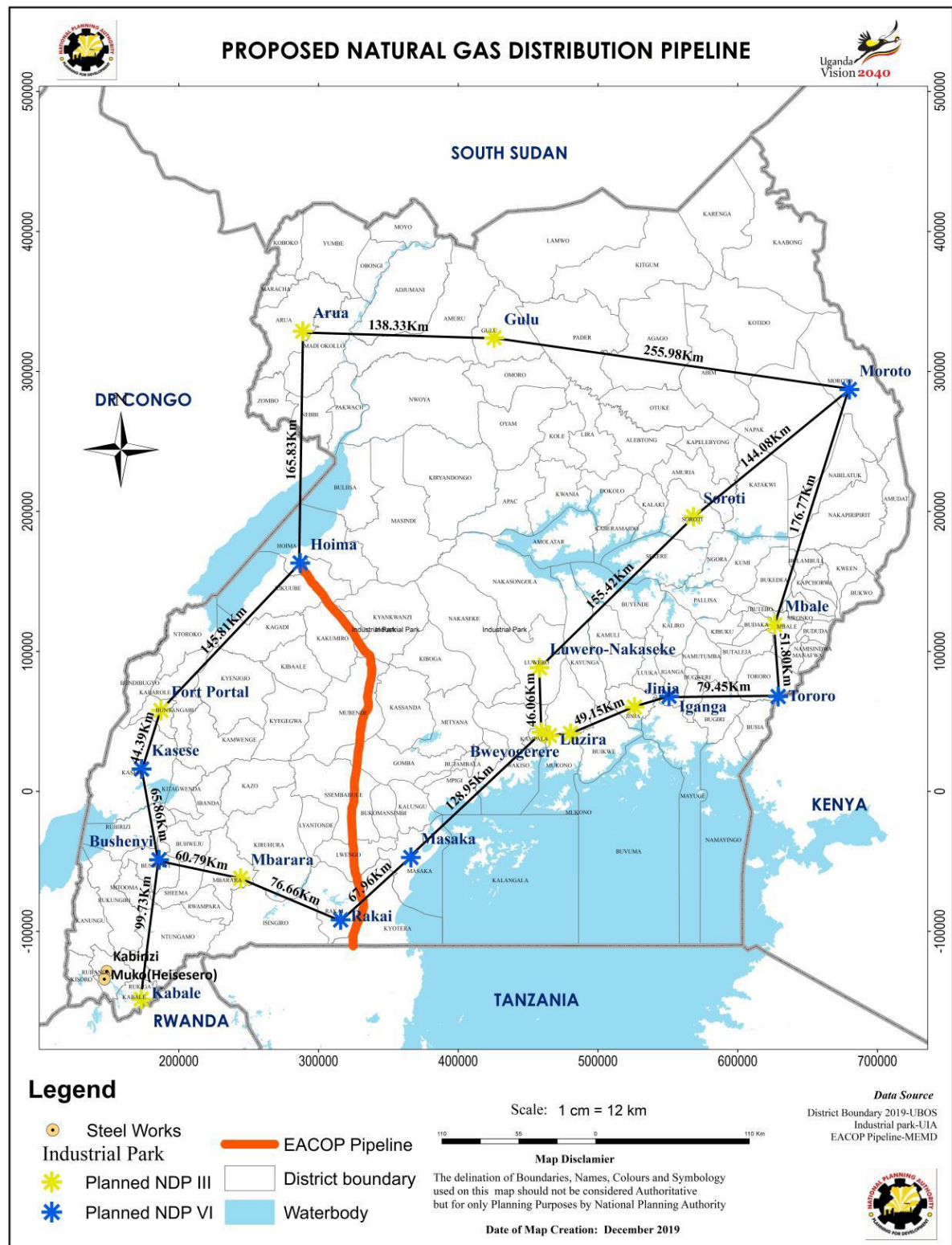


Figure 23: National Gas Pipeline and Location of Free Zones.

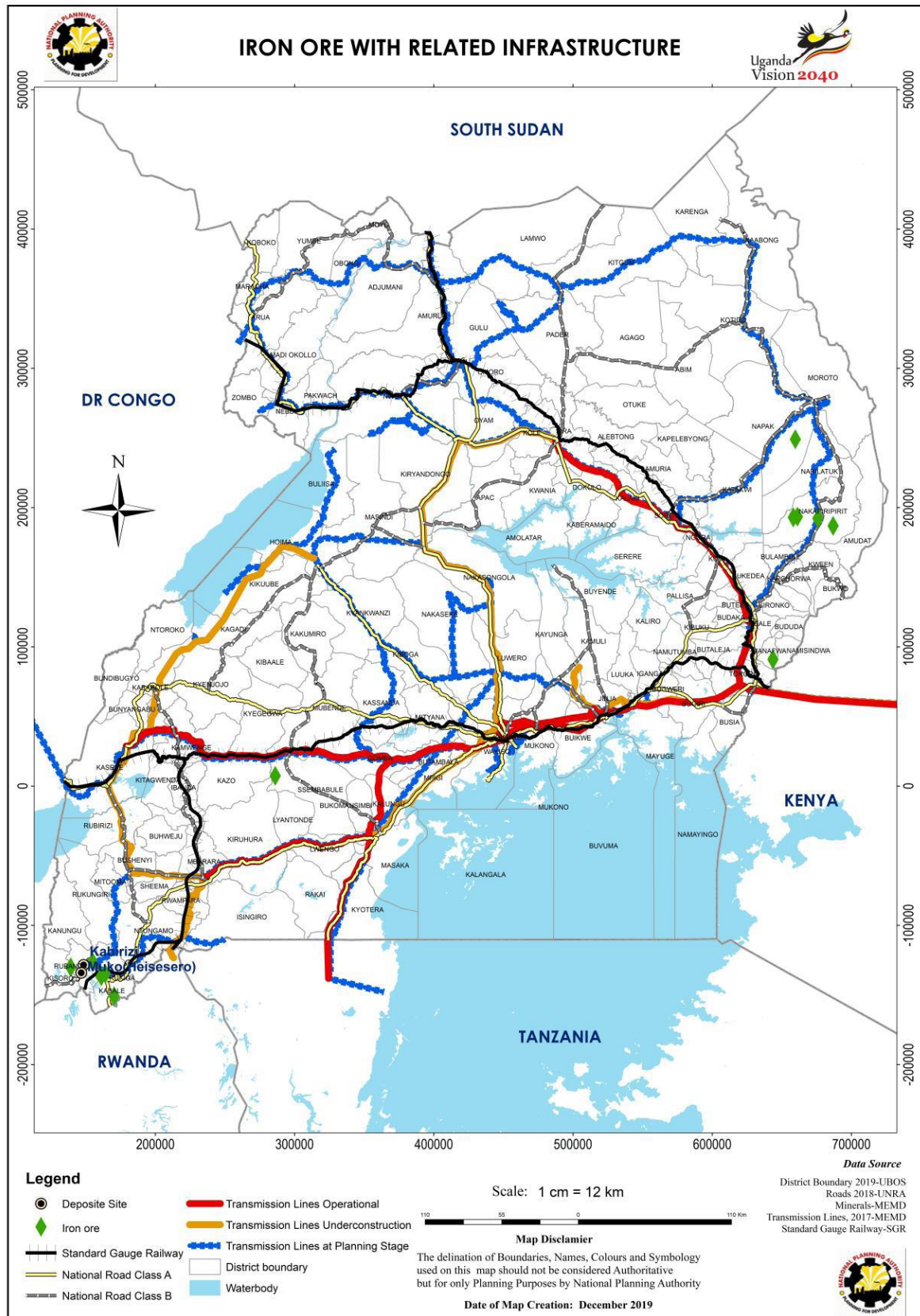


Figure 24: Planned SGR relative to Proposed Steelworks Sites

Table 32: Justification for Proposed New Iron and Steel Works

S/No	CONSIDERATIONS FOR NEW STEELWORKS	KABALE AREA ROAD	KABALE AREA RAIL/CONVEYOR	KPL+JIN ROAD	KPL+JIN RAIL	KPL+JIN ROAD+FERRY	KPL+JIN RAIL+FERRY
1	Distance from Iron Ore Deposits 1 Rugando	39.64					
	2 Muko	7.46					
	3 Buhara	15.0					
	Proximity to limestone deposit	Local or 534km from Tororo		Jinja 125km from Tororo, Kampala 205km			
	Proximity to coal deposit	96 km		Long distance from Tanzania			
2	Distance from proposed Gas Line	CLOSE within 200 km		If gas enters from SW Uganda, 519km & 430km			
3	Proximity to Power Lines	HV lines need to be constructed		HV in place, need improvement			
4	Proximity to Road Networks	Close		close			
	Provision of rail network	currently no rail, SGR/MGR		Rail needs revamping			
5	Proximity to Local Markets Distance from Kampala Jinja (80) Kabale (430)	Industries in this area would operate on value addition of indigenous ores in Uganda and can produce semis or finished niche products or complement existing products.		Current Ugandan steel manufacturers will operate on either current platform of final processing of imported semi-finished steel or to use semi-finished products from Kabale.			
6	Proximity to Export Markets - DRC	92 km		602 & 430km			
	-Rwanda	96 km		615 & 526km			
	-Burundi	344km		863 & 774km			
	-Kenya	653 km		143 & 223km			
	-Tanzania	96 km		454 & 374km			
	-South Sudan	835 km		531 & 459km			
	Presence of logistics companies & OEMs	They need to locate to Kabale		Some are in Kampala			
7	Major environmental issues	All I&S processes have the potential to pollute the environment including the DRI route. Before going ahead with construction an EIA with acceptable process or technological mitigating strategies is required how to handle dust, SOx, NOx, SS,Oil, Cl ₂ ,SO ₄ , Fe ions, slag, spent refractories, mill scale etc. These can report to the air, water or land. For Kabale site Lake Bunyonyi and surrounding prestine environs including tourist areas like the Bwindi Impenetrable Forest have to be considered. For Kampala and Jinja areas consideration should be of Lake Victoria and other water bodies plus urban environs.					
8	Availability of water sources	Lake Bunyonyi		Lake Victoria			
9	Capability to make niche high value products	DRI EAF requires large land area this proposed site be further assessed for adequacy. Will transport products instead of ores which has advantages		DRI EAF requires large land area this proposed site be further assessed for adequacy. Will transport ores including gangue which has advantages			
10	FREE ZONE Possibility	YES		YES			

7.9. Support for Future Exports

7.9.1. Iron ore exports

There is need to provide rail, road and lake transportation facilities for the export activities to be competitive. Given that Uganda's iron ores are located in the South West and the port of choice is Mombasa, currently there is no rail infrastructure to link to the port. The occurrence of the ore bodies is in a discontinuous fashion given geological processes of folding and faulting that occurred in the ore deposits areas. The grade of 65%Fe, if realised, would fetch a premium price on the world market but mining and logistical costs, if not addressed, may make the venture uncompetitive.

Uganda should reconsider exporting some raw iron ore. The major hindrance before the ban was the logistics of getting the ore to the port competitively. The proposed rail if connected to Kabale would solve these logistical competitiveness issues. The government must develop necessary infrastructure to service or provide for the needs of the industry and charge rates that leave room for profit.

7.9.2. Steel Exports

The Malaba to Ruanika rail if implemented should come online. Besides linking up with Kampala/Jinja and Tororo, the line can link to Kigali dry port in Rwanda (for exports). This rail track allows goods moved over Lake Victoria to get to rail transportation. It also links with the rail to Kenya (imports and exports).

7.10. Conclusion

All forms of infrastructure; road, rail, maritime, fast internet data, high voltage transmission etc. must be revamped to facilitate industry development like iron and steel, oil and fertiliser industries with the effect of kick-starting a massive industrial revolution in Uganda. The logistics costs must be borne to a large extent by the state with private ownership of dedicated roads, dedicated wagons or ferries etc. also being done by private sector or through PPPs. The gas line project must be expedited as it does benefit a wider spectrum of the Ugandan economy, same as for rail, barges, roads etc. Several other investments will rush into Uganda once the favourable and enabling environment is created in terms of good infrastructure and logistics.

Current situation indicates that iron ore mined by artisanal miners and delivered to Steel Rolling Mill in Jinja, which is more than 450 km from Kabale, is delivered at USD 70/t (purchase, transport, and profit inclusive). The cost of mechanized mining and beneficiation for the companies that have just started (e.g. SMI Sino Minerals International, and DRIOC) is not yet computed but as of 2015, it was expected to be around USD 20 per ton according to the report **Iron, Steel and Metal Processing Value Chain for East African Community** presented in March 2015. At this sale price, therefore, iron ore mining in SW Uganda is still internationally profitable despite the falling global prices and hence the reason Uganda is able to attract the investors. What is stopping progress on the I&S industry is a lack of reductant; international coking coal is too expensive. Additionally, the natural gas line is only scheduled for 2024. Furthermore, the lack of logistics infrastructure, which is affecting even those currently operating steel manufacturing companies this very present day.

8. VALUE CHAIN ANALYSIS

8.1. Introduction

Value chain analysis points to competitiveness, sustainability and profitability of the iron and steel operations once the manufacturing model shifts towards value addition of local ores. Uganda should therefore consider taking the necessary steps to establish iron ore mining, iron ore reduction and steel production using the recommended DRI process.

As part of the feasibility study on the establishment of a comprehensive iron and steel company that utilises the iron ore resources available in Uganda, a value chain analysis on the options available was undertaken. Part of the evaluation was to assess what would be the strategic position of iron/steel in the future aspirations of Uganda. It was opined that the establishment of an iron and steel company would have the following benefits:

- **Impact on Trade deficit:** Given that steel and steel related products account for a greater part of imports on Uganda's balance of trade, an iron and steel plant will reduce trade deficit through reduced steel and steel related imports and or increased exports in steel and other steel related products that could form part of the value chain.
- **Real GDP Growth Contribution:** Real GDP growth is a factor of increased output and the contribution of a new steel industry to GDP growth is derived from its positive effect on the balance of trade, value addition and support to other economic sectors, and, employment and wages. An integrated steelwork can contribute as much as 2% or more to the annual GDP directly and/or indirectly.
- **Contribution to the stagnant manufacturing sector:** According to the *UN Economic Commission for Africa* report³ on industrialization in Uganda, Uganda's manufacturing sector has been described as stagnant and largely regional and small scale. A new iron and steel industry can aid the genesis of a robust manufacturing sector by providing the necessary raw materials and steel infrastructure locally, where it has usually been outsourced from foreign markets.
- **Infrastructural development:** The necessary infrastructure for a modern manufacturing, engineering and retail sectors, with the inclusion of households is dependent on the steel industry since it provides vital components. Infrastructural development has often become a measure of the level of economic activity whilst providing an incentive for many other potential investors.
- **Foreign currency generation, price stability and external debt:** Given that the new steel industry is strategically positioned to reduce import expenditure and also generate significant exports, foreign currency can be generated and channeled towards external debt settlement which has also been cited by the *UN Economic Commission for Africa* report as one of the distinctive features that have pulled back Uganda's economy. Price and exchange rate stability is also enhanced through sustained generation of foreign currency.
- **Key Eastern Africa Markets:** Most of Africa's growth in infrastructural development has been concentrated in the East African Community to which Uganda is a member. The new steel industry can take advantage of rapid economic and infrastructural growth in Rwanda, Tanzania, and Kenya

³ <https://www.uneca.org/publications/abc-industrialisation-uganda-achievements-bottlenecks-and-challenges>

including DRC (not a member of EAC but in the region). In the long term and with increased stability, South Sudan is also a key market to watch.

- **Employment creation, wages and poverty alleviation:** Such a strategic industry as iron and steel can employ thousands directly and indirectly driving up wages and standards of living. This has a ripple effect on other sectors like engineering and services (e.g. finance, health insurance and communications) and a huge effect on poverty alleviation (figure 25).

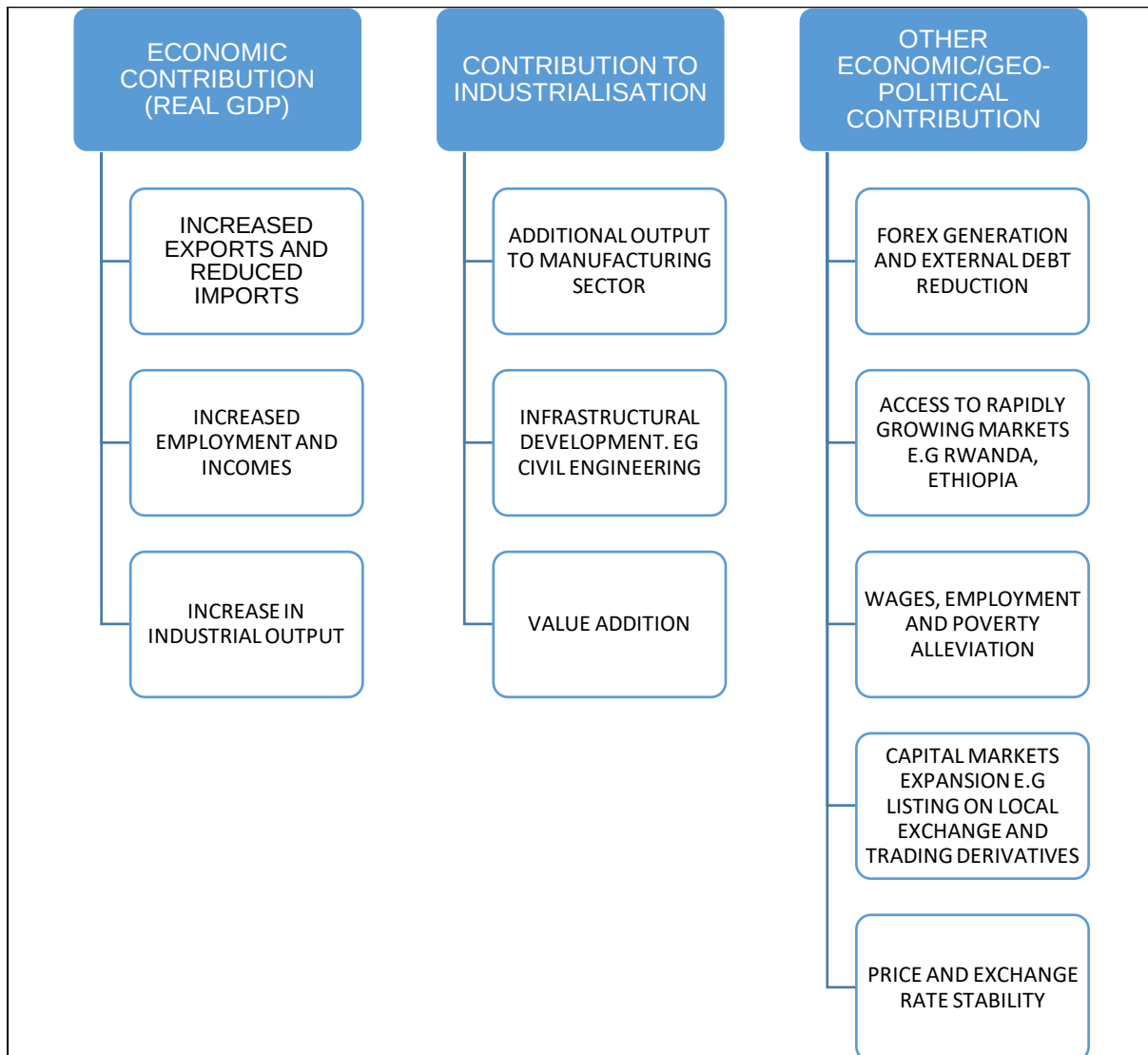


Figure 25: Expected benefits from the Iron and steel value chain

All the expected benefits alluded to in Fig 25 can only be realised if the steel manufacturing venture is established from a business point of view. This means that all areas identified by the value chain analysis should be addressed so that the business is competitive in the steel industry which has cutthroat competition from other nations and multi-national companies.

8.2. Competitive advantage

Competitive advantage is what sets a business apart from competitors. This is achieved by having a clear idea of the target market, the benefit the product provides to the target market, and a solid understanding of competitors and their offerings.

This value chain analysis was done to analyze the activities that the current and future steel industries in Uganda will perform to produce steel. The major goal is to evaluate ways to improve the competitive advantage of steelmaking in Uganda.

In Uganda it is proposed that steelmaking industries can gain a competitive advantage in one of the following areas.

8.2.1. Cost Leadership

The goal of a cost leadership strategy is to become the lowest-cost provider in your industry or market. Companies who excel with a low-cost strategy have extreme operational efficiency and use low-cost materials and resources to reduce the overall price of their product or service. Activity Based Costing (ABC) would help identify areas that need improvement e.g. whether the activity labour intensive, raw materials basket is expensive, energy, utilities, logistics, tariffs, taxes etc. This is where areas for improvement can be identified. For the 1.5 million single steelworks, economies of scale also feature as a cost advantage.

8.2.2. Differentiation

With a differentiation strategy, the competitive advantage is gained by offering a unique or highly specialized product or service. The business needs to dedicate time and resources to innovation, research, and development. A successful differentiation strategy allows the business to set a premium price for its product or service. Currently with infrastructure projects in EAC for road, rail and dams, it is advisable to start manufacturing of rail and other large sections for construction. The development of oil and gas justifies investment in pipe manufacturing. Car manufacturing is being established in Rwanda, Kenya and Uganda itself. This is creating demand for automotive plate. The consumption of ordinary plate is also the largest contributor in EAC imports so investment in slab and HRC and related hot strip or cold drawn plate mills must be pursued.

8.3. Value Chain Analysis Steps

Value chain analyses require research and can take time to develop. Below are the general steps it takes to create a value chain analysis:

1. Determine the business' primary and support activities.

Together, the primary and support activities make up the value chain. And they include each action required in the development of a product or service, from raw material to final product.

2 Analyse the value and cost of the activities.

The team tasked with creating the value chain analysis should brainstorm ways each activity provides value to customers and the business as a whole. Compare the activity to the competitive advantage you're trying to achieve (cost leadership or differentiation) and see if it supports the goal. This should be executed concurrently with brainstorming the ways each activity provides value to customers and the steel manufacturing business. After the value analysis is complete,

3 Identify opportunities to gain a competitive advantage.

Once the value chain analysis is complete, the primary stakeholders in the business can see an overview of where the business is excelling and where improvements can be made operationally.

Begin with the improvements that take minor changes and provide high-impact results. This also applies to growing the markets. After the easy wins are identified and actioned, it is time to tackle the bigger challenges that might be hindering efficiency.

The value chain analysis gives businesses a clear idea of how to adjust their actions and processes to provide the most value to their target market and increase profit margins for the company. This is partly covered by the competitiveness report for existing industries and the choice of processes and technology for the proposed new steelworks. This chapter will therefore concentrate on a clear idea of how to adjust steelmaking actions and processes in Uganda to provide the most value to their target market and increase profit margins for the company.

8.4. Local, Regional and International Market trends.

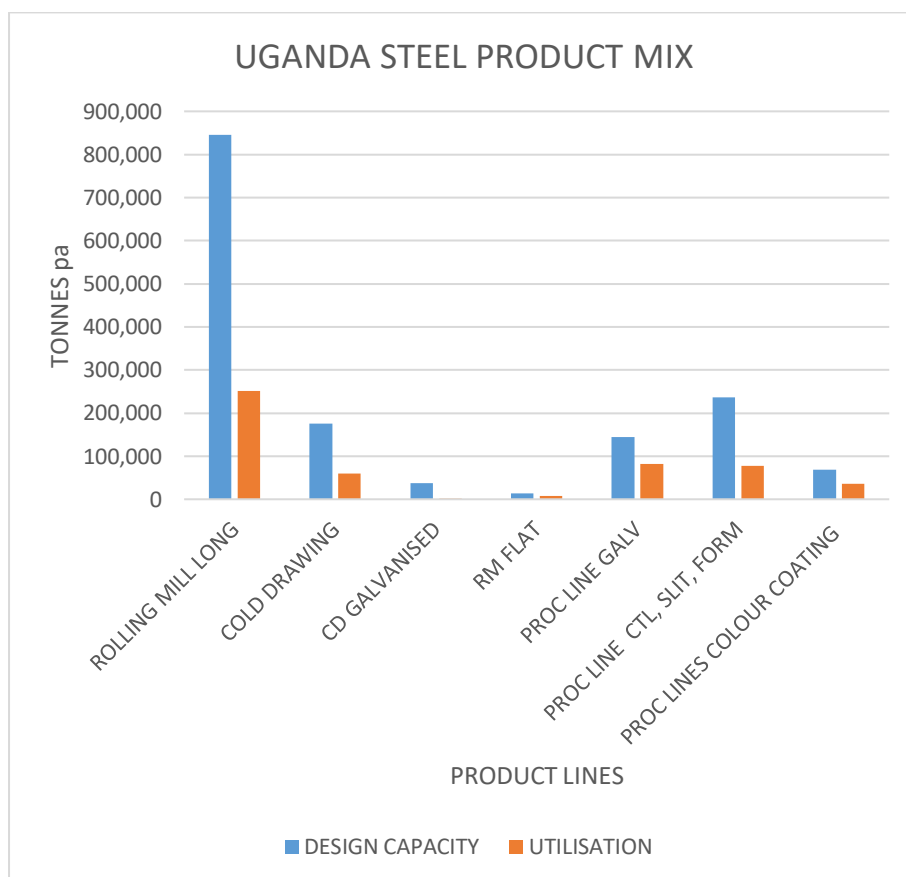
Markets are the key driving component of the value chain that will determine viability or lack of it. There is need for Uganda to carry out a comprehensive market study to assess viability starting from market fundamentals. Equally important is the optimisation of production issues raised in previous chapters so as to improve competitiveness of the DRI, SMS+CCM and Steel rolling (the iron making, steel making, and casting steel into primary and semi-finished steel products). Costs have been determined in chapter 6 and markets will be sought on the basis of cost plus margin in order to determine whether prices will be competitive or not. Improving Product Mix in response to demand will also form part of the basis whether new market share will be realised. All steel manufacturing companies must be compelled by law to produce to international quality standards in order to protect national brand and national reputation so that there is competitiveness arising from quality assurance thereby facilitating market penetration.

8.4.1. Local Market Trends

The current steel consumption levels in Uganda are very low with per capita levels at 12.8 kg per person per year. These levels are low compared to world average. This is a motivation to set up more capacity after taking more measures to stimulate more local demand and also to apply reasonable tariffs to protect the national economy and realise the benefits of utilising Uganda's local resources and realising the benefits as outlined at the beginning of this chapter. As pointed out in earlier chapters, the current total production is higher than the production of liquid steel being produced in Uganda. This means that the mills are augmenting their raw material inputs from imports. There is already demand which is being satisfied by imports. This would be understandable for sections like HRC for which currently there is no plant in Uganda capable of producing them. However, inputs such as billets and coil rods should eventually all come from liquid steel produced from the iron ores in Uganda. There is also more demand for steel and very likely players are importing finished products and selling them without any further value addition activity i.e. operators in phase VI simply selling the imported finished products. Other local steel production can also be taken by exports as long as production from high quality cheap iron ore is done competitively. Strategies are required to promote more local demand and facilitate regional and international trade. The current 2019 utilisation of 34% is an opportunity loss since assets must be sweated to give return on investment within a reasonable time period. Focusing on consolidating the local market is key to the development of a robust steel industry before efforts for the external markets can be pursued. With high competition, approaches that consolidate the local market are critical as global competition makes it hard to strive in external markets especially for high cost small producers. Figure 26 shows the products that are currently being

produced in Uganda and the export value to EAC. If these are compared with the products that are being consumed in Uganda, it becomes evident that there is need to produce construction steel which is going into infrastructure projects.

In managing a developed iron and steel industry an enabling environment plus some protections are key elements to be deployed in return for the bigger benefits of national development goals. It is necessary to strike a balance between protections and to demand innovation from the companies. The issues that are needed to support a developed steel industry, should be at prices that sustain viability of the entire iron and steel industry for everyone not just monopolies.



(KEY: Rolling Mill Long - long products, CD - cold drawn, RM FLAT -flats from hot rolling in a rolling mill with a reheating furnace, PROC - a processing activity also called finishing activities after manufacturing is complete, GALV= galvanizing, CTL= cut to length)

Figure 26: Steel products made in Uganda.

8.4.2. Regional Market

The current levels of exports to regional markets are low as demonstrated by an even high import bill by regional companies (Figure 27) compared to what Uganda is selling to them, Figure 28. It appears the EAC region imports a lot of HRC including Uganda itself. If only Uganda could partner with private companies with vast experience of this technology and establish a plant that produces HRC, surely this market is there for Uganda to take. In South Africa AMSA produces HRC at both Saldanha Bay and Vanderbijil Park. This would be a great leap forward into the EAC markets. Rails are the other niche products likely to present a huge market share given a lot of projects to resuscitate existing Meter Gauge Rail as well as construction of new Standard Gauge Rail throughout the whole region and other parts of East Africa (Ref: Corridor Diagnostic Study of The Northern and Central Corridors of East Africa). These are issues that can be addressed during the establishment of a sustainable iron and steel industry in Uganda.

From feedback obtained from Uganda steel manufacturers, they have penetrated markets in Eastern DRC and northern Angola. From literature available, there is huge steel demand in the said markets. The coming in of African Continental Free Trade Area (AfCFTA) presents further opportunities if Uganda steelworks goes into production before possible competition from Tanzania and Mozambique. Growing these markets will be driven by developing and maintaining competitive and comparative advantages along the value chain.

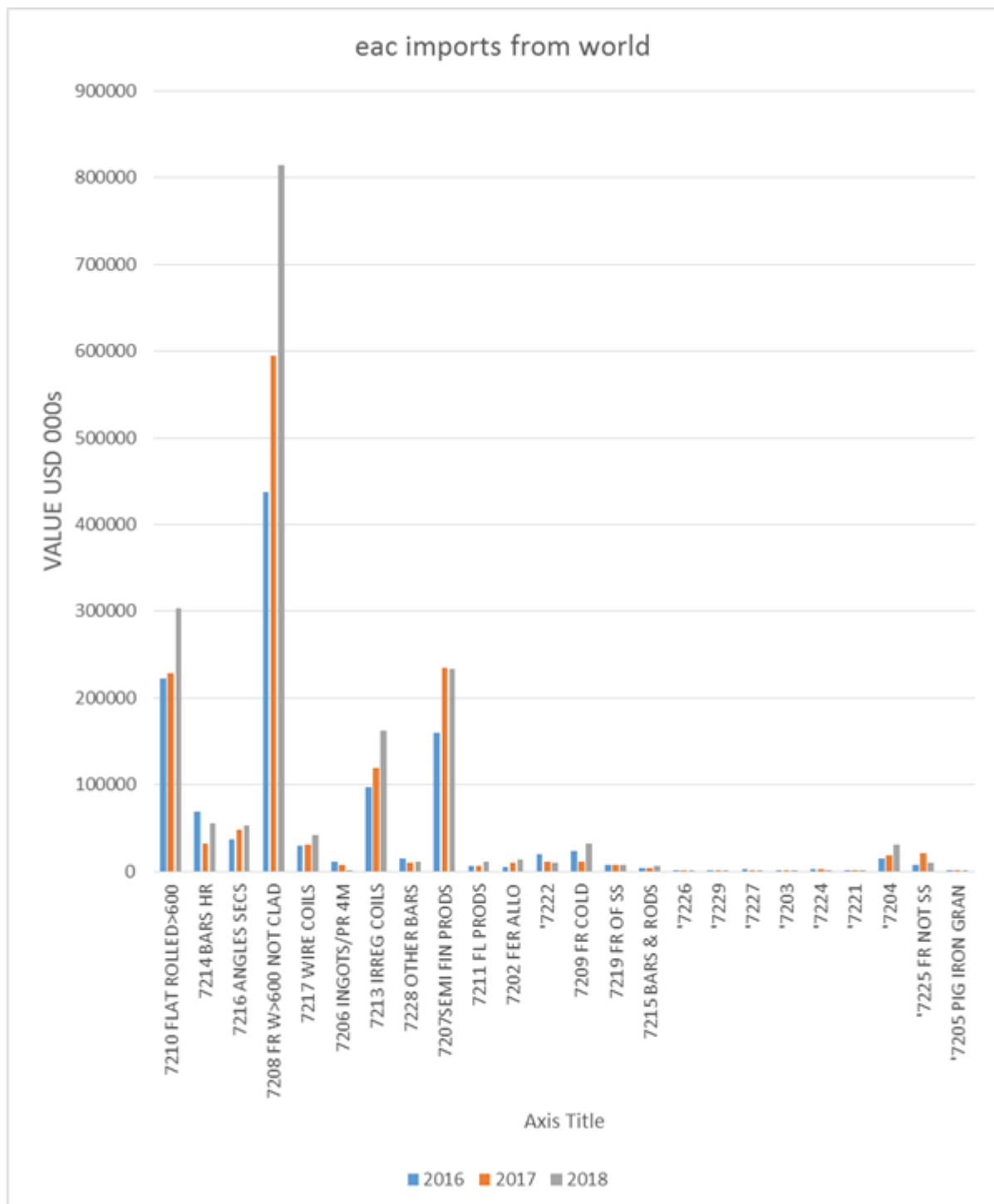


Figure 27: EAC Steel Imports from the World
(Source : <https://www.trademap.org/Index.aspx>)

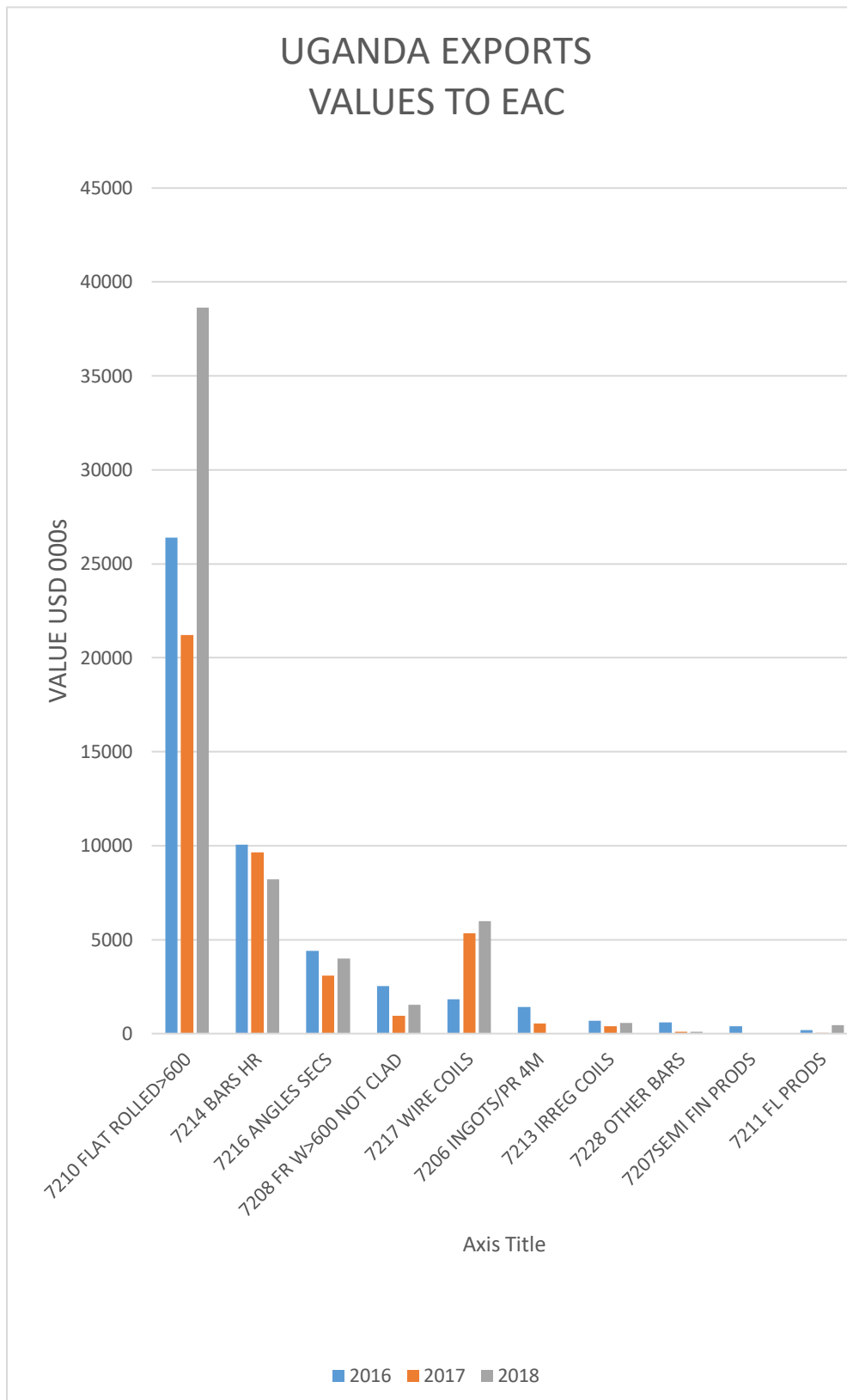


Figure 28: Uganda Export Values to the EAC

Figure 29 shows steel demand in selected SADC and EAC countries. There is potential for Uganda to capture part of this market as long as steel production along its value chain remains competitive.

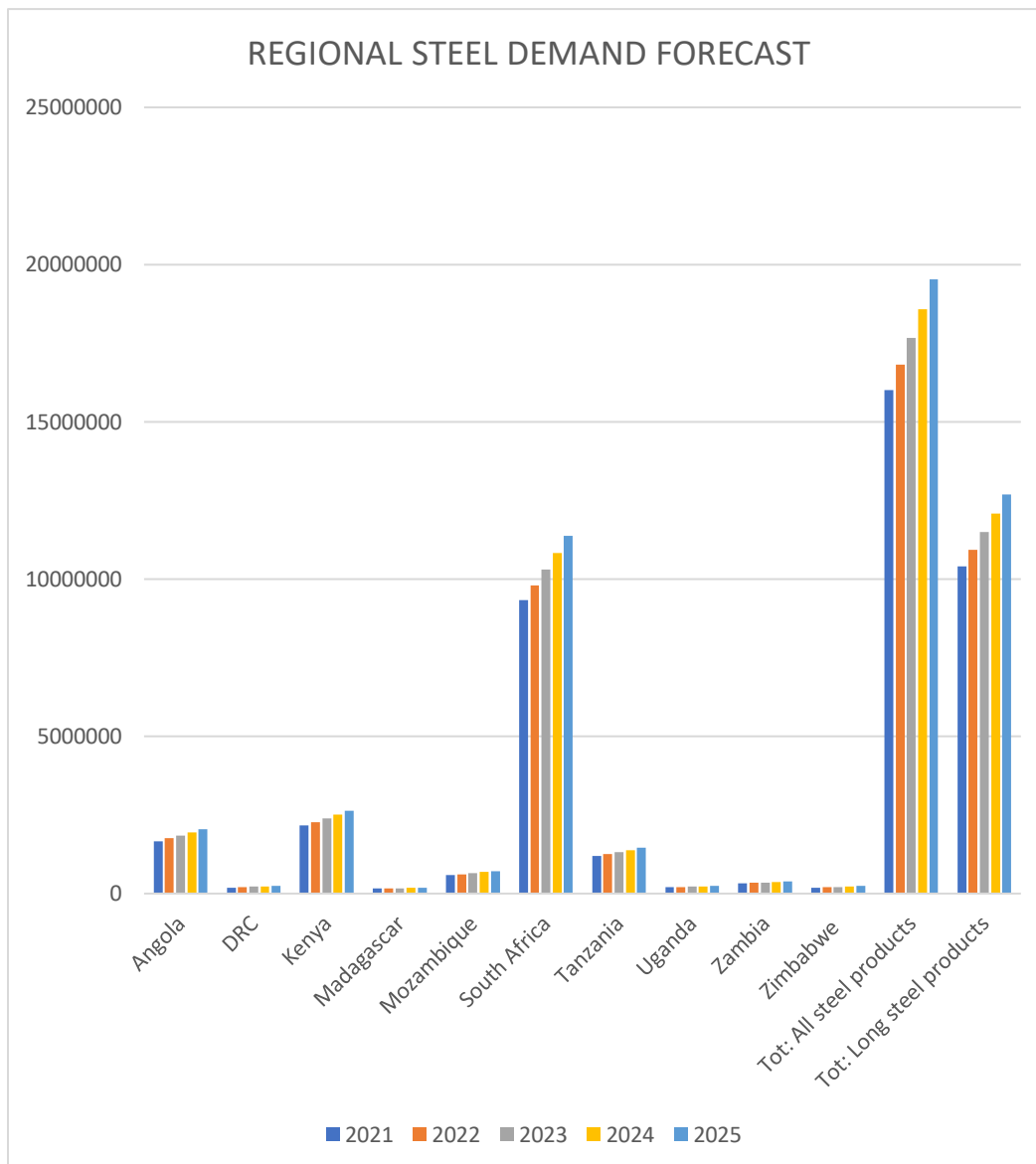


Figure 29: Steel Demand Forecast in Selected SADC and EAC Countries

8.4.3. International Markets

On the international market a company must have the necessary muscle and stamina to be competitive against the big players such as China, India, and ArcelorMittal. As an example, China has 500 million capacity producing in excess to their requirements, with the balance being dumped on the international market including USA. The USA has now responded by raising tariffs on imported steel from China and other countries, which is impacting on all steel companies in the world. Many countries have put tariff barriers to protect their own companies too and this has affected worldwide trade. Uganda might need to think very deeply whether venturing on the international market will create value or cause its steel industry to lose value. Its land locked position and poor infrastructure makes the task even more difficult. Figure 30 shows Uganda's low volumes of steel trade on the international market. It therefore follows that for Uganda's purposes of fulfilling the objective of earning foreign currency for the country, the EAC and other continental markets should be the targets for now with intermittent orders being transacted on the international arena.

Figure 30 shows the current levels of Uganda's international trade in steel.

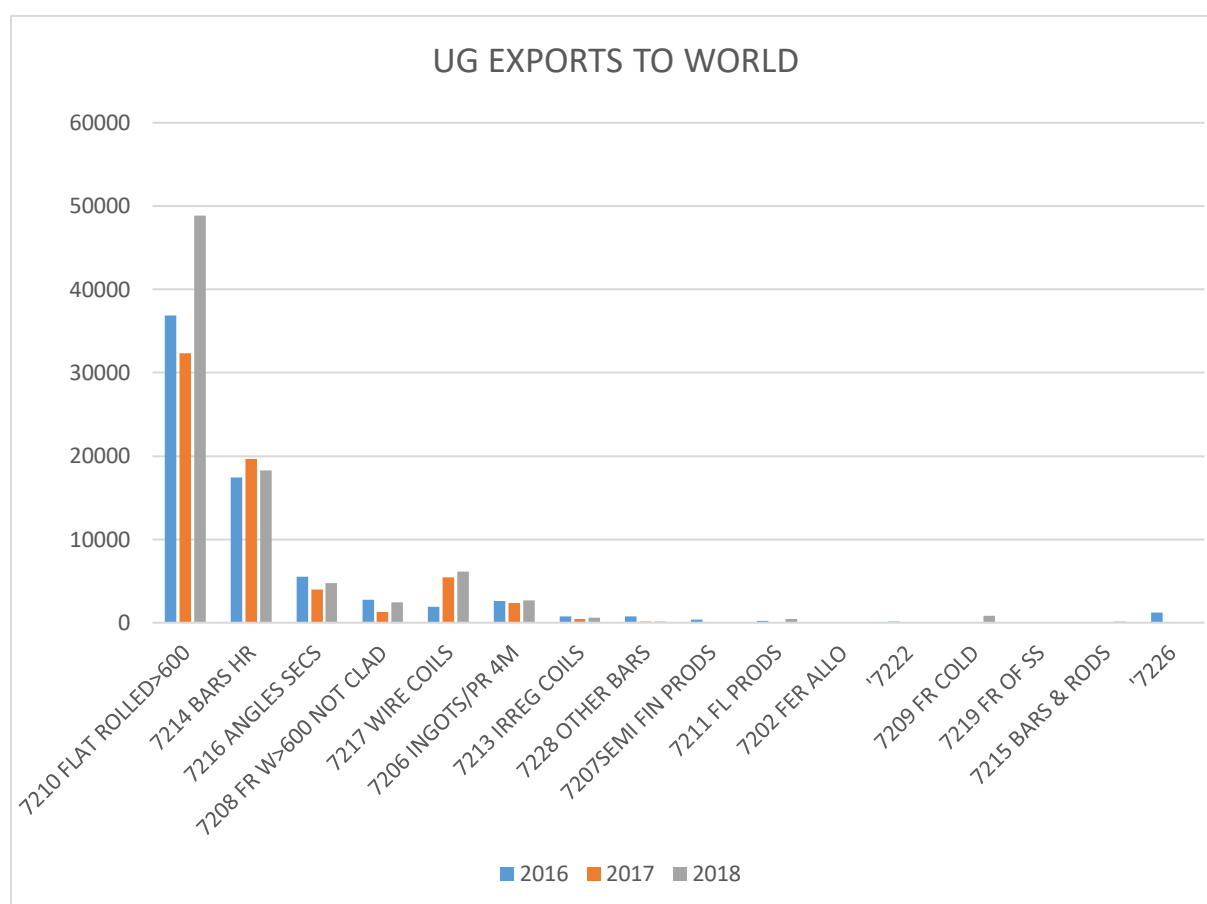


Figure 30: Uganda Exports to the World
(Source: <https://www.trademap.org/Index.aspx>)

8.5. Uganda's Positions/Choices

Uganda is expecting many spill overs arising from the establishment of a sustainable iron and steel industry. Uganda should exploit its iron ore deposits and have a percentage local and percentage of exported iron ore and steel having properly assessed profitability in each of these market segments. There should be a well guided study in public investment decision on iron ore. Regarding steelmaking, Uganda has first move advantage if it can invest ahead of the EAC and supply both domestic and regional markets.

The value chain analysis points to rich pickings for Uganda. It is presented here as advice to NPA. If there is belief in going ahead of which both technical and political will are there, then the hard-core issues must go into implementation phase, especially the infrastructural issues highlighted in chapter 7. Concurrent with these hard-core issues, some soft issues need immediate redress as well. These soft issues are presented below as advice to NPA:

8.5.1. Skills Development Program

For an organisation with good infrastructure, good finances, good technology and excellent raw materials to be competitive, high skills of its human capital as well as their dedication to smart work are what will place the organisation far ahead of competition. Recommendations on skills

development were given (Annex 5). The current analysis is that there is a skills gap. Uganda needs to implement the skills development program immediately so that in five years' time all is set for the establishment of a sustainable and viable iron and steel industry.

8.5.2. Supplier development program

Iron mining and iron and steel works require various supplies in order to carry out the transformation processes and create value in the form of finished iron ore, iron and steel products. In order to sustain these activities and be viable, a well-established program resulting in supply chain of diverse and dependable suppliers is a prerequisite. In an increasingly dynamic marketplace, businesses are expected to change faster and often these innovations come from suppliers. Supplier development is about generating a new capability or competency in suppliers. By developing suppliers, organisations can generate competitive advantage. This can manifest itself in a new product for sale, a new streamlined process or the implementation of new standard activities that benefit both supplier and buyer. The Government of Uganda has through a number of development policies and strategies highlighted the need for Local Content. It is in this spirit that the Iron and Steel project has developed a local content strategy to be enhanced by a Supplier Development Program to enable reliable and able players of the private sector and or individuals of Uganda to effectively participate. Areas of participation are found in the development, maintenance and operations of the Iron and Steel Industry. The program shall follow a systematic means that gives preference to domestically produced/ sourced goods and components or substantial value addition without compromising outcomes of the project's inception to commissioning and subsequent operations and maintenance. Suppliers are an enormous – and often untapped – pool for innovation, continuous improvement and cost reduction. By enabling this potential, companies can enjoy an enormous impact on the bottom-line. A report on supplier development to improve competitiveness of the value chain is available for consideration (Annex 4).

9. ANNEXES

In compliance with the Terms of Reference (ToR), the following annexes are included separately as part of the Feasibility Study Report:

9.5. Annex 1: Evaluation Questions and Answers

The concerns and issues arising from stakeholders, which are listed in the form of 30 evaluation questions in the ToR were addressed and incorporated in this study. However, for easy of reference, a separate document listing the questions and their corresponding answers is provided as Annex 1 of the Feasibility Report.

9.6. Annex 2: Technical Competitiveness Report

The Technical Competitiveness report is a compilation of key stakeholder visits, outlining details of steel companies' operational efficiencies and deficits, thereby identifying opportunity areas for development to enhance the steel industry in Uganda. It also includes stakeholder issues and concerns arising within the industry and how the government can develop synergies with the stakeholders to make the industry more competitive.

9.7. Annex 3: Economic Model

Results of the economic model are incorporated in section 5.4 of Chapter 5, with details of the methodology presented separately in Annex 2.

9.8. Annex 4: Supplier Development Program

The supplier development report provides further details on how to build capabilities and capacities of diverse suppliers to enhance their competitiveness, support growth in their communities and improve their effectiveness as suppliers to meet the needs of procuring companies. The program creates conditions to competitively seek contracts and business opportunities in the upcoming iron and steel value chain and on the global marketplace.

9.9. Annex 5: Skills Development Program

The report provides details on key skills required in the iron and steel value chain and the responsibilities of the various stakeholders to develop capacity and country readiness for the industry.

9.10. Annex 6: South Africa Trip

The report provides details of lessons learnt from seasoned establishments in the industry and how government policies creates competitive edge for local steel players.

9.11. Annex 7: Iron and Steel Policy Review

A review of the draft policy on iron and steel industry prepared by the Uganda Government was conducted with the aim to provide additions and adjustments to the policy document content in order to align it to going international practice that makes the industry competitive and investor friendly as provided in Annex 7

9.12. Annex 8: I&S Feasibility and Gas Demand Study Presentation

The completion of the project was followed by a presentation (Annex 8) to key stakeholders on the results of the study and the recommendations arrived at to drive the iron and steel industry forward and to utilise natural gas resources that is available in the region.

9.13. Annex 9: Workshop Validation Report

The Annex provides proceedings of the stakeholder workshop events held on the 24th and the 28th of January 2020 at the NPA offices and Golden Tulip Hotel respectively. In addition to proceedings of the events, it also outlines stakeholder's questions and feedback after the expert presentations and answers to some of the stakeholder's questions provided by the experts during the question and answer time. Questions and recommendations that could not be answered during the event period were addressed at a later stage and incorporated in the respective reports.

9.14. Annex 10: NDP III Recommendations

Recommendations arrived at following detailed analysis of the steel industry and gas demand in Uganda have been provided as part of the conclusions of each chapter in the main feasibility report. Summaries of the key recommendations are also provided in the executive summary and conclusions sections of the feasibility report.

10. CONCLUSIONS AND RECOMMENDATIONS

Results of the research indicate positive scope for the development of an I&S value chain. Several steel companies are already at play in the production of steel at the downstream end of the value chain while a few are pursuing exploration and mineral development on the upstream end for the purposes of producing iron ore for steel plants raw material feed.

Iron ore deposits exist in South Western and in Eastern Uganda, but little is mined to feed into the existing steel industry, which is currently importing semi processed steel from other countries for its raw material feed. The country is also short on scrap metal to fully develop EAF plants that can feed the steel industry. It is therefore coming as no surprise that Uganda is actively pursuing the development of its iron ore deposits to close the gap between the upstream side and the downstream end of the value chain.

Whilst iron ore deposits are available in Uganda, quantification and characterization into mineral resource/reserve are yet to be done, an important part of the value chain development process which would aid in the determination of technologies to be employed on mining, processing, iron making and steel production, as well as infrastructural and logistical requirements of the whole process.

In light of these opportunity areas and the constraints associated with them, this research has come up with the following recommendations as viable options that can eventually deliver the development of the value chain.

- The government is encouraged to consider the exploration of the deposits as a national agenda and therefore invest in exploration to fully evaluate the deposits and delineate a mineral resource that can be used to plan the path the I&S value chain development should take. Exploration should be carried out by a professional organization that can produce results that conform to the ongoing international reporting standards like the NI43-101 of Canada, the JORC code of Australia and the SAMREC of South Africa. There is significant value addition that is gained by using these standards especially when seeking investment from private/listed corporations, be it PPP, JVs or open tender.
- Iron ore, being a bulk commodity requires well planned sustainable exploitation strategy, which can only be delivered by well experienced corporations with enough capital to invest in the industry. It is common practice for a government investment arm to take position either as a standalone investor or as a partner in the business. The numerous small-scale miners that have taken position in the area, with either exploration or mining licenses issued to them, are a potential hindrance to large scale mining investment due to the risks associated with their artisanal operations unless the government re-aligns its policies on the criteria for investment in bulk commodities like iron ore.
- Based on outcrops observed in the iron formations, there is high variability in hematite ore geometry, with discontinuous beds of varying thicknesses - Geological mapping of the entire ironstone region is highly recommended as a first step ahead of any other exploration activity in order to fully understand the extent and structural controls of the mineralization. Geological mapping and the geophysical survey information available will then pave the way for targeting and subsequent exploration strategy.
- The available geophysical data should be further analysed by an expert geophysicist who can provide leads on the likely geometry and the minimum expected down-dip extension of the

deposit. This information from the analysis can provide direction and scope on the minimum required exploration activity in the early stages of the program.

- The different hematite types (massive and friable specularite) should be sampled for metallurgical tests in the early stages of development of the iron and steel value chain. The results will inform the stakeholders on the likely scenarios required to prepare the ore for the reduction processes – Hematite fines from mining activity and the platy specularite ores may require sintering or pelletisation before feeding into the iron making processes.
- Coal based DRP represents a feasible starting point to the development of the steel industry in Uganda and in accordance with steel market the RKF seems the most suitable solution to produce liquid steel. The utilization of a gas based DRP (MIDREX and HYL) is not suitable for the present needs of Uganda's development of iron and steel value chain. It is however the best available process in the order of 500,000t/y DRI production both for efficiency and environmental reasons. It is therefore the best available technology for Uganda once the following pre-conditions are fulfilled: natural gas availability, stable electric energy grid, proper infrastructures and increased market for steel products.
- In parallel to the geological work that must be undertaken, work on infrastructure and logistics must start in-order to support the private sector that has already made tangible moves regardless of a few grey areas. This work will bring national benefit even before the bigger plant is finally brought online after geological confirmation.
- For the enabling activities that are required for realization of the iron and steel value chain, a schedule of priority tasks in their sequence should be prepared where the critical path is defined. This schedule should then be distributed to identified ministries and institutions. There is need for a multi-sectorial approach as alluded to during discussion with the stakeholders visited. The current level of collaboration with national research and development institutions is low and there is need to have coordinated multi sectorial approach.
- The governance structures would identify which institutions will do which tasks and who shall have overall responsibility to deliver the result. As much as the multi-sectorial committee shall jointly coordinate the agenda for the iron and steel companies, a fully dedicated working group or task force for the business of establishing an iron and steel industry should be put together with all relevant structures and clear terms, with targets for deliverables: DGSM shall continue its mandate of delivering a state level geological quantification and characterisation of the iron ore deposits, the ministries in the infrastructure and logistics cluster will deliver on all the requirements highlighted, MoTIC and UDC to follow up on relevant issues whilst MoFPED does the Economic and Financial Planning plus implementation.
- For purposes of capacity building and promotion of local content, an immediate set up of supplier development programs and institutions of training for skills development will be the highest level of demonstrating that there is seriousness to uphold the survival of the existing steel manufacturing industry and the eventual commissioning of a comprehensive end to end iron and steel value chain that will be dependent on Uganda's iron ore as the ferrous raw material for steelmaking.
- A proper sequencing of the development of the value chain should be followed as provided in the recommendations in the respective chapters of the report, with exploration and ore characterisation as the starting point. The development of ironmaking and steel plants can then follow based on the market dynamics as predicted by the economic model and/or government strategy for the steel industry.